

Principles of Quantum Mechanics – R. Shankar

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Contents

I Basics	4
1 Mathematical Introduction	5
1.1 The Delta function	5
1.2 The Derivative Operator	6
1.2.1 The Hermitian Operator $K = -iD$	6
1.2.2 Eigenvalues of K	7
1.2.3 Conjugate actions of X and K	8
2 Review of Classical Mechanics	9
2.1 Lagrangian Mechanics	9
2.1.1 Procedure	10
2.1.2 The Euler-Lagrange Equation	10
2.1.3 Conservation Laws from the Lagrangian	11
2.1.4 The Electromagnetic Lagrangian	12
2.1.5 The Two-Body Problem	13
2.2 Hamiltonian Mechanics	13
2.2.1 Derivation	13
2.2.2 The Electromagnetic Hamiltonian	14
2.2.3 Conservation Laws from the Hamiltonian	14
2.2.4 Canonical Transformations	15
2.2.5 Active Transformations	16
2.2.6 Symmetries	16
2.2.7 A Relation Between S and E	17
2.3 A Problem: Wave-Particle Duality	18
2.3.1 Plane Waves	18
2.3.2 Double-Slit Experiment	18
3 Postulates of Quantum Mechanics	19
3.1 The Postulates	19
3.1.1 Ensembles	20
3.1.2 Compatible Variables	21
3.1.3 Density Matrices	21
3.2 Example: Gaussian Wavefunction	22

3.3	Generalization to N Degrees of Freedom	23
3.4	The Schrödinger Equation	24
3.4.1	Hamiltonian Setup	24
3.4.2	Time-Independent Schrödinger Equation	24
3.4.3	Time-Dependent Schrödinger Equation	25
4	Simple Problems in 1D	25
4.1	The Free Particle	25
4.1.1	Allowed Energy States	25
4.1.2	Propagator	26
4.2	The Particle in a Box	26
4.2.1	Wavefunction	26
4.2.2	Allowed Energies	27
4.3	The Continuity Equation of Probability	28
4.4	Quantum Tunneling	28
4.5	Ehrenfest's Theorem	29
5	The Harmonic Oscillator	29
5.1	The Classical Oscillator	29
5.2	Quantum Oscillator in the X Basis	30
5.3	Quantum Oscillator in the Energy Basis	31
6	The Path Integral Formulation	31
6.1	Overview	31
6.2	The Free Particle	32
6.2.1	Classical Approximation	32
6.2.2	Full Solution	32
7	The Heisenberg Uncertainty Relations	34
7.1	Derivation	34
7.2	Equality	34
7.3	Energy-Time Relation	35
8	Systems with N Degrees of Freedom	35
8.1	The Direct Product Space $\mathbb{V}_{1\otimes 2}$	35
8.2	The Two-Particle State	36
8.2.1	Separable Hamiltonians	36
8.3	Identical Particles	37
8.3.1	Bosons and Fermions	37
8.3.2	Spin Statistics	37
8.3.3	Symmetrization/Antisymmetrization of N Particles	38
II	Angular Momentum	38
9	Transformations and Their Generators	38
9.1	Translations	39
9.1.1	Active vs. Passive Transformations	39
9.1.2	Momentum as a Generator of Translations	39

9.1.3	Finite Translations	40
9.2	Invariance Under Transformations	40
9.2.1	Time-Translation Invariance	40
9.2.2	Parity Invariance	41
9.2.3	Time-Reversal Symmetry	41
9.3	Rotations	42
9.4	Summary: J, L, S	43
10	Orbital Angular Momentum	43
10.1	Angular Momentum in 3 Dimensions	43
10.2	Relation between L^2 and L_Z	44
10.3	Total Angular Momentum: $J = L + S$	45
10.4	Case Study: Free Particle in Spherical Coordinates	45
10.4.1	Schrodinger Equation in Spherical Coordinates	45
10.4.2	Free Particle in Spherical Coordinates	46
10.4.3	Bessel and Neumann Wavefunctions	46
10.5	The Isotropic Oscillator	47
10.6	The Hydrogen Atom	48
10.6.1	Schrodinger Equation	48
11	Spin	49
11.1	The Spin Matrix	49
11.1.1	Pauli Matrices	49
11.2	Spin Dynamics	50
11.2.1	Orbital Magnetic Moment	50
11.2.2	Quantum Spin Magnetic Moment	51
11.2.3	Parametric Resonance	51
11.3	The Zeeman Effect	51
11.4	Addition of Spin	52
11.4.1	Product Basis	52
11.4.2	Total- s basis	53
11.4.3	Construction of Fermion Wavefunctions	53
11.5	Addition of Total Angular Momenta	54
III	Further Topics	54
12	The WKB Method	54
12.1	Methodology	54
13	Perturbation Theory	55
13.1	Time-Independent Perturbation Theory	55
13.2	Ex. Harmonic Oscillator in Electric Field	56
13.3	Time-Dependent Perturbation Theory	56
13.3.1	Ex. Harmonic Oscillator in Electric Field	57
13.3.2	The Adiabatic Theorem	57
13.3.3	Fermi's Golden Rule	57
13.4	Electrodynamics	58

13.4.1 The Aharonov-Bohm Effect	59
13.4.2 The Photoelectric Effect in Hydrogen	60
14 Field Quantization	61
14.1 The Need for Quantized Fields	61
14.2 Sketch of Procedure	61
15 Scattering Theory	63
15.1 Procedure	63
15.2 Alternative: Green's Function	64
15.3 Born Approximation	66
15.3.1 Perturbation Theory	66
15.3.2 Yukawa & Coulomb Potentials	66
16 The Dirac Equation	67
16.1 Derivation	67
16.1.1 Klein-Gordon Equation	67
16.1.2 Dirac Equation	68
16.2 Positive/Negative Energy Solutions	68
17 Path Integrals: Part II	69
17.1 Different Path Integrals	69
17.1.1 Configuration Space Path Integral	69
17.1.2 Phase Space Path Integral	70
17.2 Landau Levels	71
17.3 The Berry Phase	72
17.3.1 Accumulation of Geometric Phase	72
17.3.2 Vector and Scalar Berry Potentials	73
17.3.3 The Magnetic Monopole Argument	74
17.4 Imaginary-Time Path Integrals	75
17.4.1 Formalism	75
17.4.2 Spontaneous Symmetry Breaking	75
17.5 Quantum Statistical Mechanics	77
17.5.1 The Classical Limit	77
17.5.2 Coupled Oscillators	77
17.5.3 The Ising Model	78
17.6 Path Integrals with Spin	79
17.6.1 A Brief Introduction	79
17.6.2 Grassman Variables and Berezin Integration	80
17.6.3 The Fermionic Path Integral	81

Part I

Basics

1 Mathematical Introduction

1.1 The Delta function

In discrete space with basis $|x_i\rangle$,

- Orthogonality: $\langle x_i | x_j \rangle = \delta_{ij}$
- Completeness: $\sum_{i=1}^n |x_i\rangle \langle x_i| = I$

Natural generalization to continuous functions.

- Orthogonality: $\langle x | x' \rangle = \delta(x - x')$
- Completeness: $\int_a^b |x'\rangle \langle x| dx = I$

δ is the Dirac delta function, defined as

$$\begin{aligned} \delta(x - x') &= 0, \quad x \neq x' \\ \int_a^b \delta(x - x') dx' &= 1, \quad a < x < b \end{aligned}$$

Sometimes we write $\delta(0) = \infty$, but this isn't well defined except in the context of an integral. The δ also works like a "sampling function":

$$\int \delta(x - x') f(x') dx' = f(x)$$

Derivative: The derivative of the delta function acts like an operator

$$\delta'(x - x') = \delta(x - x') \frac{d}{dx}$$

To see this, note that

$$\int \delta'(x - x') f(x') dx' = \frac{d}{dx} \int \delta(x - x') f(x') dx' = \frac{df(x)}{dx}$$

Fourier Series: We can also represent the delta function using a fourier series. Recall that

$$\text{Fourier transform: } f(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} f(x) dx$$

$$\text{Inverse transform: } f(x') = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx'} f(k) dk$$

$$f(x') = \int_{-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ik(x'-x)} dk \right) f(x) dx$$

Reasoning from the use of delta as a sampling function,

$$\delta(x' - x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ik(x'-x)} dk$$

1.2 The Derivative Operator

We can have operators in infinite dimensions, e.g.

$$\Omega |f\rangle = |\tilde{f}\rangle$$

The derivative is one such operator:

$$D |f\rangle = \left| \frac{df}{dx} \right\rangle$$

The elements of D in the x basis are

$$D_{xx'} \equiv \langle x | D | x' \rangle = \delta'(x - x') = \delta(x - x') \frac{d}{dx'}$$

To see this, recall the use of δ as a sampling function.

$$\begin{aligned} \langle x | D | f \rangle &= \langle x | \frac{df}{dx} \rangle = \frac{df(x)}{dx} \\ \int \langle x | D | x' \rangle \langle x' | f \rangle dx' &= \frac{df}{dx} \end{aligned}$$

1.2.1 The Hermitian Operator $K = -iD$

However, D is not Hermitian: $D_{xx'} \neq D_{x'x}^*$.

$$\begin{aligned} D_{xx'} &= \delta'(x - x') \\ D_{x'x}^* &= \delta'(x' - x)^* = -\delta'(x - x') \end{aligned}$$

Instead, we can make D Hermitian using a constant complex term: $K = -iD$. Then

$$\begin{aligned} K_{xx'} &= -i\delta'(x - x') \\ K_{x'x}^* &= [-i\delta'(x' - x)]^* = i\delta'(x' - x) = -i\delta(x - x') \end{aligned}$$

Warning: For operators in infinite dimensions, $K_{xx'} = K_{x'x}^*$ doesn't actually guarantee Hermiticity. To see this, note that a Hermitian operator must satisfy (for functions f, g)

$$\begin{aligned} \langle g | K | f \rangle &= \langle f | K | g \rangle^* \\ \int_a^b \int_a^b \langle g | x \rangle \langle x | K | x' \rangle \langle x' | f \rangle dx dx' &\stackrel{?}{=} \left(\int_a^b \int_a^b \langle f | x \rangle \langle x | K | x' \rangle \langle x' | g \rangle dx dx' \right)^* \\ \int_a^b g^*(x) \left[-i \frac{df(x)}{dx} \right] dx &\stackrel{?}{=} \left(\int_a^b f^*(x) \left[-i \frac{dg(x)}{dx} \right] dx \right)^* \\ -ig^*(x)f(x) \Big|_a^b + i \int_a^b \frac{dg^*(x)}{dx} f(x) dx &\neq i \int_a^b \frac{dg^*(x)}{dx} f(x) dx \end{aligned}$$

In order to be Hermitian, the **surface term** of K must be 0. K is Hermitian in the space of functions where this holds. $K_{xx'} = K_{x'x}^*$ is no longer a sufficient condition.

$$-ig^*(x)f(x) \Big|_a^b = 0$$

Examples where

- String clamped at one end, $f(0) = f(L) = 0$. Derivative is $K = -i \frac{d}{dx}$
 - Periodic function, $f(\phi) = f(\phi + 2\pi)$. Derivative is $K = -i \frac{d}{d\phi}$
-

In QM, we study functions $-\infty < x < \infty$. There are two classes of such functions.

1. f vanishes as $x \rightarrow -\infty$ and $x \rightarrow \infty$. Here, $K = -i \frac{d}{dx}$ is Hermitian.
2. f doesn't vanish, like e^{ikx} . If K is Hermitian, it must satisfy

$$e^{ikx} e^{ik'x} \Big|_{-\infty}^{\infty} = 0$$

If $k = k'$, this holds. Otherwise, we define the limit as $|x| \rightarrow \infty$ to be the average over a large interval. Then if $k \neq k'$,

$$\lim_{x \rightarrow \infty} e^{ikx} e^{ik'x} = \lim_{L, \Delta \rightarrow \infty} \frac{1}{\Delta} \int_L^{L+\Delta} e^{i(k-k')x} dx = 0$$

Thus K is also Hermitian for class 2 functions.

1.2.2 Eigenvalues of K

For eigenvalue k ,

$$\begin{aligned} K |k\rangle &= k |k\rangle \\ \langle x | K |k\rangle &= k \langle x | k\rangle \\ \int \langle x | K |x'\rangle \langle x' | k\rangle dx' &= k \langle x | k\rangle \end{aligned}$$

Let $\langle x | k\rangle = \psi_k(x)$. Using the definition $K = -iD$,

$$-i \frac{d}{dx} \psi_k(x) = k \psi_k(x)$$

The solution is $\psi_k(x) = A e^{ikx}$. Thus, any real number k is an eigenvalue of K with eigenvector $A e^{ikx}$. Since A is a free parameter, set it to $1/\sqrt{2\pi}$ so that

$$\begin{aligned} |k\rangle &\leftrightarrow \frac{1}{\sqrt{2\pi}} e^{ikx} \\ \langle k | k'\rangle &= \int_{-\infty}^{\infty} \langle k | x\rangle \langle x | k'\rangle dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(k'-k)x} dx = \delta(k' - k) \end{aligned}$$

Thus $\langle k | k'\rangle = 0$ if $k \neq k'$, and blows up if $k = k'$.

We require a real eigenvalue k because if k is imaginary, e^{ikx} blows up as $x \rightarrow \infty$. The choice of a real eigenvalue restricts us to the **physical Hilbert space**.

1.2.3 Conjugate actions of X and K

In finite dimensions, the eigenfunctions of a Hermitian operator form a complete basis. Assume this holds for infinite dimensions.

K is Hermitian. Expand in both X and K basis. Recall $|k\rangle = \frac{1}{\sqrt{2\pi}} e^{ikx}$.

$$\begin{aligned} f(k) &= \langle k|f\rangle = \int_{-\infty}^{\infty} \langle k|x\rangle \langle x|f\rangle dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} f(x) dx \\ f(x) &= \langle x|f\rangle = \int_{-\infty}^{\infty} \langle x|k\rangle \langle k|f\rangle dk = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx} f(k) dk \end{aligned}$$

The Fourier transform pass f from one complete basis to another.

Additionally, note the matrix elements of X and K in the X and K basis, respectively:

$$\begin{aligned} \langle x|X|x'\rangle &= x'\delta(x-x') \\ \langle k|K|k'\rangle &= k'\delta(k-k') \end{aligned}$$

In finite dimensions, we had (in the X basis) $X|x\rangle = x|x\rangle$. To see the action of X in infinite dimensions, let $X|f\rangle = |\tilde{f}\rangle$.

$$\begin{aligned} \langle x|X|f\rangle &= \int_{-\infty}^{\infty} \langle x|X|x'\rangle \langle x'|f\rangle dx' = xf(x) \\ \langle x|X|f\rangle &= \langle x|\tilde{f}\rangle = \tilde{f}(x) \end{aligned}$$

Thus $\tilde{f}(x) = xf(x)$, and the effect of X in infinite dimensions is $X|f(x)\rangle = |xf(x)\rangle$. That is, multiplying $f(x)$ by x .

An interesting phenomenon occurs if we calculate the matrix elements of X in the K basis. Recall $|k\rangle = \frac{1}{\sqrt{2\pi}} e^{ikx}$.

$$\begin{aligned} \langle k|X|k'\rangle &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \langle k|x\rangle \langle x|X|x'\rangle \langle x'|k'\rangle dx dx' \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x'\delta(x-x') \langle k|x\rangle \langle x'|k'\rangle dx dx' \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ikx} e^{ik'x} dx \\ &= i \frac{d}{dk} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(k'-k)x} dx \right) = i\delta'(k-k') \end{aligned}$$

Thus, if g is a function with image $g(k)$ in the K basis,

$$X|g(k)\rangle = |i \frac{dg(k)}{dk}\rangle$$

To summarize,

$$\begin{aligned} \text{In } X \text{ basis } & \begin{cases} X \equiv x \\ K \equiv -i \frac{d}{dx} \end{cases} \\ \text{In } K \text{ basis } & \begin{cases} X \equiv i \frac{d}{dx} \\ K \equiv k \end{cases} \end{aligned}$$

Operators like these are **conjugate** to each other.

Note that X and K do not commute.

$$\begin{aligned} XK |f\rangle &= -ix \frac{df(x)}{dx} \\ KX |f\rangle &= -i \frac{d}{dx} (xf(x)) = -ix \frac{df(x)}{dx} - if(x) \end{aligned}$$

Thus $[X, K] |f\rangle = if = iI |f\rangle$. Since f was arbitrary, we have $[X, K] = iI$.

In QM, X and K are the main operators we'll be seeing (X gives position, K gives derivative). However, they take different forms: $\hat{X} = X$ and $\hat{P} = \hbar K$.

2 Review of Classical Mechanics

2.1 Lagrangian Mechanics

Newtonian mechanics can find the path of a particle at an infinitesimal time slice:

$$m\ddot{x} = -\frac{dV}{dx}$$

Given $x(t_i)$ and $\dot{x}(t_i)$ at a point t_i , we can obtain the classical path's $x_{\text{cl}}(t_i + \Delta t)$ and $\dot{x}_{\text{cl}}(t_i + \Delta t)$ after some infinitesimal time Δt using the equation above. By continuing this process, we recreate the classical path x_{cl} of the particle.

$$\begin{aligned} x_{\text{cl}}(t_i + \Delta t) &= x(t_i) + \dot{x}(t_i)\Delta t \\ \dot{x}_{\text{cl}}(t_i + \Delta t) &= \dot{x}(t_i) + \ddot{x}(t_i)\Delta t \end{aligned}$$

Lagrangian mechanics seeks to find the classical path immediately. Given two points (x_i, t_i) and (x_f, t_f) , we can find the path taken by the particle between them. But what distinguishes the classical path from all other possible ones?

2.1.1 Procedure

1. Define the Lagrangian $L = T - V$. Assume time-independent potential, thus $L = L(x, \dot{x})$.
2. For each path from (x_i, t_i) to (x_f, t_f) , define the action as a function of the entire path:

$$S[x(t)] = \int_{t_i}^{t_f} L(x, \dot{x}) dt$$

3. **Principle of least action:** the classical path is that on which S is minimized.

NB: assumption of a time-independent potential doesn't always hold. For example, $F_B = qv \times B$. So $L = T - V$ isn't always the correct form.

2.1.2 The Euler-Lagrange Equation

For finite variables, the function was at a extremal if the 1st order change δf vanishes:

$$f(\mathbf{x}^0 + \eta) = f(\mathbf{x}^0) + \sum_{i=1}^n \left. \frac{\partial f}{\partial x_i} \right|_{\mathbf{x}^0} \eta_i + O(\eta^2)$$

$$\delta f = \sum_{i=1}^n \left. \frac{\partial f}{\partial x_i} \right|_{\mathbf{x}^0} \eta_i = 0$$

Assume this holds for functions of infinite variables, like $S[x(t)]$. Let $x_{cl}(t)$ be the classical path, and $x_{cl}(t) + \eta(t)$ a nearby path. Endpoints must match, so $\eta(t_i) = \eta(t_f) = 0$. Now find the extremal where 1st order change δS vanishes:

$$\begin{aligned} S[x_{cl}(t) + \eta(t)] &= \int_{t_i}^{t_f} L(x_{cl}(t) + \eta(t), \dot{x}_{cl}(t) + \dot{\eta}(t)) dt \\ &= \int_{t_i}^{t_f} \left[L(x_{cl}(t), \dot{x}_{cl}(t)) + \left. \frac{\partial L}{\partial x(t)} \right|_{x_{cl}} \cdot \eta(t) + \left. \frac{\partial L}{\partial \dot{x}(t)} \right|_{x_{cl}} \cdot \dot{\eta}(t) + \dots \right] dt \\ &\approx S[x_{cl}(t)] + \delta S \end{aligned}$$

Find where $\delta S = 0$ for an arbitrary deviation η :

$$\begin{aligned} 0 = \delta S &= \int_{t_i}^{t_f} \left[\left. \frac{\partial L}{\partial x(t)} \right|_{x_{cl}} \cdot \eta(t) + \left. \frac{\partial L}{\partial \dot{x}(t)} \right|_{x_{cl}} \cdot \dot{\eta}(t) \right] dt \\ &= \int_{t_i}^{t_f} \left[\left. \frac{\partial L}{\partial x(t)} \right|_{x_{cl}} \cdot \eta(t) \right] dt + \left[\left. \frac{\partial L}{\partial \dot{x}(t)} \right|_{x_{cl}} \cdot \eta(t) \right]_{t_i}^{t_f} - \int_{t_i}^{t_f} \left[\frac{d}{dt} \left. \frac{\partial L}{\partial \dot{x}(t)} \right|_{x_{cl}} \cdot \eta(t) \right] dt \\ &= \int_{t_i}^{t_f} \left[\left. \frac{\partial L}{\partial x(t)} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}(t)} \right|_{x_{cl}} \cdot \eta(t) \right] dt \end{aligned}$$

Thus we get the Euler-Lagrange equation, which must be satisfied by a classical path:

$\left[\frac{\partial L}{\partial x(t)} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}(t)} \right]_{x_{cl}} = 0$
--

2.1.3 Conservation Laws from the Lagrangian

A benefit of the Lagrangian is that unlike the Newtonian, coordinates don't have to be Cartesian. We can have an arbitrary q_i and $\dot{q}_i$.

Further note:

- Canonical momentum conjugate: $p_i = \partial L / \partial \dot{q}_i$
- Canonical force conjugate: $F_i = \partial L / \partial q_i$

$$F_i = \frac{d}{dt} p_i$$

E.g. if $q_i = \theta$, then p_i is angular momentum and F_i is torque.

Call q_i a **cyclic coordinate** if L depends on $\dot{q}_i$, but not q_i . Then p_i is conserved:

$$\frac{dp_i}{dt} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = \frac{\partial L}{\partial q_i} = 0$$

Conservation laws fall out naturally out of the Lagrangian, while it may be opaque in the Newtonian. For example, consider a Lagrangian in polar coordinates $L = L(\rho, \phi)$. The metric is $ds^2 = d\rho^2 + \rho^2 d\phi^2$. Thus the infinitesimal velocity is

$$v = \frac{ds}{dt} = \sqrt{(\dot{\rho})^2 + (\rho\dot{\phi})^2}$$

The Lagrangian is

$$L = \frac{1}{2} m [\dot{\rho}^2 + \rho^2 \dot{\phi}^2] - V(\rho, \phi)$$

Applying Euler-Lagrange for ρ and ϕ respectively gives

$$\begin{aligned} \frac{d}{dt} (m\dot{\rho}) &= -\frac{\partial V}{\partial \rho} + m\rho\dot{\phi}^2 \quad \longrightarrow \quad m\ddot{\rho} = -\frac{\partial V}{\partial \rho} + m\rho(\dot{\phi})^2 \\ \frac{d}{dt} (m\rho^2\dot{\phi}) &= -\frac{\partial V}{\partial \phi} \quad \longrightarrow \quad m\ddot{\phi} = -\frac{1}{\rho^2} \frac{\partial V}{\partial \phi} - \frac{1}{\rho} (2m\dot{\rho}\dot{\phi}) \end{aligned}$$

Note how easily we can derive the centrifugal and coriolis forces. Also note that conservation laws for cyclic coordinates $V(\rho, \phi) = V(\rho)$ and $V(\rho, \phi) = V(\phi)$ fall out naturally, and the corresponding equations of motion are that of $\partial V / \partial \phi = 0$ and $\partial V / \partial \rho = 0$.

In sum:

- Equations of motion are simpler to find with Lagrangian
- Conservation laws are easy to see from Lagrangian
- Path integral formalism uses Lagrangian (later)

2.1.4 The Electromagnetic Lagrangian

Electromagnetic force:

$$\vec{F} = q \left(\vec{E} + \frac{\vec{v}}{c} \times \vec{B} \right)$$

Note: this is Gaussian units, hence the $1/c$ factor for magnetic force.

The electromagnetic force is velocity-dependent, so $L = T - V$ doesn't work. Instead, we construct L to reduce to the expected equations. The EM lagrangian is

$$L_{\text{EM}} = \frac{1}{2} m \vec{v} \cdot \vec{v} - q\phi + \frac{q}{c} \vec{v} \cdot \vec{A}$$

ϕ and $\vec{A}$ are the scalar and vector potentials, related to E and B by

$$\begin{aligned} \vec{E} &= -\nabla\phi - \frac{1}{c} \frac{\partial \vec{A}}{\partial t} \\ \vec{B} &= \nabla \times \vec{A} \end{aligned}$$

The Euler-Lagrange equation is

$$\begin{aligned} \frac{d}{dt} \left(m\dot{x}_i + \frac{q}{c} A_i \right) &= -q \frac{\partial \phi}{\partial x_i} + \frac{q}{c} \frac{\partial (\vec{v} \cdot \vec{A})}{\partial x_i}, \quad i = 1, 2, 3 \\ \frac{d}{dt} \left(m\vec{v} + \frac{q}{c} \vec{A} \right) &= -q \nabla \phi + \frac{q}{c} \nabla (\vec{v} \cdot \vec{A}) \\ \frac{d}{dt} (m\vec{v}) &= -q \nabla \phi + \frac{q}{c} \left[-\frac{d\vec{A}}{dt} + \nabla (\vec{v} \cdot \vec{A}) \right] \end{aligned} \tag{1}$$

By time-dependence of the magnetic vector potential $\vec{A}$,

$$\frac{d\vec{A}}{dt} = \frac{\partial \vec{A}}{\partial t} + \sum_i \frac{\partial \vec{A}}{\partial x_i} \frac{dx_i}{dt} = \frac{\partial \vec{A}}{\partial t} + (\vec{v} \cdot \nabla) \vec{A}$$

Thus equation (1) becomes

$$\frac{d}{dt} (m\vec{v}) = -q \nabla \phi - \frac{q}{c} \frac{\partial \vec{A}}{\partial t} + \frac{q}{c} \underbrace{\left[\nabla (\vec{v} \cdot \vec{A}) - (\vec{v} \cdot \nabla) \vec{A} \right]}_{\vec{v} \times (\nabla \times \vec{A})} \tag{2}$$

This reduces to $\vec{F} = q\vec{E} + \frac{q}{c} \vec{v} \times \vec{B}$.

The time-dependent EM field is usually not conservative, so there's no path-independent work function to use as the potential. Thus L_{EM} is made to fit the situation and reduce as expected. $L = T - V$ only works for conservative forces.

2.1.5 The Two-Body Problem

Assume 2 masses m_1, m_2 that exert equal and opposite forces on each other. We can reduce this to a one-body problem with Lagrangian mechanics.

The potential only depends on relative position, $V(\vec{r}_1, \vec{r}_2) = V(\vec{r}_1 - \vec{r}_2)$. Thus $6 \rightarrow 3$ cyclic coordinates in 3D.

The two-body Lagrangian is

$$L = \frac{1}{2}m_1|\dot{\vec{r}}_1|^2 + \frac{1}{2}m_2|\dot{\vec{r}}_2|^2 - V(r_1 - r_2) \quad (1)$$

Define $\vec{r}_{\text{CM}}$ and $\vec{r}$ such that

$$\begin{aligned} \vec{r}_{\text{CM}} &= \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}, \quad \vec{r} = \vec{r}_1 - \vec{r}_2 \\ \vec{r}_1 &= \vec{r}_{\text{CM}} + \frac{m_2\vec{r}}{m_1 + m_2} \\ \vec{r}_2 &= \vec{r}_{\text{CM}} - \frac{m_1\vec{r}}{m_1 + m_2} \end{aligned}$$

The two-body Lagrangian becomes

$$L = \frac{1}{2}(m_1 + m_2)|\dot{\vec{r}}_{\text{CM}}|^2 + \frac{1}{2}\frac{m_1m_2}{m_1 + m_2}|\dot{\vec{r}}|^2 - V(\vec{r}) \quad (2)$$

This can be thought of as the Lagrangian of two fictitious particles that don't interact: one "CM" particle with mass $M = m_1 + m_2$, and one "reduced mass" particle with mass $\mu = (m_1m_2)/(m_1 + m_2)$. Note that $L(r, \dot{r}, r_{\text{CM}}, \dot{r}_{\text{CM}}) = L(r, \dot{r}) + L(r_{\text{CM}}, \dot{r}_{\text{CM}})$.

By going to the CM frame, we can eliminate $\dot{r}_{\text{CM}}$ and just solve for the motion of μ . We've reduced the two-body problem to a one-body problem.

2.2 Hamiltonian Mechanics

2.2.1 Derivation

The Lagrangian uses $L(q, \dot{q})$. The Hamiltonian reverses the role of p and $\dot{q}$, using $H(q, p)$. This "role reversal" is achieved with a **Legendre transform**. Suppose $u(x) = df/dx$ and $x(u) = dg/du$. The relationship between f and g is a Legendre transform:

$$g(u) = x(u)u - f(x(u))$$

Define the Hamiltonian $H(q, p)$ as a Legendre transform of the Lagrangian $L(q, \dot{q})$:

$$H(q, p) = \sum_{i=1}^n p_i \dot{q}_i - L(q, \dot{q}) \quad , \quad p_i = \frac{\partial L}{\partial \dot{q}_i}$$

Hamilton's **canonical equations** can be found:

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \quad , \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Lagrangian	Hamiltonian
n coordinates q_i and n velocities $\dot{q}_i$	n coordinates q_i and n momenta p_i
State of system is point in n -dimensional configuration space	State of system is point in $2n$ -dimensional configuration space
n coordinates evolve according to n 2nd-order differential equations	$2n$ coordinates and momenta evolve according to $2n$ 1st-order differential equations
For given L , multiple trajectories can pass through a point in configuration space depending on $\dot{q}_i$.	For given H , only one trajectory passes through a point in configuration space.

For conservative forces with $T = \frac{1}{2}m\dot{x}_i^2$, we get

$$p_i = \frac{\partial L}{\partial \dot{x}_i} = m_i \dot{x}_i$$

$$H = \sum_i p_i \dot{x}_i - L(x, \dot{x}) = T + V$$

But this isn't always the right form. It's only for conservative forces.

2.2.2 The Electromagnetic Hamiltonian

If L_{EM} generates the correct equations of motion, then H_{EM} will as well. But it won't take the form $H = T + V$ since now the force isn't conservative.

Start from the electromagnetic Lagrangian:

$$L_{\text{EM}} = \frac{1}{2}m\vec{v} \cdot \vec{v} - q\phi + \frac{q}{c}\vec{v} \cdot \vec{A}$$

Now find the canonical momentum $\vec{p}$ and the Hamiltonian H :

$$\vec{p} = \frac{\partial L}{\partial \vec{v}} = m\vec{v} + \frac{q}{c}\vec{A}$$

$$H_{\text{EM}} = \vec{p} \cdot \vec{v} - L_{\text{EM}} = \frac{1}{2}m\vec{v} \cdot \vec{v} + q\phi = \frac{|(\vec{p} - q\vec{A}/c)|^2}{2m} + q\phi$$

2.2.3 Conservation Laws from the Hamiltonian

If q_i is a cyclic coordinate in the Hamiltonian, then

$$\dot{p}_i = -\frac{\partial H}{\partial q_i} = 0$$

The associated momentum is conserved. But other quantities are too, and this is often easier to see in the Hamiltonian than in the Lagrangian.

Let $\omega(p, q)$ be a state function with **no explicit dependence on t** . Define the **Poisson bracket**

$$\{\omega, \lambda\} = \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial \lambda}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial \lambda}{\partial q_i} \right)$$

The time derivative of ω is:

$$\begin{aligned} \frac{d\omega}{dt} &= \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial q_i}{\partial t} + \frac{\partial \omega}{\partial p_i} \frac{\partial p_i}{\partial t} \right) \\ &= \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial H}{\partial q_i} \right) \\ &= \{\omega, H\} \end{aligned}$$

Any state function ω whose Poisson bracket with H vanishes is conserved. H itself is conserved if there's no explicit t dependence.

The Poisson bracket between q and p satisfy

$$\begin{aligned} \{q_i, q_j\} &= \{p_i, p_j\} = 0 \\ \{q_i, p_j\} &= \delta_{ij} \end{aligned}$$

From these, the canonical equations can be derived in terms of Poisson brackets:

$$\begin{aligned} \dot{q}_i &= \{q_i, H\} \\ \dot{p}_i &= \{p_i, H\} \end{aligned}$$

2.2.4 Canonical Transformations

Euler-Lagrange equations are invariant under a coordinate change $(q, \dot{q}) \rightarrow (\bar{q}, \dot{\bar{q}})$, which implies the invariance of Hamilton's equations under $(q, p) \rightarrow (\bar{q}, \bar{p})$.

General transformations of the form $q \rightarrow \bar{q}(q, p)$ and $p \rightarrow \bar{p}(q, p)$ are called **canonical transformations** if they obey the canonical equations. $(\bar{q}, \bar{p})$ are called **canonical coordinates**.

Theorem: If $(\bar{q}(q, p), \bar{p}(q, p))$ are canonical coordinates iff.

$$\begin{aligned} \{\bar{q}_j, \bar{q}_k\} &= 0 = \{\bar{p}_j, \bar{p}_k\} \\ \{\bar{q}_j, \bar{p}_k\} &= \delta_{jk} \end{aligned}$$

Proof: For state functions $\omega(q, p)$,

$$\dot{\omega} = \{\omega, H\} = \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial H}{\partial q_i} \right)$$

Thus for $\dot{\bar{q}}_j$,

$$\dot{\bar{q}}_j = \sum_i \left(\frac{\partial \bar{q}_j}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial \bar{q}_j}{\partial p_i} \frac{\partial H}{\partial q_i} \right) \quad (1)$$

Using the chain rule, we can also get

$$\begin{cases} \frac{\partial H(q, p)}{\partial p_i} = \frac{\partial H(\bar{q}, \bar{p})}{\partial p_i} = \sum_k \left(\frac{\partial H}{\partial \bar{q}_k} \frac{\partial \bar{q}_k}{\partial p_i} + \frac{\partial H}{\partial \bar{p}_k} \frac{\partial \bar{p}_k}{\partial p_i} \right) \\ \frac{\partial H(q, p)}{\partial q_i} = \frac{\partial H(\bar{q}, \bar{p})}{\partial q_i} = \sum_k \left(\frac{\partial H}{\partial \bar{q}_k} \frac{\partial \bar{q}_k}{\partial q_i} + \frac{\partial H}{\partial \bar{p}_k} \frac{\partial \bar{p}_k}{\partial q_i} \right) \end{cases} \quad (2)$$

Plugging (2) into (1) and simplifying gives (following a similar procedure for $\dot{\bar{p}}_j$),

$$\begin{cases} \dot{\bar{q}}_j = \sum_k \left(\frac{\partial H}{\partial \bar{q}_k} \{ \bar{q}_j, \bar{q}_k \} + \frac{\partial H}{\partial \bar{p}_k} \{ \bar{q}_j, \bar{p}_k \} \right) \\ \dot{\bar{p}}_j = \sum_k \left(\frac{\partial H}{\partial \bar{q}_k} \{ \bar{p}_j, \bar{q}_k \} + \frac{\partial H}{\partial \bar{p}_k} \{ \bar{p}_j, \bar{p}_k \} \right) \end{cases} \quad (3)$$

For (3) to reduce to the canonical equations, we must have $\{ \bar{q}_j, \bar{q}_k \} = 0 = \{ \bar{p}_j, \bar{p}_k \}$ and $\{ \bar{q}_j, \bar{p}_k \} = \delta_{jk}$.

*Note that the Poisson bracket is independent of parametrization. Whether $\omega(q, p)$ or $\omega(\bar{q}, \bar{p})$,

$$\{ \omega, \lambda \}_{q, p} = \{ \omega, \lambda \}_{\bar{q}, \bar{p}}$$

2.2.5 Active Transformations

The transformation $q \rightarrow \bar{q}(q, p)$ and $p \rightarrow \bar{p}(q, p)$ is passive if (q, p) and $(\bar{q}, \bar{p})$ refer to the same point in phase space.

If the transformation $(q, p) \rightarrow (\bar{q}, \bar{p})$ preserves the range of variables, it's called **regular**.

- E.g. transformation from a Cartesian plane to a translated/rotated one is regular.
- But a transformation from a Cartesian plane to spherical coordinates isn't. Some variables are now bounded by 2π instead of ∞ .

A transformation that "shifts" the phase space so that (q, p) and $(\bar{q}, \bar{p})$ no longer refer to the same point is called **active**. Active transformations are a subset of regular ones.

We call ω invariant under the regular transformation $(q, p) \rightarrow (\bar{q}, \bar{p})$ if

$$\omega(q, p) = \omega(\bar{q}, \bar{p})$$

2.2.6 Symmetries

Two results:

1. If H is invariant under the following infinitesimal transformations, then $g(q, p)$ is conserved. We call g a **generator of the transformation**.

$$\begin{aligned} q_i &\rightarrow \bar{q}_i + \varepsilon \frac{\partial g}{\partial p_i} \equiv q_i + \delta q_i \\ p_i &\rightarrow \bar{p}_i - \varepsilon \frac{\partial g}{\partial q_i} \equiv p_i + \delta p_i \end{aligned}$$

2. If H is invariant under the regular, canonical, but not necessarily infinitesimal transformation $(q, p) \rightarrow (\bar{q}, \bar{p})$, and if (q, p) is a solution to the equations of motion, so is $(\bar{q}, \bar{p})$.

Proof of (1): Note that

$$\begin{aligned}\delta H &= \sum_i \frac{\partial H}{\partial q_i} (\delta q_i) + \frac{\partial H}{\partial p_i} (\delta p_i) \\ &= \sum_i \frac{\partial H}{\partial q_i} \left(\varepsilon \frac{\partial g}{\partial p_i} \right) + \frac{\partial H}{\partial p_i} \left(-\varepsilon \frac{\partial g}{\partial q_i} \right) \\ &= \varepsilon \{H, g\}\end{aligned}$$

For H to be invariant (to first order), $\delta H = 0$ and $\{H, g\} = 0$. Thus g is conserved.

In general for state variables ω ,

$$\delta \omega = \varepsilon \{\omega, g\}$$

2.2.7 A Relation Between S and E

Let x_{cl} be a classical path from (x_i, t_i) to (x_f, t_f) , $H(t_f)$ be the Hamiltonian at t_f , and S the classical action. A neat relation is

$$\frac{\partial S_{\text{cl}}(x_f, t_f; x_i, t_i)}{\partial t_f} = -H(t_f)$$

If E is conserved, this becomes

$$\frac{\partial S_{\text{cl}}(x_f, t_f; x_i, t_i)}{\partial t_f} = -E$$

Proof: Hold x_i, x_f fixed and vary with $\eta(t)$. For convenience, $x_i = t_i = 0$ and $\eta(0) = 0$.

$$x(t) = x_{\text{cl}}(t) + \eta(t)$$

The variation in action can be split into (1) the contribution from difference in path, (2) the contribution from the new “end” time $t = t_f + \Delta t$.

$$\begin{aligned}\delta S_{\text{cl}} &= \int_0^{t_f} \frac{\partial L}{\partial t} dt + L(t_f) \Delta t \\ &= \int_0^{t_f} \left[\frac{\partial L}{\partial x} \eta(t) + \frac{\partial L}{\partial \dot{x}} \dot{\eta}(t) \right] dt + L(t_f) \Delta t \\ &= \int_0^{t_f} \left[-\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} + \frac{\partial L}{\partial x} \right] \eta(t) dt + \int_0^{t_f} \frac{d}{dt} \left[\frac{\partial L}{\partial \dot{x}} \eta(t) \right] dt + L(t_f) \Delta t \\ &= 0 + \left[\frac{\partial L}{\partial \dot{x}} \eta(t) \right]_0^{t_f} + L(t_f) \Delta t\end{aligned}$$

Substitute $\eta(t_f) = -\dot{x}(t_f) \Delta t$ to get

$$\delta S = \left[-\frac{\partial L}{\partial \dot{x}} \dot{x} + L \right]_{t_f} \Delta t = -H(t_f) \Delta t$$

2.3 A Problem: Wave-Particle Duality

2.3.1 Plane Waves

An electromagnetic wave ψ that travels at c obey the wave equation

$$\nabla^2 \psi = \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2}$$

A special case is the plane wave:

$$\psi(x, t) = Ae^{i\phi} = A \exp \left(i \left(\frac{2\pi}{\lambda} x - \frac{2\pi}{T} t \right) \right)$$

- Wave number $k = 2\pi/\lambda$ (phase change per unit length at fixed t)
- Angular frequency $\omega = 2\pi/T$ (phase change per unit time at fixed x)
- Wave speed $v = \omega/T$

This can be extended to plane waves in 3D, using the **wave vector** $\vec{k}$.

$$\psi(\vec{r}, t) = Ae^{i(\vec{k} \cdot \vec{r} - \omega t)} \quad , \quad \omega = |\vec{k}|v$$

2.3.2 Double-Slit Experiment

The wave prediction is $I_{1+2} \neq I_1 + I_2$, due to interference effects.

The classical particle prediction is $I_{1+2} = I_1 + I_2$.

By opening one slit and lowering the intensity, we find that light energy comes in bundles. Each bundle has energy and momentum that satisfies

$$E = \hbar\omega \quad , \quad p = \hbar k$$

Thus light seems to be made up of particles called photons. But if both slits are opened at I , we get a wave effect: $I_{1+2} \neq I_1 + I_2$. The difference can't be reconciled classically.

Born suggested that each photon is associated with a wave ψ called the **probability amplitude**. $|\psi(x)|^2$ is the probability of finding the particle at x . The wave is described by

$$\omega = \frac{E}{\hbar} \quad , \quad k = \frac{p}{\hbar}$$

The wave interferes with itself (after being split by the double-slit) to form the overall pattern according to $|\psi(x)|^2$.

De Broglie suggested that every particle of momentum p has a wave associated with it, not just massless photons. The wave number is

$$k = \frac{p}{\hbar}$$

3 Postulates of Quantum Mechanics

3.1 The Postulates

Classical:

1. State of particle is described by $x(t)$ and $p(t)$ in 2D phase space.
2. Every dynamical variable $\omega(x, p)$ is a function of x, p .
3. If a particle is in the state $(x; p)$, a perfect measurement of ω yields $\omega(x, p)$ without changing the state.
4. Time evolution described by

$$\dot{x} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial x}$$

Quantum:

1. State of particle is a vector $|\psi(t)\rangle$ in Hilbert space.
2. Independent variables x, p of classical mechanics become Hermitian operators X, P such that

$$\begin{aligned}\langle x|X|x'\rangle &= x\delta(x - x') \\ \langle x|P|x'\rangle &= -\hbar\delta'(x - x')\end{aligned}$$

(Note that this is just X and $P = \hbar K$ from before). Then $\omega(x, p)$ becomes

$$\Omega(X, P) = \omega(x \rightarrow X, p \rightarrow P)$$

3. If a particle is in the state $|\psi\rangle$, a perfect quantum measurement of Ω yields an eigenvalue ω with probability

$$P(\omega) \propto |\langle \omega|\psi\rangle|^2$$

The state of the system changes:

$$|\psi\rangle \rightarrow |\omega\rangle$$

Note that Ω is Hermitian, so the eigenvalues are real. In terms of the projection operator $\mathbb{P}_\omega = |\omega\rangle\langle\omega|$,

$$\begin{aligned}P(\omega) \propto |\langle \omega|\psi\rangle|^2 &= \langle \psi|\omega\rangle\langle\omega|\psi\rangle \\ &= \langle \psi|\mathbb{P}_\omega|\psi\rangle \\ &= \langle \psi|\mathbb{P}_\omega\mathbb{P}_\omega|\psi\rangle \\ &= \langle \mathbb{P}_\omega\psi|\mathbb{P}_\omega\psi\rangle\end{aligned}$$

4. Time evolution described by

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle$$

$H(X, P) = H(x \rightarrow X, p \rightarrow P)$ from the classical Hamiltonian.

Potential problems:

1. $\Omega(x \rightarrow X, p \rightarrow P)$ is ambiguous. E.g. for $\omega = xp$, is $\Omega = XP$ or PX ? Solution is to symmetrize with $\Omega = (XP + PX)/2$.
2. Ω can be degenerate, e.g. two eigenvalues $\omega_1 = \omega_2 = \omega$. Then what is $P(\omega)$? Solution: find some basis $|\omega_1\rangle, |\omega_2\rangle$ in the eigenspace $\mathbb{V}_\omega$. Then

$$P(\omega) = |\langle\omega_1|\psi\rangle|^2 + |\langle\omega_2|\psi\rangle|^2$$

This is the same result we get by assuming ω_1, ω_2 are infinitesimally distinct, then asking for $P(\omega_1 \text{ or } \omega_2)$.

3. Eigenvalue spectrum of Ω can be continuous, e.g. X . Solution: expand

$$|\psi\rangle = \int |\omega\rangle \langle\omega|\psi\rangle d\omega$$

Introduce the **wave function in ω space** $\psi(\omega) = \langle\omega|\psi\rangle$ and the **probability density** $P(\omega) = |\langle\omega|\psi\rangle|^2$. $P(\omega) d\omega$ is the probability of measuring between ω and $\omega + d\omega$. Note $\int P(\omega) d\omega = 1$.

3.1.1 Ensembles

Used to test quantum theory. A quantum ensemble is a large number N of particles all in state $|\omega\rangle$.

A single particle, after measurement, could give any of the eigenvalues. Measure all N to get a statistical result.

Difference with classical:

- In classical ensemble of N particles, if we measure λ_1 in $N/3$ trials and λ_2 in $2N/3$ trials, we expect there were $N/3$ amount of λ_1 and $2N/3$ amount of λ_2 before measurement.
- In quantum ensemble, all particles are assumed to be $|\omega\rangle$ before measurement.

To calculate average over ensemble, use the **expectation value**:

$$\begin{aligned}\langle\Omega\rangle &= \sum_i P(\omega_i) \omega_i = \sum_i |\langle\omega_i|\psi\rangle|^2 \omega_i \\ &= \sum_i \langle\psi|\omega_i\rangle \langle\omega_i|\psi\rangle \omega_i \\ &= \sum_i \langle\psi|\Omega|\omega_i\rangle \langle\omega_i|\psi\rangle \\ &= \langle\psi|\Omega|\psi\rangle\end{aligned}$$

To measure the uncertainty (fluctuation around the mean), use the **standard deviation**:

$$\Delta\Omega = \langle(\Omega - \langle\Omega\rangle)^2\rangle^{1/2}$$

Classically, this is called “root-mean-square” deviation. In QM, it’s “uncertainty in Ω .”

$$(\Delta\Omega)^2 = \begin{cases} \sum_i P(\omega_i) (\omega_i - \langle\Omega\rangle)^2 & \text{(discrete spectrum)} \\ \int P(\omega) (\omega - \langle\Omega\rangle)^2 d\omega & \text{(continuous spectrum)} \end{cases}$$

E.g. if a particle has $\langle X \rangle = a$ and $\Delta X = \Delta$, then we know it's likely to be spotted near a with deviation of order Δ .

3.1.2 Compatible Variables

Is there a filtering process by which we take $|\psi\rangle$ and produce a state with well-defined ω, λ for 2 variables Ω, Λ ?

- Not generally, because measurement changes the state. I.e. (if we do Ω then Λ)

$$\text{first: } |\psi\rangle \rightarrow \sum |\omega\rangle \langle \omega|\psi\rangle$$

$$\text{second: } |\omega\rangle \rightarrow \sum |\lambda\rangle \langle \lambda|\omega\rangle$$

But if Ω and Λ have simultaneous eigenkets, then

$$\Omega |\omega\lambda\rangle = \omega |\omega\lambda\rangle$$

$$\Lambda |\omega\lambda\rangle = \lambda |\omega\lambda\rangle$$

$$(\Omega\Lambda - \Lambda\Omega) |\omega\lambda\rangle = 0$$

We can “know” both variables at the same time, if it has a simultaneous eigenket. Variables such that $[\Omega, \Lambda] = 0$ are called **compatible**.

E.g. The canonical commutation relation $[X, P] = i\hbar \neq 0$. Thus, a state can't have well-defined eigenkets of X and P simultaneously. This is the Heisenberg uncertainty relation.

3.1.3 Density Matrices

Instead of an ensemble of N particles all in $|\psi\rangle$, we can consider an ensemble of N systems, n_i of which are $|i\rangle$.

Let $p_i = n_i/N$ be the probability of measuring a $|i\rangle$. Define the density matrix:

$$\rho = \sum_i p_i |i\rangle \langle i|$$

- “Pure”: all $n_i = 0$ except for one
- “Mixed”: otherwise.

The ensemble average of Ω is $\langle \bar{\Omega} \rangle = \sum_i p_i \langle i|\Omega|i\rangle$. Note that

$$\begin{aligned} \text{Tr}(\Omega\rho) &= \sum_j \langle j|\Omega\rho|j\rangle = \sum_i \sum_j \langle j|\Omega\rho|i\rangle \langle i|j\rangle p_i \\ &= \sum_i \sum_j \langle i|j\rangle \langle j|\Omega\rho|i\rangle p_i = \sum_i \langle i|\Omega\rho|i\rangle p_i \\ &= \langle \bar{\Omega} \rangle \end{aligned}$$

E.g. if we want the probability of obtaining ω , $\overline{P(\omega)}$, then

$$\begin{aligned} P(\omega) &= |\langle \omega|\psi\rangle|^2 = \langle \psi|\omega\rangle \langle \omega|\psi\rangle = \langle \psi|\mathbb{P}_\omega|\psi\rangle \\ \overline{P(\omega)} &= \text{Tr}(\mathbb{P}_\omega\rho) \end{aligned}$$

Features:

- $\rho^\dagger = \rho$
- $\rho^2 = \rho$ for pure state
- $\rho = \frac{1}{k}I$ for mixed state uniformly distributed over k sub-states
- $\text{Tr}(\rho) = 1$
- $\text{Tr}(\rho^2) \leq 1$, equality if ρ is pure.

3.2 Example: Gaussian Wavefunction

We'll find the Heisenberg uncertainty relation (sort of) for this model.

Put the wavefunction in the X -basis:

$$|\psi\rangle = \int_{-\infty}^{\infty} |x\rangle \langle x|\psi\rangle dx = \int_{-\infty}^{\infty} |x\rangle \psi(x) dx$$

Assume $\psi(x)$ is Gaussian and normalize to unity:

$$\psi(x) = \frac{1}{(\pi\Delta^2)^{1/4}} e^{-(x-a)^2/(2\Delta^2)}$$

In the X -basis, the effect of X is to multiply by x . I.e.

$$\begin{aligned} \langle x|X|\psi\rangle &= \int_{-\infty}^{\infty} \langle x|X|x'\rangle \langle x'|\psi\rangle dx' = \int_{-\infty}^{\infty} x\delta(x-x')\psi(x') dx' \\ &= x\psi(x) \end{aligned}$$

Expectation value is $\langle X \rangle = a$ (which makes sense; $\psi(x)$ is centered around a).

Also, $\langle X^2 \rangle = \Delta^2/2 + a^2$

Then the uncertainty is

$$\begin{aligned} \Delta X &= [\langle \psi|(X - \langle X \rangle)^2|\psi\rangle]^{1/2} = [\langle X^2 \rangle - 2a\langle X \rangle + a^2]^{1/2} \\ &= [\langle X^2 \rangle - a^2]^{1/2} \\ &= \frac{\Delta}{\sqrt{2}} \end{aligned}$$

Now for P . The action in the X basis is $-i\hbar\frac{d}{dx}$. I.e.

$$\begin{aligned} \langle x|P|\psi\rangle &= \int_{-\infty}^{\infty} \langle x|P|x'\rangle \langle x'|\psi\rangle dx' = \int_{-\infty}^{\infty} -i\hbar\delta'(x-x')\psi(x') dx' \\ &= -i\hbar\frac{d\psi}{dx} \end{aligned}$$

Now find the eigenvalues. Note that $\langle x|p\rangle = \psi_p(x) \neq \psi(x)$.

$$\begin{aligned} P|p\rangle &= p|p\rangle \longrightarrow \langle x|P|p\rangle = p\langle x|p\rangle \\ -i\hbar \frac{d\psi_p(x)}{dx} &= p\psi_p(x) \\ \psi_p(x) &= \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} \end{aligned}$$

By finding $\psi(p) = \langle p|\psi\rangle$, we get a Gaussian of width $\hbar/(\Delta\sqrt{2})$ centered at 0. Thus $\langle P\rangle = 0$ and $\Delta P = \hbar/(\Delta\sqrt{2})$.

Put together: for this Gaussian wavefunction model, we get

$$\Delta X \cdot \Delta P = \frac{\hbar}{2}$$

This is a lower bound on the Heisenberg uncertainty relation.

3.3 Generalization to N Degrees of Freedom

Use mutually commuting operators $X_1, \dots, X_N$. In their simultaneous eigenbasis $|x_1, x_2, \dots, x_N\rangle$,

$$\langle x_1, x_2, \dots, x_N | x'_1, x'_2, \dots, x'_N \rangle = \delta(x_1 - x'_1) \cdots \delta(x_N - x'_N)$$

And

$$\begin{aligned} |\psi\rangle &\longrightarrow \langle x_1, \dots, x_N | \psi \rangle = \psi(x_1, \dots, x_N) \\ X_i |\psi\rangle &\longrightarrow \langle x_1, \dots, x_N | X_i | \psi \rangle = x_i \psi(x_1, \dots, x_N) \\ P_i |\psi\rangle &\longrightarrow \langle x_1, \dots, x_N | P_i | \psi \rangle = -i\hbar \frac{\partial}{\partial x_i} \psi(x_1, \dots, x_N) \end{aligned}$$

Then follow the procedure of replacing classical variables with quantum operators. E.g. for the classical operator

$$\omega = \frac{p_x^2 + p_y^2 + p_z^2}{2m} + x^2 + y^2 + z^2$$

We associate the quantum operator such that (with $X_1 = X$, $X_2 = Y$, and $X_3 = Z$)

$$\Omega |\omega\rangle = \omega |\omega\rangle \quad \Omega = \frac{P_x^2 + P_y^2 + P_z^2}{2m} + X^2 + Y^2 + Z^2$$

Then substitute $|\omega\rangle = \psi_\omega(x, y, z)$, $X \rightarrow x$, and $P_x \rightarrow -i\hbar \frac{d}{dx}$.

$$\left[-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + x^2 + y^2 + z^2 \right] \psi_\omega(x, y, z) = \omega \psi_\omega(x, y, z) \quad (1)$$

Once we have this form, we can change to other coordinates, e.g. spherical. Use

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \equiv \nabla^2 \equiv \frac{1}{r^2} \left(\frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right)$$

And substitute into (1) to get

$$-\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi_\omega}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi_\omega}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi_\omega}{\partial \phi^2} \right] + r^2 \psi_\omega = \omega \psi_\omega \quad (2)$$

3.4 The Schrödinger Equation

3.4.1 Hamiltonian Setup

Simply substitute $x \rightarrow X$, $p \rightarrow P$ in the classical Hamiltonian. E.g. for (1) harmonic oscillator, (2) constant force f , and (3) particle of charge q in an EM field,

$$\begin{aligned} 1) \quad H &= \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2 & \longrightarrow & \quad H = \frac{P^2}{2m} + \frac{1}{2} m \omega^2 X^2 \\ 2) \quad H &= \frac{p^2}{2m} - f x & \longrightarrow & \quad H = \frac{P^2}{2m} - f X \\ 3) \quad H &= \frac{|\vec{p} - \frac{q}{c} \vec{A}(\vec{r}, t)|^2}{2m} + q\phi(\vec{r}, t) & \longrightarrow & \quad H = \frac{1}{2m} \left(P \cdot P - \frac{q}{c} A \cdot P - \frac{q}{c} P \cdot A + \frac{q^2}{c^2} A \cdot A \right) + q\phi \end{aligned}$$

Note: P doesn't commute with A (which depends on X, Y, Z).

3.4.2 Time-Independent Schrödinger Equation

Start from Postulate IV, $i\hbar |\dot{\psi}\rangle = H |\psi\rangle$. We seek to find a propagator of the form

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle$$

Use the energy basis, since $H |E\rangle = E |E\rangle$.

$$|\psi(t)\rangle = \sum_E |E\rangle \langle E | \psi(t) \rangle \equiv \sum_E a_E(t) |E\rangle$$

Apply $(i\hbar \frac{\partial}{\partial t} - H)$ and use $(i\hbar \frac{\partial}{\partial t} - H) |\psi(t)\rangle = 0$ to get

$$\begin{aligned} 0 &= \sum_E (i\hbar \frac{\partial}{\partial t} - H) a_E(t) |E\rangle = \sum_E (i\hbar \dot{a}_E - E a_E) |E\rangle \\ a_E(t) &= a_E(0) e^{-iEt/\hbar} \\ \langle E | \psi(t) \rangle &= \langle E | \psi(0) \rangle e^{-iEt/\hbar} \end{aligned}$$

Now plug back in to $|\psi\rangle$ to get the propagator:

$$\begin{aligned} |\psi(t)\rangle &= \sum_E e^{-iEt/\hbar} |E\rangle \langle E | \psi(0) \rangle \\ U(t) &= \sum_E e^{-iEt/\hbar} |E\rangle \langle E| = e^{-iHt/\hbar} \end{aligned} \quad (1)$$

Because H is unitary, $U(t)$ is unitary, thus time evolution is unitary and the norm is preserved: $\langle \psi(t) | \psi(t) \rangle = \langle \psi(0) | \psi(0) \rangle$.

3.4.3 Time-Dependent Schrödinger Equation

To approximate a time-dependent Hamiltonian, split $(0, t)$ into N intervals of width $\Delta = t/N$. Then

$$\begin{aligned} |\psi(t)\rangle &= |\psi(0)\rangle + \Delta \left. \frac{d|\psi\rangle}{dt} \right|_0 = |\psi(0)\rangle - \frac{i\Delta}{\hbar} H(0) |\psi(0)\rangle \\ &= \left[1 - \frac{i\Delta}{\hbar} H(0) \right] |\psi(0)\rangle \\ &\approx \exp \left[-\frac{i\Delta}{\hbar} H(0) \right] |\psi(0)\rangle \end{aligned}$$

The overall propagator is found by summing the changes:

$$\begin{aligned} |\psi(t)\rangle &= \prod_{n=0}^{N-1} e^{-i\Delta H(n\Delta)/\hbar} |\psi(0)\rangle \\ U(t) &= \exp \left[-\frac{i}{\hbar} \int_0^t H(t') dt' \right], \quad N \rightarrow \infty \end{aligned} \tag{2}$$

4 Simple Problems in 1D

4.1 The Free Particle

4.1.1 Allowed Energy States

Start from the Schrodinger equation:

$$i\hbar \frac{d}{dt} |\psi\rangle = H |\psi\rangle = \frac{P^2}{2m} |\psi\rangle$$

The normal modes (in energy basis) are

$$|\psi\rangle = e^{-iEt/\hbar} |E\rangle$$

We seek to find allowed E states. $|p\rangle$ is an eigenvector of both P and P^2 , so

$$\begin{aligned} \frac{P^2}{2m} |p\rangle = E |p\rangle &\longrightarrow \frac{p^2}{2m} |p\rangle = E |p\rangle \\ p &= \pm \sqrt{2mE} \end{aligned}$$

Thus for each eigenvalue E , we have 2 eigenvectors:

$$\begin{aligned} |E, +\rangle &= |p = \sqrt{2mE}\rangle \\ |E, -\rangle &= |p = -\sqrt{2mE}\rangle \end{aligned}$$

I.e. a particle with energy E may be found moving left or right with momentum $\sqrt{2mE}$. So far, this agrees with a classical analysis. The difference is that in QM, we can have a state like

$$|E\rangle = \beta |p = \sqrt{2mE}\rangle + \gamma |p = -\sqrt{2mE}\rangle$$

4.1.2 Propagator

Use the propagator from the time-independent Schrodinger equation:

$$U(t) = \int_{-\infty}^{\infty} |p\rangle \langle p| e^{-iE(p)t/\hbar} dp = \int_{-\infty}^{\infty} |p\rangle \langle p| e^{-ip^2 t/2m\hbar} dp$$

Note that in the X basis, the propagator for a time slice is

$$\begin{aligned} U(x, t; x') &\equiv \langle x|U(t)|x'\rangle = \int_{-\infty}^{\infty} \langle x|p\rangle \langle p|x'\rangle e^{-ip^2 t/2m\hbar} dp \\ &= \frac{1}{2m\hbar} \int_{-\infty}^{\infty} e^{ip(x-x')/\hbar} \cdot e^{-ip^2 t/2m\hbar} dp \\ &= \sqrt{\frac{m}{2\pi\hbar it}} e^{im(x-x')^2/2\hbar t} \end{aligned}$$

In terms of the propagator, given the initial state $\psi(x', 0)$, we can find an arbitrary state $\psi(x, t)$ as

$$\psi(x, t) = \int U(x, t; x') \psi(x', 0) dx'$$

4.2 The Particle in a Box

4.2.1 Wavefunction

Consider a box potential:

$$V(x) = \begin{cases} 0 & |x| < L/2 \\ \infty & |x| \geq L/2 \end{cases}$$

Let I and III be the regions outside the box, while II is the region within the box. Use the eigenvalue equation in the X basis:

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

In III, we have

$$\begin{aligned} \frac{d^2\psi_{\text{III}}}{dx^2} - \frac{2m(V_0 - E)}{\hbar^2} &= 0 \quad , \quad V_0 \rightarrow \infty \\ \psi_{\text{III}} &= Ae^{-\kappa x} + Be^{\kappa x} \quad , \quad \kappa = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}} \end{aligned}$$

We must have $B = 0$ because otherwise the wavefunction blows up as $x \rightarrow \infty$. Then as $V_0 \rightarrow \infty$, $\psi_{\text{III}} = 0$. This is as expected. Similar results hold for I: $\psi_{\text{I}} = 0$.

In II, the solution is that of a free particle:

$$\psi_{\text{II}} = Ae^{ikx} + Be^{-ikx} \quad , \quad k = \sqrt{\frac{2mE}{\hbar^2}}$$

Further, it must obey the boundary conditions:

$$\begin{aligned}\psi_I(-L/2) = \psi_{II}(-L/2) = 0 &\longrightarrow Ae^{ikL/2} + Be^{ikL/2} = 0 \\ \psi_{III}(L/2) = \psi_{II}(L/2) = 0 &\longrightarrow Ae^{ikL/2} + Be^{-ikL/2} = 0 \\ \begin{bmatrix} e^{-ikL/2} & e^{ikL/2} \\ e^{ikL/2} & e^{-ikL/2} \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}\end{aligned}$$

For this to happen, the determinant must be 0, i.e. $e^{ikL} - e^{-ikL} = 0$, thus $k = n\pi/L$, where $n = 0, \pm 1, \pm 2, \dots$. We can find that $A = -e^{in\pi}B$, leading to two families of solutions:

$$\psi_n(x) = \begin{cases} \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) & n \text{ even} \\ \sqrt{\frac{2}{L}} \cos\left(\frac{n\pi x}{L}\right) & n \text{ odd} \end{cases}$$

4.2.2 Allowed Energies

Note that the energy is quantized:

$$E = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2 \pi^2 n^2}{2mL^2}$$

And further, the lowest bound on energy (for non-trivial solutions) is nonzero: the ground state energy E_g is

$$E_g = \frac{\hbar^2 \pi^2}{2mL^2}$$

An “intuitive” explanation for this is that Δx is bounded in the well, so by the uncertainty principle, we cannot have a well-defined momentum of 0.

The uncertainty principle also provides a quick estimate of the ground energy. Start with $H = P^2/2m$. From the Ehrenfest theorem, $\langle H \rangle = \langle P^2 \rangle / 2m$. Since $\langle P \rangle = 0$ inside the box, we can further write

$$\langle H \rangle = \frac{\langle (P - \langle P \rangle)^2 \rangle}{2m} = \frac{(\Delta P)^2}{2m}$$

From the uncertainty relations, $\Delta X \cdot \Delta P \geq \hbar/2$, so we have

$$\langle H \rangle \geq \frac{\hbar^2}{8m(\Delta x)^2} \geq \frac{\hbar^2}{2mL^2}$$

In energy eigenstates, $\langle H \rangle = E$. Thus we have

$$E \geq \frac{\hbar^2}{2mL^2}$$

Which is the estimated order of the ground state, accurate up to a factor of π^2 .

4.3 The Continuity Equation of Probability

From classical electrodynamics:

$$\begin{aligned} \text{(global)} \quad Q_{\text{total}} &= \text{const.} \\ \text{(local)} \quad \frac{d}{dt} \int_V \rho(\mathbf{r}, t) d^3\mathbf{r} &= - \int_{S_V} \mathbf{j} \cdot d\mathbf{S} \end{aligned}$$

In QM:

$$\begin{aligned} \text{(global)} \quad \iiint P(\mathbf{r}, t) d^3\mathbf{r} &= \text{const.} \\ \text{(local)} \quad \frac{d}{dt} \int P(\mathbf{r}, t) d^3\mathbf{r} &= - \int_{S_\infty} \mathbf{j} \cdot d\mathbf{S} \quad , \quad \mathbf{j} = \frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*) \end{aligned}$$

In a nutshell, the total probability to find a particle anywhere in the universe is conserved. Locally, the reduction in probability of a region must correspond to a change in the probability current $\mathbf{j}$ is the **probability current density**.

4.4 Quantum Tunneling

The details are complicated, see Shankar chp. 5 for more. The main idea: instead of an infinite potential, we have a region of 0 potential I, and a region of *finite* potential II. Let $V = 0$ in I and $V = V_0$ in II.

$$\begin{aligned} \text{I:} \quad \psi_E(x) &= Ae^{ik_1x} + Be^{-ik_1x} \quad , \quad k_1 = \sqrt{\frac{2mE}{\hbar^2}} \\ \text{II:} \quad \psi_E(x) &= Ce^{ik_2x} + De^{-ik_2x} \quad , \quad k_2 = \sqrt{\frac{2m(E - V_0)}{\hbar^2}} \end{aligned}$$

Set $D = 0$ for the right-traveling transmitted waves. At $x = 0$, boundary conditions require that

$$\begin{aligned} A + B &= C \\ ik_1(A - B) &= ik_2C \end{aligned}$$

After some calculations, we find the reflection probability (R) and transmission probability (T) to be:

$$\begin{aligned} R &= \left| \frac{B}{A} \right|^2 = \left| \frac{\sqrt{E_0} - \sqrt{E_0 - V_0}}{\sqrt{E_0} + \sqrt{E_0 - V_0}} \right|^2 \\ T &= 1 - R = \left| \frac{4\sqrt{E_0}\sqrt{E_0 - V_0}}{(\sqrt{E_0} + \sqrt{E_0 - V_0})^2} \right| \end{aligned}$$

The key is that even if $E_0 < V_0$ (energy not enough to overcome barrier), there's still a nonzero probability for the particle to be transmitted. This is called **quantum tunneling**.

4.5 Ehrenfest's Theorem

QM applied to many particles should reproduce the macroscopic world.

Consider the time evolution of expectation values:

$$\frac{d}{dt}\langle\Omega\rangle = \frac{d}{dt}\langle\psi|\Omega|\psi\rangle = \langle\dot{\psi}|\Omega|\psi\rangle + \langle\psi|\Omega|\dot{\psi}\rangle + \langle\psi|\dot{\Omega}|\psi\rangle$$

Assume Ω has no time dependence, so $\dot{\Omega} = 0$. Further, from the Schrodinger equation,

$$\begin{aligned} |\dot{\psi}\rangle &= -\frac{i}{\hbar}H|\psi\rangle \\ \langle\dot{\psi}| &= \frac{i}{\hbar}\langle\psi|H \end{aligned}$$

Combining the two results gives **Ehrenfest's theorem**:

$$\frac{d}{dt}\langle\Omega\rangle = -\frac{i}{\hbar}\langle[\Omega, H]\rangle$$

Note the similarity with classical mechanics:

$$\frac{d\omega}{dt} = \{\omega, H\}$$

For a Hamiltonian of the form $H = P^2/2m + V(X)$, we can also get similar forms to the canonical equations:

$$\langle\dot{X}\rangle = \left\langle\frac{\partial H}{\partial P}\right\rangle \quad , \quad \langle\dot{P}\rangle = \left\langle-\frac{\partial H}{\partial X}\right\rangle$$

Thus we can regain classical physics if the fluctuation about the mean is small. But the above equations depend on the choice of H .

5 The Harmonic Oscillator

5.1 The Classical Oscillator

Hamiltonian:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2x^2$$

Equations of motion:

$$\begin{aligned} \dot{x} &= \frac{\partial H}{\partial p} = \frac{p}{m} \\ \dot{p} &= -\frac{\partial H}{\partial x} = -m\omega^2x \end{aligned}$$

From here, the overall motion is apparent and can be solved:

$$\ddot{x} + \omega^2x = 0$$

5.2 Quantum Oscillator in the X Basis

To quantize the oscillator, we treat it as a particle $|\psi\rangle$ obeying the Schrodinger equation.

$$i\hbar \frac{d}{dt} |\psi\rangle = H |\psi\rangle$$

$$H = \frac{P^2}{2m} + \frac{1}{2}m\omega^2 X^2$$

Solving the eigenvalue equation for energy eigenstates,

$$H |E\rangle = E |E\rangle$$

$$\left(\frac{P^2}{2m} + \frac{1}{2}m\omega^2 X^2 \right) |E\rangle = E |E\rangle$$

In the X basis, we make the substitutions $P \rightarrow -i\hbar \frac{d}{dx}$, $X \rightarrow x$, and $|E\rangle \rightarrow \psi_E$. Then

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} \left(E - \frac{1}{2}m\omega^2 x^2 \right) \psi = 0$$

To solve this, we substitute $x = by$ and $\varepsilon = E/\hbar\omega$, where $b = \sqrt{\hbar/m\omega}$. Then $d^2\psi/dy^2 + (2\varepsilon - y^2)\psi = 0$, which has solution $\psi(y) = u(y)e^{-y^2/2}$, where $u(y)$ goes to $A = cy$ as $y \rightarrow 0$ and y^m as $y \rightarrow \infty$. By requiring that the solution be in Hilbert space (normalizable to unity or the delta function), $\varepsilon = (2n + 1)/2$, and the energy is quantized:

$$E_n = \varepsilon_n \hbar\omega = \left(n + \frac{1}{2} \right) \hbar\omega$$

We can find the explicit wavefunction as $\psi_E(y) = AH_n(y)e^{-y^2/2}$, where A is a normalization factor and H_n is a Hermite polynomial of degree n .

$$\psi_n(x) = \underbrace{\left(\frac{m\omega}{\pi\hbar e^{2n}(n!)^2} \right)^{1/4}}_A \cdot H_n \left[\sqrt{\frac{m\omega}{\hbar}} x \right] \cdot \exp \left(-\frac{m\omega x^2}{2\hbar} \right)$$

The propagator can be found as

$$U(x, t; x', t') = \sum_{n=0}^{\infty} |\psi_n\rangle \langle \psi_n| e^{-iEt/\hbar}$$

$$= \sum_{n=0}^{\infty} A_n^2 \exp \left(-\frac{m\omega x^2}{2\hbar} \right)^2 H_n(x) H_n(x') \exp \left(-i \left(n + \frac{1}{2} \right) (t - t')\omega \right)$$

5.3 Quantum Oscillator in the Energy Basis

A slightly easier method is to use the commutator relation $[X, P] = i\hbar$, which is basis-independent, in the energy basis. Define the operators:

$$\begin{aligned} a &= \sqrt{\frac{m\omega}{2\hbar}}X + i\sqrt{\frac{1}{2m\omega\hbar}}P \\ a^\dagger &= \sqrt{\frac{m\omega}{2\hbar}}X - i\sqrt{\frac{1}{2m\omega\hbar}}P \\ [a, a^\dagger] &= 1 \end{aligned}$$

So that the Hamiltonian $H = P^2/2m + m\omega^2 X^2/2$ becomes

$$H = \left(a^\dagger a + \frac{1}{2}\right) \hbar\omega$$

Let $\hat{H} = H/\hbar\omega$. Using a and $a^\dagger$, an eigenstate $|n\rangle$ of $\hat{H}$ can be used to generate another eigenstate of $\hat{H}$.

$$\begin{aligned} \hat{H}a|n\rangle &= (n-1)a|n\rangle \longrightarrow a|n\rangle = \sqrt{n}|n-1\rangle \\ \hat{H}a^\dagger|n\rangle &= (n+1)a^\dagger|n\rangle \longrightarrow a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \end{aligned}$$

with appropriate normalization. a and $a^\dagger$ are called the lowering/raising operators, or destruction/creation operators. The number operator is $N = a^\dagger a$, such that $N|n\rangle = n|n\rangle$. The vacuum state is easily identified as $a|n_0\rangle = 0$, or

$$n_0 = \frac{1}{2}, \quad E = n_0\hbar\omega = \frac{1}{2}\hbar\omega$$

For example, to calculate $\langle 3|X^3|2\rangle$,

$$\langle 3|X^3|2\rangle = \left(\frac{\hbar}{2m\omega}\right)^{3/2} \langle 3|(a + a^\dagger)^3|2\rangle$$

After expanding and only taking contributions from $a^\dagger a^\dagger a$, $a^\dagger a a^\dagger$, and $a a^\dagger a^\dagger$, we have

$$\langle 3|X^3|2\rangle = 9\sqrt{3} \left(\frac{\hbar}{2m\omega}\right)^{3/2}$$

6 The Path Integral Formulation

6.1 Overview

The Schrodinger formalism uses the Hamiltonian, while the path integral formalism (from Feynman) uses the Lagrangian. The path integral formalism gives the full propagator more easily.

To find $U(x, t; x', t')$,

1. Draw all paths from (x', t') to (x, t) .
2. Find the action $S[x(t)]$ for each path.
3. $U(x, t; x', t') = A \sum e^{iS[x(t)]/\hbar}$

6.2 The Free Particle

6.2.1 Classical Approximation

Note that paths can destructively interfere. Using just the classical path is an okay approximation for macroscopic objects.

$$\begin{aligned}x_{\text{cl}}(t'') &= x' + \frac{x - x'}{t - t'}(t'' - t') \\L = T - V &= \frac{1}{2}m \left(\frac{x - x'}{t - t'} \right)^2 = \text{const.} \\S_{\text{cl}} &= \int_{t'}^t L dt'' = \frac{1}{2}m \left(\frac{x - x'}{t - t'} \right)^2 (t - t')\end{aligned}$$

The propagator:

$$U(t) = A' e^{iS_{\text{cl}}/\hbar} \approx A' \exp \left(\frac{im(x - x')^2}{2\hbar(t - t')} \right)$$

To find A' , the normalization constant, we require $U \rightarrow \delta(x - x')$ as $(t - t') \rightarrow 0$. Recalling the Gaussian definition of the delta function,

$$\begin{aligned}\delta(x - x') &= \lim_{\Delta \rightarrow 0} \frac{1}{\sqrt{\pi\Delta^2}} \exp \left(-\frac{(x - x')^2}{\Delta^2} \right), \quad \Delta = \sqrt{\frac{2i\hbar(t - t')}{m}} \\A' &= \sqrt{\frac{m}{2\pi\hbar i(t - t')}}\end{aligned}$$

Thus the propagator, to a classical approximation, is

$$U(x, t; x', t') \equiv U(x, t; x') = \sqrt{\frac{m}{2\pi\hbar i(t - t')}} \exp \left(\frac{im(x - x')^2}{2\hbar(t - t')} \right)$$

Which agrees with the one found classically, through the Schrodinger equation.

6.2.2 Full Solution

Without approximations, the solution is a little harder. Let $(x_N; t_N)$ be the endpoint and $(x_0; t_0)$ be the starting point. The path integral is

$$U(x_N, t_N; x_0, t_0) = \int_{x_0}^{x_N} e^{iS[x(t)]/\hbar} \mathcal{D}[x(t)]$$

$\mathcal{D}[x(t)]$ means “integrate over all paths connecting x_0 and x_N in the interval (t_0, t_N) .” Approximate by splitting the path into $(N + 1)$ intervals $x_0, x_1, \dots, x_N$ corresponding to $t_0, t_1, \dots, t_N$, where $t_n = t_0 + n\varepsilon$, with $\varepsilon = (t_N - t_0)/N$. Interpolate with straight lines. Then set $N \rightarrow \infty$ at the end.

Thus

$$S = \int_{t_0}^{t_N} L dt \rightarrow \sum_{i=0}^{N-1} \frac{1}{2} m \left(\frac{x_{i+1} - x_i}{\varepsilon} \right)^2 \varepsilon$$

$$U(x_N, t_N; x_0, t_0) = \int_{x_0}^{x_N} e^{iS[x(t)]/\hbar} \mathcal{D}[x(t)] = \lim_{N \rightarrow \infty} A \int \int \cdots \int \exp \left[\frac{im}{2\hbar} \sum_{i=0}^{N-1} \frac{(x_{i+1} - x_i)^2}{\varepsilon} \right] dx_1 dx_2 \cdots dx_{N-1}$$

By substituting $y_i = \sqrt{m/2\hbar\varepsilon} x_i$,

$$U(x_N, t_N; x_0, t_0) = A \left(\frac{2\hbar\varepsilon}{m} \right)^{(N-1)/2} \lim_{N \rightarrow \infty} \int \int \cdots \int \exp \left[- \sum_{i=0}^{N-1} \frac{(y_{i+1} - y_i)^2}{\varepsilon} \right] dy_1 dy_2 \cdots dy_{N-1}$$

By finding the integral portion of dy_1 , then dy_2 , and so on, the total integral is found to be

$$\sqrt{\frac{(i\pi)^{N-1}}{N}} e^{-(y_N - y_0)^2 / iN}$$

Thus the propagator is

$$\begin{aligned} U(x_N, t_N; x_0, t_0) &= A \left(\frac{2\hbar\varepsilon}{m} \right)^{(N-1)/2} \cdot (i\pi)^{(N-1)/2} \cdot \frac{1}{\sqrt{N}} \cdot \exp \left(\frac{im(x_N - x_0)^2}{2\hbar N\varepsilon} \right) \\ &= A \left(\frac{2\pi\hbar\varepsilon i}{m} \right)^{N/2} \left(\frac{m}{2\pi\hbar i N\varepsilon} \right)^{1/2} \exp \left(\frac{im(x_N - x_0)^2}{2\hbar N\varepsilon} \right) \end{aligned}$$

As $N \rightarrow \infty$ and $N\varepsilon \rightarrow t_N - t_0 \equiv t$, we recover the full propagator given a specific value of A , which agrees with the ones before.

$$\begin{aligned} A &= \left(\frac{2\pi\hbar\varepsilon i}{m} \right)^{-N/2} \\ U(x_N, t_N; x_0, t_0) &= \sqrt{\frac{m}{2\pi\hbar i t}} \exp \left(\frac{im(x_N - x_0)^2}{2\hbar t} \right) \end{aligned}$$

By convention, let $A \equiv B^{-N}$, such that

$$\begin{aligned} B &= \sqrt{\frac{2\pi\hbar\varepsilon i}{m}} \\ \int \mathcal{D}[x(t)] &= \lim_{N \rightarrow \infty} \frac{1}{B} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \frac{dx_1}{B} \frac{dx_2}{B} \cdots \frac{dx_{N-1}}{B} \end{aligned}$$

7 The Heisenberg Uncertainty Relations

7.1 Derivation

Let Ω, Λ be Hermitian and $[\Omega, \Lambda] = i\Gamma$. Also define $\hat{\Omega} = \Omega - \langle \Omega \rangle$ and $\hat{\Lambda} = \Lambda - \langle \Lambda \rangle$.

$$\begin{aligned} (\Delta\Omega)^2(\Delta\Lambda)^2 &= \langle \psi | (\Omega - \langle \Omega \rangle)^2 | \psi \rangle \langle \psi | (\Lambda - \langle \Lambda \rangle)^2 | \psi \rangle \\ &= \langle \psi | \hat{\Omega}^2 | \psi \rangle \langle \psi | \hat{\Lambda}^2 | \psi \rangle \\ &= \langle \hat{\Omega}\psi | \hat{\Omega}\psi \rangle \langle \hat{\Lambda}\psi | \hat{\Lambda}\psi \rangle \end{aligned}$$

Recall the Schwartz inequality $|V_1|^2|V_2|^2 \geq |\langle V_1 | V_2 \rangle|^2$. Thus

$$\begin{aligned} (\Delta\Omega)^2(\Delta\Lambda)^2 &\geq |\langle \hat{\Omega}\psi | \hat{\Lambda}\psi \rangle|^2 = |\langle \psi | \hat{\Omega}^\dagger \hat{\Lambda} | \psi \rangle|^2 \\ &= |\langle \psi | \hat{\Omega} \hat{\Lambda} | \psi \rangle|^2 \\ &= \left| \langle \psi | \frac{\{\hat{\Omega}, \hat{\Lambda}\}}{2} + \frac{[\hat{\Omega}, \hat{\Lambda}]}{2} | \psi \rangle \right|^2 \\ &= \frac{1}{4} \left| \langle \psi | \{\hat{\Omega}, \hat{\Lambda}\} | \psi \rangle + i \langle \psi | \Gamma | \psi \rangle \right|^2 \\ &= \frac{1}{4} \langle \psi | \{\hat{\Omega}, \hat{\Lambda}\} | \psi \rangle^2 + \frac{1}{4} \langle \psi | \Gamma | \psi \rangle^2 \end{aligned}$$

For the canonically conjugate operators like X and P , $\Gamma = \hbar$, so

$$\begin{aligned} (\Delta\Omega)^2(\Delta\Lambda)^2 &\geq \frac{1}{4} \langle \psi | \{\hat{\Omega}, \hat{\Lambda}\} | \psi \rangle^2 + \frac{1}{4} \hbar^2 \\ \Delta\Omega \cdot \Delta\Lambda &\geq \hbar/2 \end{aligned}$$

7.2 Equality

Schwartz inequality is an equality iff. $|V_1\rangle = c|V_2\rangle$. So the minimum case occurs when

$$\begin{aligned} \hat{\Omega} | \psi \rangle &= c \hat{\Lambda} | \psi \rangle \\ \langle \psi | \{\hat{\Omega}, \hat{\Lambda}\} | \psi \rangle &= 0 \end{aligned}$$

For X, P , this becomes

$$\begin{aligned} (P - \langle P \rangle) | \psi \rangle &= c (X - \langle X \rangle) | \psi \rangle \\ \langle \psi | (P - \langle P \rangle) (X - \langle X \rangle) + (X - \langle X \rangle) (P - \langle P \rangle) | \psi \rangle &= 0 \end{aligned}$$

Project onto the X basis and shift frames so that $\langle X \rangle = 0$ for now. We get

$$\begin{aligned} \left(-i\hbar \frac{d}{dx} - \langle P \rangle \right) \psi(x) &= c x \psi(x) \quad \longrightarrow \quad \psi(x) = \psi(0) e^{i\langle P \rangle x / \hbar} e^{icx^2 / 2\hbar} \\ \langle \psi | c^* X^2 + c X^2 | \psi \rangle &= 0 \quad \longrightarrow \quad (c + c^*) \langle \psi | X^2 | \psi \rangle = 0 \end{aligned}$$

So c must be imaginary, $c = i|c|$. Let $\Delta^2 = \hbar/|c|$. Then

$$\psi(x) = \psi(0)e^{i\langle P \rangle x/\hbar}e^{-x^2/2\Delta^2}$$

For arbitrary X with $\langle X \rangle \neq 0$, we have

$$\psi(x) = \psi(\langle X \rangle)e^{i\langle P \rangle(x-\langle X \rangle)/\hbar}e^{-(x-\langle X \rangle)^2/2\Delta^2}$$

I.e. a Gaussian function of arbitrary width provides the lower-bound solution for the Heisenberg uncertainty relation.

7.3 Energy-Time Relation

A lesser-known but important relation:

$$\Delta E \cdot \Delta t \geq \hbar/2$$

One interpretation: violations in energy conservation by ΔE are possible over times $\Delta t \sim \hbar/\Delta E$.

E.g. nucleons are held together by an exchange of pions. A nucleon can produce pions out of thin air, as long as it's caught by the receiver within $\Delta t \sim \hbar/\Delta E = \hbar/\mu c^2$. If the pion travels at c , it can travel at most $r = c\Delta t = \hbar/\mu c$, the **Compton wavelength**. This is the maximal range of the nuclear force.

8 Systems with N Degrees of Freedom

8.1 The Direct Product Space $\mathbb{V}_{1 \otimes 2}$

Let system 1 have $|x_1\rangle, |p_1\rangle$ in $\mathbb{V}_1$ and system 2 have $|x_2\rangle, |p_2\rangle$ in $\mathbb{V}_2$. The **direct product** (or **tensor product**) $|x_1\rangle \otimes |x_2\rangle$ is in $\mathbb{V}_{1 \otimes 2}$.

- **Linear:** $(\alpha_1 |x_1\rangle + \alpha_2 |x_2\rangle) \otimes \beta |x_3\rangle = \alpha_1 \beta |x_1\rangle \otimes |x_3\rangle + \alpha_2 \beta |x_2\rangle \otimes |x_3\rangle$
- **Inner product:** $(\langle x'_1| \otimes \langle x'_2|)(|x_1\rangle \otimes |x_2\rangle) = \langle x'_1|x_1\rangle \langle x'_2|x_2\rangle = \delta(x'_1 - x_1)\delta(x'_2 - x_2)$.

Operators on either $\mathbb{V}_1$ or $\mathbb{V}_2$ can be promoted to operators on $\mathbb{V}_{1 \otimes 2}$, e.g.

$$\begin{aligned} X_1^{(1)} &\longrightarrow X_1^{(1)} \otimes I^{(2)} \equiv X_1^{(1) \otimes (2)} \\ P_2^{(2)} &\longrightarrow I^{(1)} \otimes P_2^{(2)} \equiv P_2^{(1) \otimes (2)} \end{aligned}$$

This obeys the following relation:

$$(\Omega_1^{(1)} \otimes \Gamma_2^{(2)}) (\Theta_1^{(1)} \otimes \Lambda_2^{(2)}) = (\Omega\Theta)_1^{(1)} \otimes (\Gamma\Lambda)_2^{(2)}$$

The commutation relations are:

$$\begin{aligned} [X_i^{(1) \otimes (2)}, P_j^{(1) \otimes (2)}] &= i\hbar \delta_{ij} I^{(1) \otimes (2)} \\ [X_i^{(1) \otimes (2)}, X_j^{(1) \otimes (2)}] &= [P_i^{(1) \otimes (2)}, P_j^{(1) \otimes (2)}] = 0 \end{aligned}$$

From now on, notation will be simplified:

$$\begin{aligned} |x_1\rangle \otimes |x_2\rangle &\longrightarrow |x_1 x_2\rangle \\ X_1^{(1)\otimes(2)} &\longrightarrow X_1 \end{aligned}$$

8.2 The Two-Particle State

8.2.1 Separable Hamiltonians

We know $H = H_1 + H_2$.

In the energy basis, let $|E\rangle = |E_1\rangle \otimes |E_2\rangle$ and $E = E_1 + E_2$.

$$\begin{aligned} H |E\rangle &= (E_1 + E_2) |E_1\rangle \otimes |E_2\rangle = E |E\rangle \\ |\psi(t)\rangle &= e^{-iEt/\hbar} |E\rangle = e^{-iE_1 t/\hbar} |E_1\rangle \otimes e^{-iE_2 t/\hbar} |E_2\rangle \end{aligned}$$

We can also use the explicit X basis and substitute for P and X .

$$\begin{aligned} H |E\rangle &= E |E\rangle \\ \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x_1^2} + V(x_1) - \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x_2^2} + V(x_2) \right] \psi_E(x_1, x_2) &= E \psi_E(x_1, x_2) \end{aligned}$$

If the states are separable, $\psi_E(x_1, x_2) = \psi_{E_1}(x_1)\psi_{E_2}(x_2)$. By plugging in and solving, we also get

$$\psi_E(x_1, x_2) = e^{-iE_1 t/\hbar} \psi_{E_1}(x_1) \cdot e^{-iE_2 t/\hbar} \psi_{E_2}(x_2)$$

Alternatively, if the potential is of the form $V(x_1, x_2) = V(x_1 - x_2)$, the problem can be solved. Consider:

$$\begin{aligned} H &= \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} + V(x_1, x_2) \\ V(x_1, x_2) &\neq V_1(x_1) + V_2(x_2) \end{aligned}$$

Define the following coordinates:

$$\begin{aligned} x_{\text{CM}} &= \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} \\ x_{\text{rel}} &= x_1 - x_2 \end{aligned}$$

The Hamiltonian then reduces to

$$\begin{aligned} H &= H_{\text{CM}} + H_{\text{rel}} = \frac{p_{\text{CM}}^2}{2M} + \frac{p_{\text{rel}}^2}{2\mu} + V(x_{\text{rel}}) \\ M &= m_1 + m_2 \quad , \quad p_{\text{CM}} = M \dot{x}_{\text{CM}} \\ \mu &= \frac{m_1 m_2}{m_1 + m_2} \quad , \quad p = \mu \dot{x}_{\text{rel}} \end{aligned}$$

The eigenfunctions of H factorize:

$$E = \frac{p_{\text{CM}}^2}{2M} + E_{\text{rel}}$$

$$\psi_E(x_1, x_2) = \psi_{E_{\text{CM}}}(x_{\text{CM}}) \cdot \psi_{E_{\text{rel}}}(x_{\text{rel}}) = \frac{e^{ip_{\text{CM}} \cdot x_{\text{CM}}/\hbar}}{\sqrt{2\pi\hbar}} \cdot \psi_{E_{\text{rel}}}(x_{\text{rel}})$$

By studying the system in the CM frame, we can further simplify to $E \rightarrow E_{\text{rel}}$, $\psi_E \rightarrow C\psi_{E_{\text{rel}}}(x_{\text{rel}})$.

8.3 Identical Particles

8.3.1 Bosons and Fermions

Unlike classical mechanics, in QM, we don't know the exact history of a system. The following situations:

- Particle a at x_1 , particle b at x_2
- Particle b at x_1 , particle a at x_2

are indistinguishable if a and b are identical.

We do know that they should be eigenvectors of $X_1 + X_2$ with eigenvalue $a + b$. In the eigenstate of $X_1 + X_2$, there are 2 options:

1. Symmetric: $|\psi\rangle = \frac{1}{\sqrt{2}}(|ab\rangle + |ba\rangle) \equiv |ab, S\rangle$
 - **Bosons:** pion, photon, graviton
 - Spin is integer multiple of $\hbar$
2. Antisymmetric: $|\psi\rangle = \frac{1}{\sqrt{2}}(|ab\rangle - |ba\rangle) \equiv |ab, A\rangle$
 - **Fermions:** electron, proton, neutron
 - Spin is half-integer multiple of $\hbar$

From this we get the **Pauli exclusion principle**: 2 fermions can't be in the same quantum state. If they were, then

$$|\omega_1\omega_2, A\rangle = |\omega_1\omega_2\rangle - |\omega_2\omega_1\rangle \equiv |\omega_1\omega_1\rangle - |\omega_1\omega_1\rangle = 0$$

8.3.2 Spin Statistics

The probability to find half a symmetric/antisymmetric particle pair in x_1 or x_2 , given energy measurements of $n_1 = 3$ and $n_2 = 4$:

$$P_{S/A}(x_1, x_2) = 2 \left| \psi_{S/A}(x_1, x_2) \right|^2 = 2 \left| \frac{1}{\sqrt{2}} [\psi_3(x_1)\psi_4(x_2) \pm \psi_4(x_1)\psi_3(x_2)] \right|^2$$

$$= |\psi_3(x_1)|^2 |\psi_4(x_2)|^2 + |\psi_4(x_1)|^2 |\psi_3(x_2)|^2$$

$$\pm \left[\psi_3^*(x_1)\psi_4(x_1)\psi_4^*(x_2)\psi_3(x_2) + \psi_4^*(x_1)\psi_3(x_1)\psi_3^*(x_2)\psi_4(x_2) \right]$$

The 3rd term (**interference term**) shows that quantum statistics is different from classical statistics. There's something more than just summing both probabilities of $(x_1, 3; x_2, 4)$ or $(x_2, 3; x_1, 4)$.

If $x_1 = x_2 = x$, then

- Bosons tend to **clump together**. Bose-Einstein statistics. Note that the probability is twice that of the classical probability to find either $(x_1, 3; x_2, 4)$ or $(x_1, 4; x_2, 3)$.

$$P_S = 2 \left[|\psi_3(x_1)|^2 |\psi_4(x_2)|^2 + |\psi_4(x_1)|^2 |\psi_3(x_2)|^2 \right]$$

- Fermions tend to **avoid each other**. Fermi-Dirac statistics.

$$P_A = 0$$

8.3.3 Symmetrization/Antisymmetrization of N Particles

Assume $n = 3$ and n_1, n_2, n_3 all distinct. To find a symmetric state, use the **totally symmetric state**.

$$|n_1 n_2 n_3, S\rangle = \frac{1}{\sqrt{3!}} \left[|n_1 n_2 n_3\rangle + |n_1 n_3 n_2\rangle + |n_2 n_1 n_3\rangle \right. \\ \left. + |n_2 n_3 n_1\rangle + |n_3 n_1 n_2\rangle + |n_3 n_2 n_1\rangle \right]$$

For an antisymmetric state, use the **totally antisymmetric state**. Signs depend on the Levi-Civita symbol of $\epsilon_{n_1 n_2 n_3}$.

$$|n_1 n_2 n_3, A\rangle = \frac{1}{\sqrt{3!}} \left[|n_1 n_2 n_3\rangle - |n_1 n_3 n_2\rangle + |n_2 n_3 n_1\rangle \right. \\ \left. - |n_2 n_1 n_3\rangle + |n_3 n_1 n_2\rangle - |n_3 n_2 n_1\rangle \right]$$

Alternatively, use a **Slater determinant**. This also makes it easier to see that the fermion wavefunction vanishes if multiple x 's or n 's coincide.

$$\psi_{n_1 n_2 n_3}(x_1, x_2, x_3, A) = \frac{1}{\sqrt{3!}} \begin{vmatrix} \psi_{n_1}(x_1) & \psi_{n_2}(x_1) & \psi_{n_3}(x_1) \\ \psi_{n_1}(x_2) & \psi_{n_2}(x_2) & \psi_{n_3}(x_2) \\ \psi_{n_1}(x_3) & \psi_{n_2}(x_3) & \psi_{n_3}(x_3) \end{vmatrix}$$

Part II

Angular Momentum

9 Transformations and Their Generators

Classical mechanics:

- Translation: $x \rightarrow x + \varepsilon$ and $p \rightarrow p$
- Translational invariance: $H \rightarrow H$
- Conservation law: $\dot{p} = 0$

Quantum mechanics:

- Translation: $\langle X \rangle \rightarrow \langle X \rangle + \varepsilon$ and $\langle P \rangle \rightarrow \langle P \rangle$
- Translational invariance: $\langle H \rangle \rightarrow \langle H \rangle$
- Conservation law: $\langle \dot{P} \rangle = 0$

9.1 Translations

9.1.1 Active vs. Passive Transformations

Under an infinitesimal translation, each state $|\psi\rangle$ goes to $|\psi_\varepsilon\rangle$ such that

$$\begin{aligned}\langle\psi_\varepsilon|X|\psi_\varepsilon\rangle &= \langle\psi|X|\psi\rangle + \varepsilon \\ \langle\psi_\varepsilon|P|\psi_\varepsilon\rangle &= \langle\psi|P|\psi\rangle\end{aligned}$$

In the **active transformation picture**, define a translation operator $T(\varepsilon)$ such that $T(\varepsilon)|\psi\rangle = |\psi_\varepsilon\rangle$. This corresponds to actually moving the particle by ε .

$$\begin{aligned}\langle\psi_\varepsilon|T^\dagger(\varepsilon)XT(\varepsilon)|\psi_\varepsilon\rangle &= \langle\psi|X|\psi\rangle + \varepsilon \\ \langle\psi_\varepsilon|T^\dagger(\varepsilon)PT(\varepsilon)|\psi_\varepsilon\rangle &= \langle\psi|P|\psi\rangle\end{aligned}$$

In the **passive translation picture**, the operators X and P themselves are replaced. This corresponds to shifting the frame of reference by ε .

$$\begin{aligned}X &\rightarrow T^\dagger(\varepsilon)XT(\varepsilon) = X + \varepsilon I \\ P &\rightarrow T^\dagger(\varepsilon)PT(\varepsilon) = P\end{aligned}$$

For definite position kets $|x\rangle$, we have $T(\varepsilon)|x\rangle = |x + \varepsilon\rangle$. Thus for a general $|\psi\rangle$,

$$\begin{aligned}T(\varepsilon)|\psi\rangle &= T(\varepsilon)\int_{-\infty}^{\infty}|x\rangle\langle x|\psi\rangle dx \\ &= \int_{-\infty}^{\infty}\langle x + \varepsilon\rangle\langle x|\psi\rangle dx \\ &= \int_{-\infty}^{\infty}\langle x'\rangle\langle x' - \varepsilon|\psi\rangle dx'\end{aligned}$$

Thus

$$\langle x|T(\varepsilon)|\psi\rangle = \psi(x - \varepsilon)$$

E.g. if $\psi(x) = \langle x|\psi\rangle \sim e^{-x^2}$, then $\psi_\varepsilon(x) \sim e^{-(x-\varepsilon)^2}$.

9.1.2 Momentum as a Generator of Translations

First, find the explicit form of $T(\varepsilon)$. Use $T(0) = I$. Assume the form

$$T(\varepsilon) = I - \frac{i\varepsilon}{\hbar}G$$

call G the **generator** of the transformation. Then

$$\begin{aligned}\langle x|T(\varepsilon)|\psi\rangle &= \psi(x - \varepsilon) \\ \langle x|\psi\rangle - \frac{i\varepsilon}{\hbar}\langle x|G|\psi\rangle &\approx \psi(x) - \frac{d\psi}{dx}\varepsilon \\ \langle x|G|\psi\rangle &\approx -i\hbar\frac{d\psi}{dx}\end{aligned}$$

Thus $G = P$, and **momentum is a generator of translations**, just like classical mechanics. The explicit form of $T(\varepsilon)$ is

$$T(\varepsilon) = I - \frac{i\varepsilon}{\hbar}P$$

We can also find a quantum analogue of the conservation law. Note that $\langle H \rangle$ must remain the same, so

$$\begin{aligned}\langle \psi | H | \psi \rangle &= \langle \psi | T^\dagger(\varepsilon) H T(\varepsilon) | \psi \rangle \\ &= \langle \psi | \left(I + \frac{i\varepsilon}{\hbar}P \right) H \left(I - \frac{i\varepsilon}{\hbar}P \right) | \psi \rangle \\ &= \langle \psi | H | \psi \rangle + \frac{i\varepsilon}{\hbar} \langle \psi | [P, H] | \psi \rangle + O(\varepsilon^2)\end{aligned}$$

Thus $\langle [P, H] \rangle = 0$, and by Ehrenfest's theorem, P is a time-independent variable when averaged, i.e.

$$\langle \dot{P} \rangle = 0$$

In general, the infinitesimal canonical transformation generated by $g(x, p)$, $\delta\omega = \varepsilon\{\omega, g\}$, corresponds to the infinitesimal unitary transformation $U_G(\varepsilon) = I - i\varepsilon/\hbar G$.

$$\delta\Omega = -\frac{i\varepsilon}{\hbar}[\Omega, G]$$

9.1.3 Finite Translations

Divide interval of length a into N segments, then take the limit as $N \rightarrow \infty$.

$$\begin{aligned}T(a/N) &= I - \frac{ia}{\hbar N}P \\ T(a) &= \lim_{N \rightarrow \infty} \left[I - \frac{ia}{\hbar N}P \right]^N = e^{-iaP/\hbar}\end{aligned}$$

In the X basis, $P = -i\hbar \frac{d}{dx}$, so

$$\begin{aligned}T(a) &= e^{-a d/dx} = 1 - a \frac{d}{dx} + \frac{a^2}{2!} \frac{d^2}{dx^2} + \frac{a^3}{3!} \frac{d^3}{dx^3} - \dots \\ \langle x | T(a) | \psi \rangle &= \psi(x) - a \frac{d\psi}{dx} + \frac{a^2}{2!} \frac{d^2\psi}{dx^2} - \dots = \psi(x - a)\end{aligned}$$

9.2 Invariance Under Transformations

9.2.1 Time-Translation Invariance

Prepare two systems at times t_1 and t_2 , both in state $|\psi\rangle$. Evolve under $U = e^{-iHt/\hbar}$.

$$\begin{aligned}|\psi(t_1 + \varepsilon)\rangle &\approx \left[I - \frac{i\varepsilon}{\hbar}H(t_1) \right] |\psi_0\rangle \\ |\psi(t_2 + \varepsilon)\rangle &\approx \left[I - \frac{i\varepsilon}{\hbar}H(t_2) \right] |\psi_0\rangle\end{aligned}$$

The outcome is the same if $H(t_2)|\psi_0\rangle = H(t_1)|\psi_0\rangle$. Since ψ_0 and t_1, t_2 are arbitrary, $dH/dt = 0$, i.e. the system has no t dependence.

By Ehrenfest's theorem, $i\hbar \frac{d}{dt}\langle\Omega\rangle = \langle[\Omega, H]\rangle$. With $\Omega = H$,

$$\langle\dot{H}\rangle = 0$$

Which is the law of conservation of energy. Experiments that conserve energy have the same outcome regardless of where it happens.

9.2.2 Parity Invariance

Define the discrete parity operator Π such that $\Pi|x\rangle = |-x\rangle$ and $\Pi|p\rangle = |-p\rangle$. In the x basis, action of the Π on an arbitrary $|\psi\rangle$ is

$$\begin{aligned}\Pi|\psi\rangle &= \Pi \int_{-\infty}^{\infty} |x\rangle \langle x|\psi\rangle dx \\ &= \int_{-\infty}^{\infty} |-x\rangle \langle x|\psi\rangle dx \\ &= \int_{-\infty}^{\infty} |x'\rangle \langle -x'|\psi\rangle dx \\ \langle x|\Pi|\psi\rangle &= \psi(-x)\end{aligned}$$

Note that $\Pi^2 = I$. Thus Π has eigenvalues ± 1 , $\Pi = \Pi^{-1}$, and Π is both Hermitian and unitary.

H is parity-invariant if 2 systems that differ by a parity, after evolution, also differ by a parity. I.e.

$$\Pi^\dagger H(X, P)\Pi = H(-X, -P) = H(X, P)$$

Thus $[\Pi, U(t)] = 0$ as well.

All natural laws are invariant under spatial translations (homogeneity of space), temporal translations (homogeneity of time), and rotations (isotropic). But the weak interaction is **not invariant under parity**. The mirror-reflected scenario does not reflect a possible physics.

9.2.3 Time-Reversal Symmetry

Let the system start in (x_0, p_0) and come to (x, p) after time T . Reverse the momentum to get $(x, -p)$, then let the system evolve for time T . If we get $(x_0, -p_0)$, the system is time-reversal invariant (TIR). Note that the **momentum is reversed**.

TIR is not possible if e.g. the potential/force depends on t or x . For a particle with time-independent H , the complex conjugate performs the momentum-reversal.

$$i\hbar \frac{\partial \psi}{\partial t} = H\psi(x, t)$$

For TIR to hold, start at $\psi(x, 0)$, evolve for T , complex conjugate, evolve for T , and we should get $\psi^*(x, 0)$.

$$\psi(x, 0) \rightarrow e^{-iHT/\hbar}\psi(x, T) \rightarrow e^{iH^*T/\hbar}\psi^*(x, T) \rightarrow e^{-iHT/\hbar}e^{-iH^*T/\hbar}\psi^*(x, 0)$$

So TIR holds iff. $H = H^*$, i.e. if H is real.

9.3 Rotations

In 2D classical space,

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} p_x \\ p_y \end{pmatrix} \rightarrow \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} p_x \\ p_y \end{pmatrix}$$

Now consider an infinitesimal rotation $\varepsilon_z \hat{k}$ around the z -axis. Let L_z be the generator of infinitesimal rotations. The rotation operator is then

$$U[R(\varepsilon_z \hat{k})] = I - \frac{i\varepsilon}{\hbar} L_z$$

From the classical case,

$$U[R] |x, y\rangle = |x \cos \phi - y \sin \phi, x \sin \phi + y \cos \phi\rangle \approx |x - y\varepsilon_z, x\varepsilon_z + y\rangle$$

In the quantum case,

$$\begin{aligned} U[R] |\psi\rangle &= U[R] \int |x, y\rangle \langle x, y | \psi \rangle dx dy \\ &= \int |x - y\varepsilon, x\varepsilon + y\rangle \langle x, y | \psi \rangle dx dy \\ &= \int |x', y'\rangle \langle x' + y\varepsilon, y' - x\varepsilon | \psi \rangle dx' dy' \\ \langle x, y | U[R] | \psi \rangle &= \psi(x + y\varepsilon_z, y - x\varepsilon_z) \end{aligned}$$

With $U[R] = I - i\varepsilon_z L_z / \hbar$, we can see that

$$L_z \rightarrow x \left(-i\hbar \frac{\partial}{\partial y} \right) - y \left(-i\hbar \frac{\partial}{\partial x} \right) \equiv X P_y - Y P_x$$

The operator has an easier form in polar coordinates:

$$L_z \rightarrow -i\hbar \frac{\partial}{\partial \phi}$$

$$U[R(\phi_0)] \rightarrow \exp \left(-\phi_0 \frac{\partial}{\partial \phi} \right)$$

Call L_z the **angular momentum operator**, which generates infinitesimal rotations about the z -axis. If the system is rotationally invariant, the experiment and its rotated version must give the same outcome.

$$U^\dagger[R] H(X, P_x; Y, P_y) U[R] = H(X, P_x; Y, P_y)$$

$$[L_z, H] = 0 \longrightarrow L_z \text{ conserved}$$

The equation $[L_z, H] = 0$ imply that L_z and H have a simultaneous eigenbasis. The eigenvectors of L_z are given by $L_z |l_z\rangle = l_z |l_z\rangle$. In polar coordinates,

$$\begin{aligned} -i\hbar \frac{\partial \psi_{l_z}}{\partial \phi} &= l_z \psi_{l_z}(\rho, \phi) \\ \psi_{l_z}(\rho, \phi) &= R(\rho) e^{il_z \phi / \hbar} \end{aligned}$$

By forcing Hermiticity, $\langle \psi_1 | L_z | \psi_2 \rangle = \langle \psi_2 | L_z | \psi_1 \rangle^*$, we get $\psi(\rho, 0) = \psi(\rho, 2\pi)$. Thus

$$l_z = m\hbar \quad , \quad m = 0, \pm 1, \pm 2, \dots$$

m is the **magnetic quantum number**.

9.4 Summary: J, L, S

Operator	Measurement outcome	Definition
L^2 L_z	$\hbar^2 l(l+1)$, $l = 0, 1, 2, \dots$ $\hbar m_l$, $m_l = -l, -l+1, \dots, l-1, l$	l = orbital/azimuthal quantum number m_l = magnetic quantum number
S^2 S_z	$\hbar^2 s(s+1)$, $s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ $\hbar m_s$, $m_s = -s, -s+1, \dots, s-1, s$	s is spin m_s = spin projection quantum number
J^2 J_z	$\hbar^2 j(j+1)$, $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ $\hbar m_j$	j = total angular momentum quantum number m_j = total angular mom. projection quantum number

10 Orbital Angular Momentum

10.1 Angular Momentum in 3 Dimensions

We follow the previous analysis and extend to 3 dimensions:

$$L_X = YP_Z - ZP_Y$$

$$L_Y = ZP_X - XP_Z$$

$$L_Z = XP_Y - YP_X$$

These are the generator of infinitesimal rotations:

$$U[R(\vec{\theta})] = \lim_{N \rightarrow \infty} \left(I - \frac{i}{\hbar} \frac{\theta}{N} \hat{\theta} \cdot \mathbf{L} \right)^N = e^{-i\vec{\theta} \cdot \mathbf{L} / \hbar}$$

Furthermore, the angular momentum operators satisfy

$$\mathbf{L} \times \mathbf{L} = i\hbar \mathbf{L} \quad \left\{ \begin{aligned} [L_x, L_y] &= i\hbar L_z \\ [L_y, L_z] &= i\hbar L_x \\ [L_z, L_x] &= i\hbar L_y \end{aligned} \right.$$

The total angular momentum operator $L^2 = L_x^2 + L_y^2 + L_z^2$ satisfies $[L^2, L_i] = 0$. Thus we can find a simultaneous eigenbasis for L^2 and L_i . Furthermore, if H is rotationally invariant under arbitrary $U[R]$, then there exists a simultaneous eigenbasis for H, L^2, L_i .

$$U^\dagger[R]HU[R] = H \quad \longrightarrow \quad [H, L_i] = 0 \quad \text{and} \quad [H, L^2] = 0$$

However, we can't find a simultaneous eigenbasis for all L_x, L_y, L_z because they don't commute.

In spherical coordinates,

$$\begin{aligned} L_z &\rightarrow -i\hbar \frac{\partial}{\partial \phi} \\ L^2 &\rightarrow -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \end{aligned}$$

10.2 Relation between L^2 and L_z

Let $|\alpha, \beta\rangle$ be the common eigenbasis of L^2 and L_z .

$$L^2 |\alpha, \beta\rangle = \alpha |\alpha, \beta\rangle$$

$$L_z |\alpha, \beta\rangle = \beta |\alpha, \beta\rangle$$

Define the lowering/raising operators $L_\pm = L_x \pm iL_y$ such that $[L^2, L_\pm] = 0$ and $[L_z, L_\pm] = \pm\hbar L_\pm$. The operators $L_\pm$ raise/lower the eigenvalue of L_z by $\hbar$. E.g.:

$$L_z L_+ |\alpha, \beta\rangle = (\hbar L_+ + L_+ L_z) |\alpha, \beta\rangle = (\beta + \hbar) L_+ |\alpha, \beta\rangle$$

$$L_+ |\alpha, \beta\rangle = C_+(\alpha, \beta) |\alpha, \beta + \hbar\rangle$$

$$L_- |\alpha, \beta\rangle = C_-(\alpha, \beta) |\alpha, \beta - \hbar\rangle$$

But there must be some bounds in place, since $|l_z| \leq \sqrt{l^2}$. In QM, $\langle \alpha\beta | L^2 - L_z^2 | \alpha\beta \rangle \geq 0$, so $\alpha \geq \beta^2$. Then there must exist a state $|\alpha\beta_{\max}\rangle$ where β can no longer be raised. From $L_+ |\alpha\beta_{\max}\rangle = 0$ and $L_- |\alpha\beta_{\min}\rangle = 0$,

$$\alpha = \beta_{\max}(\beta_{\max} + \hbar)$$

$$\alpha = \beta_{\min}(\beta_{\min} - \hbar)$$

$$\beta_{\min} = -\beta_{\max}$$

Say we go from $\beta_{\min}$ to $\beta_{\max}$ in k steps of $\hbar$. Then

$$\beta_{\max} - \beta_{\min} = 2\beta_{\max} = \hbar k$$

$$\beta_{\max}/2 = k/2, \quad k = 0, 1, 2, \dots$$

$$\alpha = \hbar^2 \left(\frac{k}{2} \right) \left(\frac{k}{2} + 1 \right)$$

We call $k/2$ the **angular momentum of the state**.

10.3 Total Angular Momentum: $J = L + S$

When rotating an arbitrary state vector ψ with L_z , we 1) reassign (x, y) to (x', y') , 2) rotate vector itself by an infinitesimal angle.

- L_z does part 1.
- S , the **spin matrix**, does part 2.

Thus the total generator of infinitesimal rotations is

$$J_i = L_i + S_i$$

$$\mathbf{J} \times \mathbf{J} = i\hbar\mathbf{J}$$

The eigenvalue of J^2 and J_z , following previous analysis, is

$$J^2 |jm\rangle = j(j+1)\hbar^2 |jm\rangle, \quad j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$$

$$J_z |jm\rangle = m\hbar |jm\rangle, \quad m = j, j-1, \dots, -j+1, -j$$

The raising and lowering operators (to appropriate normalization) can be defined:

$$J_{\pm} = J_x \pm iJ_y$$

$$J_+ |jm\rangle = \hbar\sqrt{(j-m)(j+m+1)} |j, m+1\rangle$$

$$J_- |jm\rangle = \hbar\sqrt{(j+m)(j-m+1)} |j, m-1\rangle$$

Since acting on $|j, \pm j\rangle$ results in $|0\rangle$, for each state with total angular momentum (J^2) equal to j , there's only $(2j+1)$ distinct states: one for each J_z eigenvalue $j\hbar, (j-1)\hbar, \dots$

Assume $J = L$ to disregard the spin matrix. Simultaneous eigenfunctions to H, L_z, L^2 have the form

$$\psi_{Elm}(r, \theta, \phi) = R_{Elm}(r)Y_l^m(\theta, \phi)$$

$$Y_l^m(\theta, \phi) = \text{Spherical harmonics}$$

10.4 Case Study: Free Particle in Spherical Coordinates

10.4.1 Schrodinger Equation in Spherical Coordinates

We start with the Schrodinger equation in spherical coordinates:

$$\left[-\frac{\hbar^2}{2\mu} \left(\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right) + V(r) \right] \psi_E = E\psi_E$$

$$\psi_{Elm}(r, \theta, \phi) = R_{Elm}(r)Y_l^m(\theta, \phi)$$

We note that the terms in ∇^2 look similar to the spherical-coordinate form of L^2 . After substituting,

$$\left[-\frac{\hbar^2}{2\mu} \left(\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} - \frac{l(l+1)}{r^2} \right) + V(r) \right] R(r) = ER(r)$$

We introduce $U_{El} = R_{El}/r$. This simplifies the equation to

$$\left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + V(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} \right] U_{El} = EU_{El}$$

This is just like the normal Schrodinger equation, except the range is $0 \leq r \leq \infty$ and there's a repulsive centrifugal force $l(l+1)\hbar^2/2\mu r^2$. To prevent blowups at $r \rightarrow 0$, we enforce $U_{El} \rightarrow 0$ as $r \rightarrow 0$. To make U_{El} in the Hilbert space, we enforce $U_{El} \rightarrow 0$ or $U_{El} \rightarrow e^{ikr}$ as $r \rightarrow \infty$.

If $V(r)$ is less singular than $1/r^2$, then $U_{El}'' \simeq \frac{l(l+1)}{r^2} U_{El}$. If $U_{El} \sim r^\alpha$, then

$$U_{El} \sim \begin{cases} r^{l+1} & \text{regular} \\ r^{-l} & \text{irregular} \end{cases}$$

10.4.2 Free Particle in Spherical Coordinates

We start with the Schrodinger equation in spherical coordinates, introducing $U_{El} = R_{El}/r$ as before. We get

$$\begin{aligned} \left[\frac{d^2}{dr^2} + k^2 - \frac{l(l+1)}{r^2} \right] U_{El} &= 0, \quad k^2 = \frac{2\mu E}{\hbar^2} \\ \left[-\frac{d^2}{d\rho^2} + \frac{l(l+1)}{\rho^2} \right] U_l &= U_l, \quad \rho = kr \end{aligned}$$

Define the raising and lowering operators $d_l^\dagger$ and d_l such that

$$(d_l d_l^\dagger) U_l = U_l \quad \longrightarrow \quad \begin{cases} d_l = \frac{d}{d\rho} + \frac{l+1}{\rho} \\ d_l^\dagger = -\frac{d}{d\rho} + \frac{l+1}{\rho} \end{cases}$$

We have $d_l^\dagger d_l d_l^\dagger U_l = d_l^\dagger U_l$, or $d_{l+1}^\dagger d_{l+1} (d_l^\dagger U_l) = (d_l^\dagger U_l)$. Thus we can view $d_l^\dagger U_l \sim U_{l+1}$, hence the name “raising” operator. With $d_l^\dagger$ and d_l defined, we can build a ladder of solutions from $l = 0$.

At $l = 0$, two solutions are possible from the differential equation:

$$\begin{aligned} 1) U_0^A &= \sin \rho & \longrightarrow & R_0^A = \frac{\sin \rho}{\rho} \\ 2) U_0^B &= -\cos \rho & \longrightarrow & R_0^B = \frac{-\cos \rho}{\rho} \end{aligned}$$

10.4.3 Bessel and Neumann Wavefunctions

We seek to find a relationship between R_l and R_0 . Note that

$$\begin{aligned} \rho R_{l+1} &= d_l^\dagger (\rho R_l) = \left[-\frac{d}{d\rho} + \frac{l+1}{\rho} \right] (\rho R_l) \\ R_{l+1} &= \left[-\frac{d}{d\rho} + \frac{l}{\rho} \right] R_l = \rho^l \left(-\frac{d}{d\rho} \right) \frac{R}{\rho^l} \\ \frac{R_{l+1}}{\rho_{l+1}} &= \left(-\frac{1}{\rho} \frac{d}{d\rho} \right)^2 \frac{R_l}{\rho^l} \\ &= \left(-\frac{1}{\rho} \frac{d}{d\rho} \right)^l \frac{R_0}{\rho^0} \end{aligned}$$

Thus:

$$R_l = (-\rho)^l \left(\frac{1}{\rho} \frac{\partial}{\partial \rho} \right)^l R_0$$

There are thus two families of solutions:

Spherical Bessel functions of order l : $R_l^A \equiv j_l(\rho) = (-\rho)^l \left(\frac{1}{\rho} \frac{d}{d\rho} \right)^l \left(\frac{\sin \rho}{\rho} \right)$

Spherical Neumann functions of order l : $R_l^B \equiv n_l(\rho) = (-\rho)^l \left(\frac{1}{\rho} \frac{d}{d\rho} \right)^l \left(\frac{-\cos \rho}{\rho} \right)$

As $\rho \rightarrow 0$,

$$j_l(\rho) \rightarrow \frac{\rho^l}{(2l+1)!!}$$

$$n_l(\rho) \rightarrow -\frac{(2l+1)!!}{\rho^{l+1}}$$

These correspond to the two families of U_{El} described previously. As $\rho \rightarrow 0$, $n_l(\rho)$ is singular, so if we want a solution regular in all space,

$$\psi_{Elm}(r, \theta, \phi) = j_l(kr) Y_l^m(\theta, \phi), \quad E = \frac{\hbar^2 k^2}{2\mu}$$

10.5 The Isotropic Oscillator

The isotropic oscillator is just the 1d oscillator in 3d space:

$$H = \frac{P_x^2 + P_y^2 + P_z^2}{2\mu} + \frac{1}{2}\mu\omega^2(X^2 + Y^2 + Z^2)$$

$$\psi_{Elm}(r, \theta, \phi) = j_l(kr) Y_l^m(\theta, \phi)$$

With the wavefunction for a spherically symmetric potential, we obtain the radial equation:

$$\left[\frac{dd^2}{ddr^2} + \frac{2\mu}{\hbar^2} \left(E - \frac{1}{2}\mu\omega^2 r^2 - \frac{l(l+1)\hbar^2}{2\mu r^2} \right) \right] U_{El} = 0$$

At $r \rightarrow \infty$, we have $U \sim e^{-y^2/2}$, where $y = r\sqrt{\mu\omega/\hbar}$. This is dimensionless, so let $U(y) = e^{-y^2/2}v(y)$. Furthermore, from previous analysis, we want $U(y) \rightarrow y^{l+1}$ as $r \rightarrow 0$. Together, we get the differential equation and a power series solution:

$$v'' - 2yv' + \left[2\lambda - 1 - \frac{l(l+1)}{y^2} \right] v = 0, \quad \lambda = \frac{E}{\hbar\omega}$$

$$v(y) = y^{l+1} \sum_{n=0}^{\infty} C_n y^n$$

After messy solving, we get $E = \left(n + \frac{3}{2}\right) \hbar\omega$, where $n = 2k + 1$ is the **principal quantum number** with $k = 0, 1, 2, \dots$. The first few eigenstates are:

$n = 0$	$l = 0$	$m = 0$
$n = 1$	$l = 1$	$m = \pm 1, 0$
$n = 2$	$l = 0, 2$	$m = 0; m = \pm 2, \pm 1, 0$

10.6 The Hydrogen Atom

10.6.1 Schrodinger Equation

Since $m_p \gg m_e$, consider the electron to move around a spherically symmetric potential e/r^2 generated by a stationary proton.

The Schrodinger equation for spherical coordinates with $R_{El} = U_{El}/r$ is

$$\left[\frac{d^2}{dr^2} + \frac{2m}{\hbar^2} \left(E - \frac{e^2}{r} - \frac{l(l+1)\hbar^2}{2mr^2} \right) \right] U_{El} = 0$$

$$\psi_{Elm}(r, \theta, \phi) = R_{El}(r) Y_l^m(\theta, \phi)$$

- At $r \rightarrow \infty$, $d^2U/dr^2 \sim (-2mE/\hbar^2)U$. Let $\rho = \sqrt{2mW/\hbar^2}r$, where $W = -E$ is the binding energy. Then $U_{El} \rightarrow e^{-\rho}$ as $r \rightarrow \infty$.
- At $r \rightarrow 0$, we want $U \sim r^{l+1}$ as described previously.

Let $U_{El} = e^{-\rho}v_{El}$ and plug in to the Schrodinger equation. We get

$$\frac{d^2v}{d\rho^2} - 2\frac{dv}{d\rho} + \left[\frac{e^2\lambda}{\rho} - \frac{l(l+1)}{\rho^2} \right] v = 0, \quad \lambda = \sqrt{\frac{2m}{\hbar^2 W}}$$

We try the ansatz $v = \rho^{l+1} \sum C_k \rho^k$ and after a lot of messy algebra, get

$$\frac{C_{k+1}}{C_k} = \frac{-e^2\lambda + 2(k+l+1)}{(k+l+2)(k+l+1) - l(l+1)}$$

As $k \rightarrow \infty$, $C_{k+1}/C_k \rightarrow 2/k$, which accords with $v \sim \rho^m e^\rho$. This blows up as $\rho \rightarrow \infty$, so we need the series to cut off at some k , i.e. $2(k+l+1) = e^2\lambda$. Recalling the definition of lambda, we get

$$E = -W = -\frac{me^4}{2\hbar^2(k+l+1)^2}, \quad k, l = 0, 1, 2, \dots$$

Or with the **principal quantum number** $n = k + l + 1$,

$$E = -\frac{me^4}{2\hbar^2 n^2}, \quad n = 1, 2, 3, \dots$$

Spectroscopic notation: $l = 0, 1, 2, 3, \dots$ corresponds to $s, p, d, f, \dots$. Thus $1s$ means $(n = 1; l = 0)$, $2s$ means $(n = 2, l = 0)$, and $2p$ means $(n = 2; l = 1)$. m_l is clear from l .

A **Rydberg of energy** is $Ry = me^4/2\hbar^2$, thus $E_n = -Ry/n^2$ for $n = 1, 2, 3, \dots$

11 Spin

11.1 The Spin Matrix

Even when an electron is not moving ($L^2 = 0$), it is measured to have an intrinsic angular momentum. Measurement gives $\pm \hbar/2$.

The total angular momentum is $\mathbf{J} = \mathbf{L} + \mathbf{S}$ where

$$S_i = \frac{\hbar}{2} \sigma_i$$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Note that this is a 2×2 matrix. We describe e^- with a 2-component **spinor**, not a 3-component vector.

$$\psi(x, y, z) = \begin{pmatrix} \psi_+(x, y, z) \\ \psi_-(x, y, z) \end{pmatrix} = \psi_+ \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \psi_- \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- If $(\psi_+, \psi_-) = (1, 0)$ then measuring S_z gives $\hbar/2$.
- If $(\psi_+, \psi_-) = (0, 1)$ then measuring S_z gives $-\hbar/2$.
- ψ_+ is the “amplitude” for $S_z = \hbar/2$ and ψ_- is the “amplitude” for $S_z = -\hbar/2$.

Note that the magnitude of spin cannot change. Thus the magnitude of a particle’s spin is an inherent quantity.

$$S^2 = \frac{3}{4} \hbar^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

In an alternate representation (which emphasizes the Hilbert space), $|s, s_z\rangle = |s, m\hbar\rangle \equiv |s, m\rangle$.

$$|s, m\rangle = |1/2, 1/2\rangle \longrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ in } S_z \text{ basis}$$

$$|s, m\rangle = |1/2, -1/2\rangle \longrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ in } S_z \text{ basis}$$

Thus the general spin state in S_z is $|\chi\rangle = \alpha |1/2, 1/2\rangle + \beta |1/2, -1/2\rangle$ with the normalization condition $\alpha^2 + \beta^2 = 1$. Eigenstates of S_z are (for arbitrary axis $\hat{k}$)

$$\langle 1/2, \pm 1/2 | S_{\hat{k}} | 1/2, \pm 1/2 \rangle = \pm \frac{\hbar}{2} \hat{k}$$

11.1.1 Pauli Matrices

Pauli matrices introduced above satisfy:

- Anticommutator: $\{\sigma_i, \sigma_j\} = 2I\delta_{ij}$
- Commutator: $[\sigma_i, \sigma_j] = 2i \sum_k \epsilon_{ijk} \sigma_k$ from $\sigma_i \sigma_j = i\epsilon_{ijk} \sigma_k$
- Generally, $(\hat{n} \cdot \sigma)^2 = I$ for arbitrary axis $\hat{n}$.

- If $\mathbf{A}$, $\mathbf{B}$ commute with σ , the following useful identity:

$$(\mathbf{A} \cdot \sigma)(\mathbf{B} \cdot \sigma) = (\mathbf{A} \cdot \mathbf{B})I + i(\mathbf{A} \times \mathbf{B}) \cdot \sigma$$

From this, we can get the exact form of the spin rotation operator $U[R(\vec{\theta})] = \exp[-i\vec{\theta} \cdot \vec{S}/\hbar]$. Expand and substitute to get

$$\begin{aligned} U[R(\vec{\theta})] &= \exp[-i\vec{\theta} \cdot \vec{\sigma}/2] = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{-i\theta}{2} \right)^n (\hat{\theta} \cdot \vec{\sigma})^n \\ &= I + \left(-\frac{i\theta}{2} \right) (\hat{\theta} \cdot \vec{\sigma}) + \frac{1}{2!} \left(-\frac{i\theta}{2} \right)^2 I + \frac{1}{3!} \left(-\frac{i\theta}{2} \right)^3 (\hat{\theta} \cdot \vec{\sigma}) + \dots \\ &= \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) \hat{\theta} \cdot \vec{\sigma} \end{aligned}$$

11.2 Spin Dynamics

11.2.1 Orbital Magnetic Moment

Imagine a classical planar loop with magnetic moment $\mu = I\vec{A}$ in magnetic field $\vec{B}$. The torque is $\tau = \mu \times B$. The interaction Hamiltonian is

$$H_{int} = \int \tau(\theta) d\theta = \int \mu B \sin \theta d\theta = -\mu B \cos \theta = -\mu \cdot B$$

This equation holds for tiny circular loops too. Consider a particle with mass m , charge q , and orbit radius r . The current is $I = qv/2\pi r$ and the magnetic moment is

$$\vec{\mu} = \frac{qv}{2\pi r} (\pi r^2) = \frac{q}{2m} (mvr) = \frac{q}{2m} \vec{L}$$

The ratio between μ and L is the gyromagnetic ratio, $\gamma = q/2m$.

In QM, we use the following EM Hamiltonian:

$$H = \frac{1}{2m} (\mathbf{P} - q\mathbf{A})^2 + q\Phi$$

By 1) Assuming $\mathbf{A}$ such that $\mathbf{B}$ points in $\hat{z}$, 2) Dropping last term because it's quadratic in $\mathbf{B}$, 3) Setting the Coulomb gauge $\nabla \cdot \mathbf{A} = 0$, we get the same answer from classical:

$$\vec{\mu} = \left(\frac{q}{2m} \right) \vec{L}$$

Projected onto the $\hat{z}$ axis, we get $\mu_z = q\hbar n/2m$, where $n \in \mathbb{Z}_0^+$. This $q\hbar/2m$ is called the **Bohr magneton**.

11.2.2 Quantum Spin Magnetic Moment

Spin is similar enough to orbital calculations. We assume magnetic moment $\vec{\mu} \propto \vec{S}$, specifically $\vec{\mu} = g\gamma\vec{S}$ for some constant g .

Experimentally, $g \approx 2$ (very close but not quite). The exact reason for this is in QFT. Main idea: interaction term is 2x bigger for spin magnetic moment than for orbital.

- Though this is hardly a “fair” comparison, given that L and S are different...
- Nevertheless, note that $\gamma_{\text{orbital}} = 2\gamma_{\text{spin}}$. **This is important.**

The interaction Hamiltonian for a particle with spin is:

$$H_{\text{int}} = -\vec{\mu} \cdot \vec{B} = -\gamma\vec{S} \cdot \vec{B}, \quad (\gamma \text{ is for spin now})$$

The state evolves as

$$|\psi(t)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle = e^{i\gamma t(S \cdot B)/\hbar} |\psi(0)\rangle$$

This is equivalent to a rotation by $-\gamma t \vec{B}$ around the $\vec{S}$ -axis. By symmetry, $\langle S \rangle$ precesses around $\vec{B}$ at the **Larmor frequency**:

$$\omega_0 = -\gamma B$$

11.2.3 Parametric Resonance

In a field $B\hat{k}$, the magnetic moment $\vec{\mu}$ precesses at frequency $\vec{\omega}_0 = -\gamma B\hat{k}$. In a frame **rotating** at ω , the effective field is $\vec{B}_r = \vec{B} + \vec{\omega}/\gamma$.

$$\text{Net } \omega : \vec{\omega}_0 - \vec{\omega} = -\gamma(\vec{B} + \vec{\omega}/\gamma)$$

$$\text{Net field : } \vec{B}_r = \vec{B} + \vec{\omega}/\gamma$$

Now consider an oscillating $\vec{B}$, e.g. $\vec{B} = B \cos(\omega t)\hat{i} + B \sin(\omega t)\hat{j} + B_0\hat{k}$. It's helpful to consider a rotating frame to “freeze” the direction of B .

$$\vec{B}_r = B\hat{i}_r + (B_0 - \omega/\gamma)\hat{k}$$

The magnetic moment oscillates at $\vec{\omega}_r = -\gamma\vec{B}_r$, magnitude $|\omega_r| = \gamma\sqrt{B^2 + (B_0 - \omega/\gamma)^2}$. The greatest oscillation occurs when $\omega_0 = \omega$, i.e. the frame's rotation matches the precession of $\vec{\mu}$. This is **parametric resonance**.

11.3 The Zeeman Effect

The addition of the spin Hamiltonian to the orbital Hamiltonian explains why spectral lines on hydrogen appear to “split.” Consider a separable Hamiltonian for electron (assume proton is stationary):

$$H = H_0 + H_s \longrightarrow H = H_{\text{Coulomb}} - \frac{eB}{2m}L_z - \frac{eB}{m}S_z$$

$$|\psi\rangle = |\psi_0\rangle \otimes |\chi_s\rangle$$

If $|\chi_s\rangle$ is an eigenstate of L_z , then the spin state creates 2 possibilities whereas before there was 1.

$$\begin{aligned} |nlm, m_s = 1/2\rangle &\longrightarrow \psi_{nlm}(r, \theta, \phi) \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ |nlm, m_s = -1/2\rangle &\longrightarrow \psi_{nlm}(r, \theta, \phi) \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{aligned}$$

The eigenvalues of H are:

$$H |nlm, m_s\rangle = \left[\underbrace{-\frac{\text{Ry}}{n^2}}_{\text{H atom}} + \underbrace{\frac{eB}{2m}(m\hbar + 2m_s\hbar)}_{\text{m is mass/quantum \#}} \right] |nlm, m_s\rangle$$

In hydrogen, the electron (fermion) can have $m_s = 1/2$ or $m_s = -1/2$. Thus the ground state energy level splits into $E_{n=1} = -\text{Ry} \pm e\hbar B/2m$. And so on for higher n . This splitting of spectral lines is called the **Zeeman effect**.

11.4 Addition of Spin

A system of 2 spin- $\frac{1}{2}$ particles (ignoring orbital angular momentum) is denoted $|s_1, m_1\rangle \otimes |s_2, m_2\rangle \equiv |s_1 m_1, s_2 m_2\rangle$

$$\begin{aligned} S_i^2 |s_1 m_1, s_2 m_2\rangle &= \hbar s_i(s_i + 1) |s_1 m_1, s_2 m_2\rangle \\ S_{iz} |s_1 m_1, s_2 m_2\rangle &= \hbar m_i |s_1 m_1, s_2 m_2\rangle \end{aligned}$$

Here $s = 1/2$ and $m = -1/2, +1/2$. We use the notation (throwback to quantum computing) $|+\rangle$ for $m = 1/2$ and $|-\rangle$ for $m = -1/2$.

The total angular momentum operators from $S = S_1 + S_2$ are:

$$\begin{aligned} S_z &= S_{1z} + S_{2z} \\ S^2 &= (S_1 + S_2)^2 = S_1^2 + S_2^2 + 2S_1 \cdot S_2 \end{aligned}$$

11.4.1 Product Basis

The **product basis** diagonalizes S_{1z} , S_{2z} , S_1^2 , and S_2^2 . Its eigenvalues can be found using the above formulas (and linearity of operators).

$$\begin{aligned} S_z |++\rangle &= \hbar |++\rangle \\ S_z |+-\rangle &= 0 |+-\rangle \\ S_z |-+\rangle &= 0 |-+\rangle \\ S_z |--\rangle &= -\hbar |--\rangle \end{aligned}$$

Note the degeneracy for $|\psi\rangle = \alpha |+-\rangle + \beta |-+\rangle$.

11.4.2 Total- s basis

The total- s basis diagonalizes S^2 , S_z , S_1^2 , and S_2^2 . Total S^2 and S_z now, not 1-component S_{1z} and S_{2z} .

We use $S^2 = S_x^2 + S_y^2 + S_z^2$ along with $S_z = (S_{1z} \otimes I) + (I \otimes S_{2z})$, etc. Substituting Pauli matrices is the easiest way to find that:

$$S^2 = \hbar^2 \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

(Yet more evidence that QM is just linear algebra...) We find 4 eigenstates, divided into 3 **triplets** with $s = 1$ and 1 **singlet** with $s = 0$.

$$\begin{aligned} \text{Triplet } \begin{cases} |s=1, m=1\rangle = |++\rangle \\ |s=1, m=0\rangle = \frac{1}{\sqrt{2}}(|+-\rangle + |-+\rangle) \\ |s=1, m=-1\rangle = |--\rangle \end{cases} \\ \text{Singlet } \begin{cases} |s=0, m=0\rangle = \frac{1}{\sqrt{2}}(|+-\rangle - |-+\rangle) \end{cases} \end{aligned}$$

We've found that $1/2 \otimes 1/2 = 1 \oplus 0$. I.e. direct product of 2 spin- $\frac{1}{2}$ Hilbert spaces is equivalent to direct sum of spin-1 (triplet) and spin-0 (singlet) spaces.

11.4.3 Construction of Fermion Wavefunctions

From the Pauli exclusion principle, fermion wavefunctions must be antisymmetric. To construct such:

1. (Symmetric orbital) + (Antisymmetric spin)

$$|\psi_{orb}\rangle \otimes \frac{|+-\rangle - |-+\rangle}{\sqrt{2}}$$

2. (Antisymmetric orbital) + (Symmetric spin)

$$|\psi_{orb}\rangle \otimes \begin{cases} |++\rangle \\ \frac{1}{\sqrt{2}}(|+-\rangle + |-+\rangle) \\ |--\rangle \end{cases}$$

E.g. for Helium (2 electrons) in the ground state, $|\psi_{orb}\rangle \equiv |nlm\rangle \otimes |nlm\rangle = |100\rangle \otimes |100\rangle$. This is invariant under exchange of particles (symmetric), so we tack on an antisymmetric spin wavefunction.

$$|\psi_{\text{ground}}\rangle = \left(|100\rangle \otimes \frac{1}{\sqrt{2}}(|+-\rangle - |-+\rangle) \right)_1 \otimes \left(|100\rangle \otimes \frac{1}{\sqrt{2}}(|+-\rangle - |-+\rangle) \right)_2$$

Whether you use the product basis or the total- s basis depends on the Hamiltonian. If S_{1z} , S_{2z} appear, probably product basis. If $S_1 \cdot S_2$ or S^2 appear, probably total- s basis.

- E.g. hyperfine splitting effect. Spin-coupling Hamiltonian adds onto Hydrogen's Coulomb interaction Hamiltonian.

$$H_{hf} = AS_1 \cdot S_2 = \frac{1}{2}A(S^2 - S_1^2 - S_2^2)$$

- In the ground state, this splits the energy levels:

$$\begin{aligned} \text{For triplets: } E_+ &= -R_y + \frac{1}{2}A \left(2\hbar^2 - \frac{3}{4}\hbar^2 - \frac{3}{4}\hbar^2 \right) = -R_y + \frac{1}{4}\hbar^2 A \\ \text{For singlets: } E_- &= -R_y + \frac{1}{2}A \left(0 - \frac{3}{4}\hbar^2 - \frac{3}{4}\hbar^2 \right) = -R_y - \frac{3}{4}\hbar^2 A \end{aligned}$$

11.5 Addition of Total Angular Momenta

For total angular momenta $J = L + S$, it's more complicated.

- Degeneracy: each m can be formed from multiple m_1, m_2 by selecting appropriate values of j_1, j_2 .
- Constraint: $j_1 - j_2 \leq j \leq j_1 + j_2$ for $j_1 \geq j_2$.

We use Clebsch-Gordan coefficients to expand our wavefunctions:

$$|jm, j_1 j_2\rangle = \sum_{m_1} \sum_{m_2} |j_1 m_1, j_2 m_2\rangle \underbrace{\langle j_1 m_1, j_2 m_2 | jm, j_1 j_2\rangle}_{\text{Clebsch-Gordan coeff.}}$$

Clebsch-Gordan coefficients can be found in a standard particle physics reference.

Part III

Further Topics

12 The WKB Method

12.1 Methodology

Consider a particle with energy E in a constant potential V . The eigenstates are

$$\psi(x) = \psi(0)e^{\pm ipx/\hbar}$$

If V varies very slowly, over small regions ψ still behaves like a plane wave. The accumulated phase shift between $x = 0$ and x is

$$\psi(x) = \psi(0)\exp\left[\pm \frac{i}{\hbar} \int_0^x p(x') dx'\right]$$

1. The WKB method: we have to solve

$$\left[\frac{d^2}{dx^2} + \frac{2m}{\hbar^2}(E - V) \right] \psi(x) = \left[\frac{d^2}{dx^2} + \frac{p^2(x)}{\hbar^2} \right] \psi(x) = 0$$

2. Let $\psi(x) = \exp[i\phi(x)/\hbar]$. Plug in to get

$$-\frac{1}{\hbar^2}\phi'(x)^2 + \frac{i}{\hbar}\phi''(x) + \frac{p^2(x)}{\hbar^2} = 0$$

3. Expand $\phi(x) = \phi_0 + \hbar\phi_1 + \hbar^2\phi_2 + \dots$ assuming $\hbar \ll 1$. Keep only the first two terms.

$$-\frac{(\phi'_0)^2 + p^2(x)}{\hbar^2} + \frac{i\phi''_0 - 2\phi'_1\phi'_0}{\hbar} + O(\hbar) = 0$$

4. By requiring that each of the terms must be 0, we get

$$\begin{aligned} \phi_0(x) &= \int_0^x p(x') dx' \\ \phi_1(x) &= i \ln \sqrt{p} + c \end{aligned}$$

The wavefunction according to the WKB approximation:

$$\phi(x) = \phi(0) \sqrt{\frac{p(0)}{p(x)}} \exp \left[\pm \frac{i}{\hbar} \int_0^x p(x') dx' \right]$$

13 Perturbation Theory

13.1 Time-Independent Perturbation Theory

If H_0 is known, we wish to study $H = H_0 + H_1$ where H_1 is small compared to H_0 (the “perturbation”).

- Let $|n^0\rangle$ be the eigenstates of H^0 with eigenvalue E_n^0 .
- Let $|n\rangle$ be the eigenstates of H with eigenvalue E_n .
- Assume a perturbation series expansion

$$\begin{aligned} |n\rangle &= |n^0\rangle + |n^1\rangle + \dots \\ E_n &= E_n^0 + E_n^1 + \dots \end{aligned}$$

Solve the eigenvalue equation $H|n\rangle = E_n|n\rangle$ and consider the 1st and 2nd order expansions. By clever dotting,

$$\begin{aligned} \text{1st order: } E_n^1 &= \langle n^0 | H^1 | n^0 \rangle \\ \text{2nd order: } E_n^2 &= \langle n^0 | H^1 | n^1 \rangle = \sum_{m \neq n} \frac{|\langle n^0 | H^1 | m^0 \rangle|^2}{E_n^0 - E_m^0} \end{aligned}$$

13.2 Ex. Harmonic Oscillator in Electric Field

Consider a charged particle as a harmonic oscillator in an external electric field of strength f . The potential is $V = -qfx$.

$$H = H^0 + H^1 = \frac{P^2}{2m} + \frac{1}{2}m\omega^2 X^2 - qfX$$

Use $X = \sqrt{\hbar/2m\omega}(a + a^\dagger)$. The 1st and 2nd order expansions are

$$E_n^1 = \langle n^0 | H^1 | n^0 \rangle = 0$$

$$E_n^2 = \sum_{m \neq n} \frac{|\langle m^0 | H^1 | n^0 \rangle|^2}{E_n^0 - E_m^0} = -\frac{q^2 f^2}{2m\omega^2}$$

To 1st order, the eigenstates change as

$$|n\rangle = |n^0\rangle + \sum_{m \neq n} \frac{|m^0\rangle \langle m^0| - qf\sqrt{\frac{\hbar}{2m\omega}}(a + a^\dagger)|n^0\rangle}{E_n^0 - E_m^0}$$

$$= |n^0\rangle + qf\sqrt{\frac{\hbar}{2m\omega}} \cdot \frac{1}{\hbar\omega} [\sqrt{n+1}|(n+1)^0\rangle - \sqrt{n}|(n-1)^0\rangle]$$

There's a bit of "mixing" between the $(n+1)$ -th and $(n-1)$ -th energy levels.

13.3 Time-Dependent Perturbation Theory

Now we must use the full Schrodinger equation. Let $H(t) = H^0 + H^1(t)$ where H^1 is a small time-dependent perturbation.

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = H |\psi\rangle = [H^0 + H^1(t)] |\psi\rangle$$

Let $|n^0\rangle$ be the eigenstate of H^0 so $|\psi(t)\rangle = \sum_n c_n(t) |n^0\rangle$ with $c_n(t) = c_n(0)e^{-iE_n t/\hbar}$ in the unperturbed Hamiltonian. Let $d_n(t)$ be the factor due to time-dependent perturbation.

$$|\psi(t)\rangle = \sum_n d_n(t) e^{-iE_n t/\hbar} |n^0\rangle$$

By clever dotting:

$$\text{To 0th order: } \dot{d}_f = 0$$

$$\text{To 1st order: } d_f(t) = \delta_{fi} - \frac{i}{\hbar} \int_0^t \langle f^0 | H^1(t') | i^0 \rangle e^{i\omega t'} dt'$$

$$\omega_{fi} = \frac{E_f^0 - E_i^0}{\hbar}$$

13.3.1 Ex. Harmonic Oscillator in Electric Field

Let $H^1(t) = -e\mathcal{E}Xe^{-t^2/\tau^2}$ be the perturbation applied between $t = -\infty$ and $t = +\infty$. The amplitude for $|0\rangle$ to go to $|1\rangle$ is $P_{0 \rightarrow 1} = |d_1(\infty)|^2$.

$$\begin{aligned} d_1(\infty) &= -\frac{i}{\hbar} \int_{-\infty}^{+\infty} -e\mathcal{E} \langle 1|X|0\rangle e^{-t^2/\tau^2} e^{i\omega_{10}t} dt, \quad X = \sqrt{\frac{\hbar}{2m\omega}}(a + a^\dagger) \\ &= \frac{ie\mathcal{E}}{\hbar} \sqrt{\frac{\hbar}{2m\omega}} \int_{-\infty}^{+\infty} e^{-t^2/\tau^2} e^{i\omega_{10}t} dt \\ &= \frac{ie\mathcal{E}}{\hbar} \sqrt{\frac{\hbar}{2m\omega}} \sqrt{\pi\tau^2} e^{-\omega^2\tau^2/4} \end{aligned}$$

Thus the amplitude is

$$P_{0 \rightarrow 1} = \frac{e^2 \mathcal{E}^2 \pi \tau^2}{2m\omega\hbar} e^{-\omega^2\tau^2/2}$$

13.3.2 The Adiabatic Theorem

Adiabatic Theorem: Let the system start in eigenstate $|n^0\rangle$ of H^0 . If the system changes “slowly enough,” it ends up in the eigenstate $|n^1\rangle$ of H^1 .

E.g. Hamiltonian changing from $H^0 \rightarrow H^0 + H^1$ between $t = -\infty$ and $t = 0$:

$$H(t) = H^0 + e^{t/\tau} H^1, \quad -\infty \leq t \leq 0$$

As $\tau \rightarrow \infty$, the change happens slowly enough such that an eigenstate $|n^0\rangle$ of H^0 at $t = -\infty$ transitions to an eigenstate $|n\rangle$ of H at $t = 0$.

Then as $\tau \rightarrow \infty$, the time-dependent calculation at $t = 0$ and the time-independent calculation at $t = -\infty$ must agree. And they do.

$$\begin{aligned} \text{Time-dependent: } d_m(0) &= -\frac{i}{\hbar} \int_{-\infty}^0 \langle m^0|H^1|n^0\rangle e^{t/\tau} e^{i\omega_{mn}t} dt \\ &= -\frac{i \langle m^0|H^1|n^0\rangle}{\hbar (1/\tau + i\omega_{mn})} \\ &\rightarrow \frac{\langle m^0|H^1|n^0\rangle}{E_n^0 - E_m^0} \text{ as } \tau \rightarrow \infty \end{aligned}$$

13.3.3 Fermi's Golden Rule

We consider a periodic perturbation. Model it as an exponential (which sin, cos can be constructed from):

$$H^1(t) = H^1 e^{-i\omega t}$$

The transition amplitude is:

$$\begin{aligned} d_f(t) &= -\frac{i}{\hbar} \int_0^t \langle f^0|H^1|i^0\rangle e^{i(\omega_{fi}-\omega)t'} dt' \\ &= -\frac{i}{\hbar} \langle f^0|H^1|i^0\rangle \frac{e^{i(\omega_{fi}-\omega)t} - 1}{i(\omega_{fi} - \omega)} \end{aligned}$$

The transition probability:

$$P_{i \rightarrow f} = |d_f(t)|^2 = \frac{1}{\hbar^2} |\langle f^0 | H^1 | i^0 \rangle|^2 \left(\frac{\sin\left(\frac{\omega_{fi} - \omega}{2} t\right)}{\frac{\omega_{fi} - \omega}{2}} \right)^2 t^2$$

The sinc function is peaked at the origin with width $\sim \pi$, so the system tends to transition to states with $(\omega_{fi} - \omega)t/2 \leq \pi$. I.e.

$$E_f^0 - E_i^0 = \hbar\omega \left(1 \pm \frac{2\pi}{\omega t} \right)$$

- As ω increases, the system prefers $E_f^0 = E_i^0 + \hbar\omega$.
- Or at long t exposure.

To model long t , let the perturbation be from $-T/2$ to $+T/2$ with $T \rightarrow \infty$. The Transition amplitude:

$$\begin{aligned} d_f &= \lim_{T \rightarrow \infty} -\frac{i}{\hbar} \int_{-T/2}^{+T/2} H_{fi}^1 e^{i(\omega_{fi} - \omega)t'} dt' \\ &= -\frac{2\pi i}{\hbar} H_{fi}^1 \delta(\omega_{fi} - \omega) \end{aligned}$$

The transition probability:

$$\begin{aligned} P_{i \rightarrow f} &= |d_f|^2 = \frac{4\pi^2}{\hbar^2} |H_{fi}^1|^2 \delta(\omega_{fi} - \omega) \delta(\omega_{fi} - \omega) \\ &= \frac{4\pi^2}{\hbar^2} |H_{fi}^1|^2 \lim_{T \rightarrow \infty} \delta(\omega_{fi} - \omega) \frac{1}{2\pi} \int_{-T/2}^{+T/2} e^{i(\omega_{fi} - \omega)t} dt \\ &= \frac{2\pi}{\hbar^2} |H_{fi}^1|^2 \delta(\omega_{fi} - \omega) \lim_{T \rightarrow \infty} T \end{aligned}$$

The *average* transition rate $P_{i \rightarrow f}/T$ throughout time T is

$$R_{i \rightarrow f} = \frac{P_{i \rightarrow f}}{T} = \frac{2\pi}{\hbar} |\langle f^0 | H^1 | i^0 \rangle|^2 \delta(E_f^0 - E_i^0 - \hbar\omega)$$

This is called **Fermi's Golden Rule** and it's used heavily in particle physics.

13.4 Electrodynamics

(A quick review) Maxwell's equations in Gaussian units with the continuity equation is

$$\begin{cases} \nabla \cdot E = 4\pi\rho \\ \nabla \cdot B = 0 \\ \nabla \times E = -\frac{1}{c} \frac{\partial B}{\partial t} \\ \nabla \times B = \frac{4\pi}{c} J + \frac{1}{c} \frac{\partial E}{\partial t} \end{cases}, \quad \nabla \cdot J + \frac{\partial \rho}{\partial t} = 0$$

- The identity $\nabla \cdot \nabla \times A = 0$ tells us that $B = \nabla \times A$.

- The identity $\nabla \times \nabla \times \phi = 0$ tells us that $E = -\frac{1}{c} \frac{\partial A}{\partial t} - \nabla \phi$.
- Note that Maxwell's equations are invariant under the **gauge transformation**

$$A \rightarrow A - \nabla \Lambda$$

$$\phi \rightarrow \phi + \frac{1}{c} \frac{\partial A}{\partial t}$$

In free space ($\rho = 0, J = 0$), we often set the **Coulomb gauge**

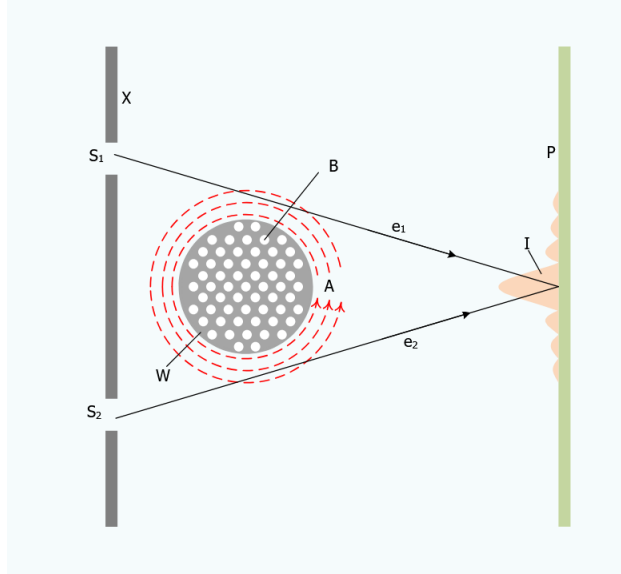
$$\nabla \cdot A = 0, \quad \phi = 0$$

13.4.1 The Aharonov-Bohm Effect

Quantum mechanics, like classical electrodynamics, is invariant under gauge transformations (gauge transforming an electrodynamic Hamiltonian shows this). Under a gauge transformation, $\psi \rightarrow e^{-iq\Lambda/\hbar c} \psi$, which doesn't change the physics.

However, the potentials A, ϕ are more “physical” than before. According to the path-integral formalism, even if a particle's classical path doesn't encounter a potential, it might be affected quantum mechanically after taking all possible paths.

The most striking example is the **Aharonov-Bohm effect**.



Setup for the Aharonov-Bohm effect. A double-slit experiment with a solenoid.

According to the path-integral approach, the total amplitude is the sum of the two paths: $\psi \cong \psi_{e_1} + \psi_{e_2}$. The classical action is

$$U \sim \sum_{\text{paths}} e^{iS/\hbar}$$

$$S = \int \mathcal{L} dt = \int_{t_i}^{t_f} \left(\frac{1}{2} m v^2 + \frac{q}{c} v \cdot A - q\phi \right) dt$$

1. Turn off the solenoid. $A = 0$ and $\phi = 0$.
2. Now turn it on (Coulomb gauge). The action changes as

$$\begin{aligned} S &\rightarrow S + \int_{t_i}^{t_f} \frac{q}{c} \mathbf{v} \cdot \mathbf{A} dt \\ &= S + \frac{q}{c} \int_{t_i}^{t_f} \mathbf{A} \cdot d\mathbf{r} \end{aligned}$$

Thus each path picks up a factor of $\exp\left(\frac{iq}{\hbar c} \int_{t_i}^{t_f} \mathbf{A} \cdot d\mathbf{r}\right)$.

3. Since the two paths (extend to common source) create a closed loop, by Stokes' theorem we should get

$$\begin{aligned} \int_{e_2} \mathbf{A} \cdot d\mathbf{r} - \int_{e_1} \mathbf{A} \cdot d\mathbf{r} &= \oint_{e_1+e_2} \mathbf{A} \cdot d\mathbf{r} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} \\ &= \int_S \mathbf{B} \cdot d\mathbf{s} = \Phi \neq 0 \end{aligned}$$

Since there are a nonzero number of magnetic flux lines now.

4. The total amplitude is thus

$$\begin{aligned} \psi &\cong \exp\left[\frac{iq}{\hbar c} \int_{e_1} \mathbf{A} \cdot d\mathbf{r}\right] \psi_{e_1} + \exp\left[\frac{iq}{\hbar c} \int_{e_2} \mathbf{A} \cdot d\mathbf{r}\right] \psi_{e_2} \\ &= \psi_{e_1} + \exp\left[\frac{iq}{\hbar c} \oint_{e_1+e_2} \mathbf{A} \cdot d\mathbf{r}\right] \psi_{e_2} \\ &= \psi_{e_1} + \exp\left[\frac{iq}{\hbar c} \Phi\right] \psi_{e_2} \end{aligned}$$

By varying the strength of B , we can move the pattern up or down. But the integer multiple of the **flux quantum** $\Phi_0 = 2\pi\hbar c/q$ doesn't change things.

Note how odd this is. Classically, the particle shouldn't even "see" the potential, but quantum mechanically, it's affected by it in a directly observable way. The potential A becomes an entity with physical influence.

13.4.2 The Photoelectric Effect in Hydrogen

Assume the ground state $|100\rangle$. Consider the EM potential to be a wave $A(\mathbf{r}) = \mathbf{A}_0 e^{i\mathbf{k}\cdot\mathbf{r}}$. Shining light onto hydrogen bombards it with photons of energy $\hbar\omega$. For large enough ω , electrons are ejected.

Let the interaction Hamiltonian be a coupling:

$$H^1(t) = \frac{e}{2mc} e^{i\mathbf{k}\cdot\mathbf{r}} \mathbf{A}_0 \cdot \mathbf{P} e^{-i\omega t}$$

From Fermi's Golden Rule,

$$\begin{aligned} R_{i\rightarrow f} &= \frac{2\pi}{\hbar} |\langle f^0 | H^1 | i^0 \rangle|^2 \delta(E_f^0 - E_i^0 - \hbar\omega) \\ H_{fi}^1 &\equiv \langle f^0 | H^1 | i^0 \rangle = \frac{e}{2mc} \left(\frac{1}{2\pi\hbar}\right)^{3/2} \left(\frac{1}{\pi a_0^3}\right)^{1/2} \int e^{-i\mathbf{P}\mathbf{r}\cdot\mathbf{r}/\hbar} e^{i\mathbf{k}\cdot\mathbf{r}} \mathbf{A}_0 \cdot (-i\hbar\nabla) e^{-r/a_0} d^3\mathbf{r} \end{aligned}$$

We assume the wave is such that it changes little over small distances – like the size of an atom. Then beyond $r > a_0$, the integral diminishes due to e^{-r/a_0} . So we can assume $r \cong a_0$ and set $e^{i\mathbf{k}\cdot\mathbf{r}} \approx 1$. This is the **electric dipole approximation**.

The integral becomes

$$\begin{aligned} \int e^{-i\mathbf{p}_f\cdot\mathbf{r}/\hbar} e^{i\mathbf{k}\cdot\mathbf{r}} \mathbf{A}_0 \cdot (-i\hbar\nabla) e^{-r/a_0} d^3\mathbf{r} &\approx -i\hbar \int e^{-i\mathbf{p}_f\cdot\mathbf{r}/\hbar} \mathbf{A}_0 \cdot \nabla e^{-r/a_0} d^3\mathbf{r} \\ &= \mathbf{A}_0 \cdot \mathbf{p}_f \int e^{-i\mathbf{p}_f\cdot\mathbf{r}/\hbar} e^{-r/a_0} d^3\mathbf{r} \end{aligned}$$

If we consider the $\mathbf{p}_f$ -direction and use spherical coordinates, eventually

$$H_{fi}^1 \approx \frac{2\pi i\hbar}{p_f} \int_0^\infty \left[e^{-(1/a_0 + ip_f/\hbar)r} - e^{-(1/a_0 - ip_f/\hbar)r} \right] r dr \longrightarrow \frac{8\pi/a_0}{[(1/a_0)^2 + (p_f/\hbar)^2]^2}$$

Then according to Fermi's Golden Rule, the transition rate for the ejection of electrons (photoelectric effect) is

$$R_{i \rightarrow f} = \frac{2\pi}{\hbar} \left(\frac{e}{2mc} \right)^2 \frac{1}{8\pi^3 \hbar^3} \frac{1}{\pi a_0^3} \frac{|\mathbf{A}_0 \cdot \mathbf{p}_f|^2 64\pi^2 a_0^6}{[1 + (p_f a_0/\hbar)^2]^4} \delta(E_f^0 - E_i^0 - \hbar\omega)$$

- In practice, we measure over a detector with angle range $d\Omega$ to see the # of ejected particles with final state between p_f and $p_f + \delta p_f$.
- Integrate over all angles to get the total **ionization rate**.
- Each ionization takes $\hbar\omega$ energy. Thus the energy absorption rate is

$$\frac{dE_{\text{abs}}}{dt} = \hbar\omega \cdot \sum_f R_{i \rightarrow f}$$

14 Field Quantization

14.1 The Need for Quantized Fields

For free space ($A = \dot{A} = 0$), the classical prediction is that no decay happens. Experimentally, this isn't the case. Just as $\langle x \rangle = \langle p \rangle = 0$ in free space but $\Delta x \Delta p \geq \hbar/2$, there are quantum fluctuations of fields in free space even if $\langle A \rangle = \langle \dot{A} \rangle = 0$.

Thus we need a full quantum theory that recasts $\mathbf{A}$ as an operator. This is **field quantization** and underlies modern quantum field theory.

1. Find a complete set of canonical coordinates & momenta to describe the classical field.
2. Promote them to operators obeying the canonical commutation relation.

14.2 Sketch of Procedure

Consider the free-space EM Lagrangian (in cgs units).

$$\begin{aligned} \mathcal{L} &= -\frac{1}{16\pi} F_{\mu\nu} F^{\mu\nu} \longrightarrow \frac{1}{8\pi} \int (|E|^2 - |B|^2) d^3\mathbf{r} \\ &= \frac{1}{8\pi} \int \left| -\frac{1}{c} \frac{\partial \mathbf{A}}{\partial t} - \nabla \phi \right|^2 - |\nabla \times \mathbf{A}|^2 d^3\mathbf{r} \end{aligned}$$

- Enact the Coulomb gauge: $\nabla \cdot \mathbf{A} = 0$ and $\phi = 0$.
- The conjugate momentum to $A_i(\mathbf{r})$ is

$$\Pi_i(\mathbf{r}) = \frac{\partial \mathcal{L}}{\partial \dot{A}_i(\mathbf{r})} = \frac{1}{4\pi c^2} \dot{A}_i(\mathbf{r}) = -\frac{\mathbf{E}}{4\pi c}$$

Maxwell's equations fall neatly from varying the Lagrangian (see Shankar 18.5 for details). Except $\nabla \cdot \mathbf{E} = 0$, which is taken as an extra condition (“**transverse gauge**”). Thus our constraints are

$$\nabla \cdot \mathbf{A} = 0, \quad \phi = 0, \quad \nabla \cdot \Pi = 0$$

To **quantize** $A(\mathbf{r})$ and $\Pi(\mathbf{r})$, we express them in the Fock space with coefficients $\mathbf{a}(\mathbf{k})$ and $\mathbf{a}^*(\mathbf{k})$ representing the annihilation/creation operators for particles corresponding to that field. Here it's a photon. Constant factors arbitrary.

$$\begin{aligned} A(\mathbf{r}) &= \int \left[\mathbf{a}(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}} + \mathbf{a}^*(\mathbf{k}) e^{-i\mathbf{k} \cdot \mathbf{r}} \right] d^3\mathbf{k} \\ \Pi(\mathbf{r}) &= \frac{1}{4\pi i c} \int k \left[\mathbf{a}(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}} - \mathbf{a}^*(\mathbf{k}) e^{-i\mathbf{k} \cdot \mathbf{r}} \right] d^3\mathbf{k} \end{aligned}$$

The constraints $\nabla \cdot \mathbf{A} = 0$ and $\nabla \cdot \Pi = 0$ lead us to conclude that (see Shankar 18.5 for details)

$$\begin{aligned} \mathbf{k} \cdot [\mathbf{a}(\mathbf{k}) + \mathbf{a}^*(-\mathbf{k})] &= 0 \\ \mathbf{k} \cdot [\mathbf{a}(\mathbf{k}) - \mathbf{a}^*(-\mathbf{k})] &= 0 \\ \therefore \mathbf{k} \cdot \mathbf{a}(\mathbf{k}) &= 0 \end{aligned}$$

$\mathbf{a}(\mathbf{k})$ along $\mathbf{k}$ is 0. Call the other two orthogonal vectors $\mathbf{a}(\mathbf{k}_1)$ and $\mathbf{a}(\mathbf{k}_2)$. Then

$$H = \sum_{\lambda=1}^2 \int \omega [\mathbf{a}^*(\mathbf{k}_\lambda) \mathbf{a}(\mathbf{k}_\lambda)] d^3\mathbf{k}$$

Now if we define the operators

$$\begin{aligned} \mathbf{q}(\mathbf{k}_\lambda) &= \frac{1}{\sqrt{2\omega}} [\mathbf{a}(\mathbf{k}_\lambda) + \mathbf{a}^*(\mathbf{k}_\lambda)] \\ \mathbf{p}(\mathbf{k}_\lambda) &= \frac{1}{i} \sqrt{\frac{\omega}{2}} [\mathbf{a}(\mathbf{k}_\lambda) - \mathbf{a}^*(\mathbf{k}_\lambda)] \end{aligned}$$

Then the Hamiltonian becomes

$$H = \sum_{\lambda=1}^2 \int \left[\frac{1}{2} \mathbf{p}^2(\mathbf{k}_\lambda) + \frac{\omega^2}{2} \mathbf{q}^2(\mathbf{k}_\lambda) \right] d^3\mathbf{k}$$

Which is just an oscillator Hamiltonian. So by “quantizing” the field, we mean reducing it to several coupled oscillators at each $\mathbf{k}$. We already know that $\mathbf{q}$, $\mathbf{p}$ are canonical variables (from SHO case) and can promote these to full operators. Hence the importance of the SHO in quantum mechanics.

Now $\mathbf{A}$ and Π can be quantum operators defined as

$$\begin{aligned}\mathbf{A} &= \sum_{\lambda} \int \sqrt{\frac{\hbar c^2}{4\pi^2 \omega}} \left[\mathbf{a}(\mathbf{k}_{\lambda}) \epsilon(\mathbf{k}_{\lambda}) e^{i\mathbf{k} \cdot \mathbf{r}} + \mathbf{a}^{\dagger}(\mathbf{k}_{\lambda}) \epsilon(\mathbf{k}_{\lambda}) e^{-i\mathbf{k} \cdot \mathbf{r}} \right] d^3\mathbf{k} \\ \Pi &= \sum_{\lambda} \int \frac{1}{i} \sqrt{\frac{\hbar \omega}{64\pi^4 c^2}} \left[\mathbf{a}(\mathbf{k}_{\lambda}) \epsilon(\mathbf{k}_{\lambda}) e^{i\mathbf{k} \cdot \mathbf{r}} - \mathbf{a}^{\dagger}(\mathbf{k}_{\lambda}) \epsilon(\mathbf{k}_{\lambda}) e^{-i\mathbf{k} \cdot \mathbf{r}} \right] d^3\mathbf{k}\end{aligned}$$

This allows for certain calculations, e.g. spontaneous emission where particle number is not conserved. Classical physics wouldn't allow such processes. For further details, see a QFT textbook.

15 Scattering Theory

15.1 Procedure

One of the best ways to learn about things is to smash them together. Method unchanged since caveman times.

1. We model particles as wave packets. The total eigenfunction of the Hamiltonian H is

$$\psi_{\mathbf{k}} = \underbrace{\psi_{\text{incoming}}}_{e^{i\mathbf{k} \cdot \mathbf{r}} \text{ plane wave}} + \psi_{\text{scattered}}$$

2. We seek the explicit form of above. A scattered wave far from the origin should be almost like a free particle since $V \rightarrow 0$ physically. Recall the free particle equation in spherical coordinates (long time ago):

$$\psi_{Elm}(r, \theta, \phi) = \begin{cases} j_l(kr) Y_l^m(\theta, \phi) & \text{Neumann function} \\ n_l(kr) Y_l^m(\theta, \phi) & \text{Bessel function} \end{cases}$$

Previously, the Neumann function was excluded because it was singular at the origin. Now we don't have that issue since $r \rightarrow \infty$. Thus (as $r \rightarrow \infty$),

$$\begin{aligned}\psi_{\text{scattered}} &\longrightarrow \sum_l \sum_m [A_l j_l(kr) + B_l n_l(kr)] Y_l^m(\theta, \phi) \\ j_l(kr) &\longrightarrow \frac{\sin(kr - l\pi/2)}{kr} \\ n_l(kr) &\longrightarrow -\frac{\cos(kr - l\pi/2)}{kr}\end{aligned}$$

3. To get the purely outgoing wave as $r \rightarrow \infty$, we set the scattered wave to 0. I.e. $A_l/B_l = -i$ as

$r \rightarrow \infty$.

$$\begin{aligned}
 \psi_{\text{scattered}} &\longrightarrow \sum_l \sum_m [-iB_l j(kr) + B_l n(kr)] Y_l^m(\theta, \phi) \\
 &= \frac{1}{kr} \sum_l \sum_m [-i \sin(kr - l\pi/2) - \cos(kr - l\pi/2)] B_l Y_l^m \\
 &= \frac{e^{ikr}}{kr} \sum_l \sum_m (-i)^l (-B_l) Y_l^m(\theta, \phi) \\
 &\equiv \frac{e^{ikr}}{r} f(k, \theta, \phi) \quad \text{as } r \rightarrow \infty
 \end{aligned}$$

f is called the **scattering amplitude**.

4. Thus the total wave state (as $r \rightarrow \infty$) is

$$\psi_k \longrightarrow e^{ikz} + f(k, \theta, \phi) \frac{e^{ikr}}{r}$$

Our objective in scattering is to find the **differential cross section** $d\sigma/d\Omega$, which tells us about the scattered area ($d\sigma$) over the scattered angle ($d\Omega$). This can be used to reconstruct the composition of the object, etc.

For the above case, this turns out to be simple:

$$\frac{d\sigma}{d\Omega} = |f(k, \theta, \phi)|^2$$

15.2 Alternative: Green's Function

We could try to find the form of $\psi_{\mathbf{k}}$ in an alternate way using Green's function and contour integration. Start from the Schrodinger equation:

$$\begin{aligned}
 -\frac{\hbar^2}{2\mu} \nabla^2 \psi + V\psi &= E\psi \\
 (\nabla^2 + k^2) \psi_{\mathbf{k}} &= \frac{2\mu}{\hbar^2} V\psi_{\mathbf{k}}
 \end{aligned}$$

We seek to find a state $\psi_{\mathbf{k}} = e^{i\mathbf{k}\cdot\mathbf{r}} + \psi_{\text{scattered}}$ such that

$$\psi_{\text{scattered}} \longrightarrow f(\theta, \phi) \frac{e^{ikr}}{r} \quad \text{as } r \rightarrow \infty$$

Green's function $G^0(\mathbf{r}, \mathbf{r}')$ is defined such that

$$(\nabla^2 + k^2)G^0(\mathbf{r}, \mathbf{r}') = \delta^3(\mathbf{r} - \mathbf{r}')$$

I.e. G^0 is the state that responds to a delta-function potential at $(\mathbf{r} - \mathbf{r}')$. Thus the full solution to Schrodinger's equation is

$$\begin{aligned}
 \psi_{\mathbf{k}} &= \psi^0(\mathbf{r}) + \frac{2\mu}{\hbar^2} \int G^0(\mathbf{r}, \mathbf{r}') V(\mathbf{r}') \psi_{\mathbf{k}}(\mathbf{r}') d^3\mathbf{r}' \\
 (\nabla^2 + k^2)\psi^0(\mathbf{r}) &= 0 \quad \text{(Free solution)}
 \end{aligned}$$

First note that since the LHS of the definition of Green's function uses $\delta^3(\mathbf{r} - \mathbf{r}')$, we can simplify G^0 to a function of 1 parameter only: the difference $\mathbf{r} - \mathbf{r}'$.

To turn the differential equation for $G^0(\mathbf{r})$ into an algebraic one, we Fourier transform to go to the momentum space.

$$\begin{aligned} (\nabla^2 + k^2)G^0(\mathbf{r}) = \delta^3(\mathbf{r}) &\longrightarrow \frac{1}{(\sqrt{2\pi})^3} \int e^{-i\mathbf{q}\cdot\mathbf{r}} (\nabla^2 + k^2)G^0(\mathbf{r}) d^3\mathbf{r} = \frac{1}{(\sqrt{2\pi})^3} \\ (k^2 - q^2) \int e^{-i\mathbf{q}\cdot\mathbf{r}} G^0(\mathbf{r}) d^3\mathbf{r} &= 1 \\ (k^2 - q^2)G^0(\mathbf{q}) &= \frac{1}{(\sqrt{2\pi})^3} \end{aligned}$$

This diverges at $\mathbf{k} = \mathbf{q}$. We introduce a fudge factor $i\varepsilon$ and use $k^2 + i\varepsilon$ instead of k^2 . At the end, we set $\varepsilon \rightarrow 0$.

$$\begin{aligned} \text{Momentum space: } G^0(\mathbf{q}) &= \frac{1}{(\sqrt{2\pi})^3} \frac{1}{k^2 + i\varepsilon - q^2} \\ \text{Coordinate space: } G_\varepsilon^0(\mathbf{r}) &= \frac{1}{(2\pi)^3} \int \frac{e^{i\mathbf{q}\cdot\mathbf{r}}}{k^2 + i\varepsilon - q^2} d\mathbf{q} \end{aligned}$$

By simplifying, we get (in the momentum space)

$$G_\varepsilon^0(\mathbf{r}) = \int_{-\infty}^{+\infty} \frac{1}{4\pi^2 i r} \frac{e^{iqr} q}{k^2 + i\varepsilon - q^2} dq$$

Now we use a neat trick from Complex Analysis: a half-circle contour in the complex plane, with path integral of the curved portion C_ρ going to 0 as $\rho \rightarrow \infty$.

The function (inside the integrand) has simple poles at $q \approx \pm(k + i\varepsilon/2k)$ to 1st order in ε . Let $\eta = \varepsilon/2k$. The residue at $q \approx k + i\eta$ is $-e^{i(k+i\eta)r}/8\pi^2 i r$ by residue formulas.

$$\begin{aligned} \oint \frac{1}{4\pi^2 i r} \frac{e^{iqr} q}{k^2 + i\varepsilon - q^2} dq &= \text{Re} \left[\int_{-\infty}^{+\infty} \frac{1}{4\pi^2 i r} \frac{e^{iqr} q}{k^2 + i\varepsilon - q^2} dq \right] + \underbrace{\int_{C_\rho} \frac{1}{4\pi^2 i r} \frac{e^{iqr} q}{k^2 + i\varepsilon - q^2} dq}_{=0 \text{ as } \rho \rightarrow \infty, \text{ substitute } q=re^{i\theta}} = 2\pi i \sum \text{Residues} \\ G_\varepsilon^0(\mathbf{r}) &= 2\pi i \left(-\frac{1}{8\pi^2 i r} e^{i(k+i\eta)r} \right) = -\frac{e^{i(k+i\eta)r}}{4\pi r} \\ G^0(\mathbf{r}) &= \lim_{\varepsilon \rightarrow 0} G_\varepsilon^0(\mathbf{r}) = -\frac{e^{ikr}}{4\pi r} \end{aligned}$$

Finally, we have that our state is approximately

$$\psi_{\mathbf{k}} = e^{i\mathbf{k}\cdot\mathbf{r}} - \frac{e^{ikr}}{4\pi r} \frac{2\mu}{\hbar^2} \int e^{-i\mathbf{k}_f\cdot\mathbf{r}'} V(\mathbf{r}') \psi_{\mathbf{k}}(\mathbf{r}') d^3\mathbf{r}'$$

Which has the same form as found in the previous section if $r \rightarrow \infty$. We've made the following approximation as $r \rightarrow \infty$ (pp. 538 – 539):

$$\frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} \approx \frac{e^{ikr}}{r} e^{-i\mathbf{k}_f \cdot \mathbf{r}'}$$

15.3 Born Approximation

15.3.1 Perturbation Theory

A wave packet broad in x can be modeled as a plane wave. Such a wave has large Δx but a “spiked” Δp . The propagator for scattering is $U(t_f, t_i)$.

Define the **S-matrix** (scattering matrix) to be

$$S = \lim_{\substack{t_f \rightarrow \infty \\ t_i \rightarrow -\infty}} U(t_f, t_i)$$

For a detector with opening angle $d\Omega$, the probability to be scattered is

$$P(\mathbf{p}_i \rightarrow d\Omega) = \sum_{\mathbf{p}_f \text{ in } d\Omega} |\langle \mathbf{p}_f | S | \mathbf{p}_i \rangle|^2$$

If the potential is small enough compared to the kinetic energy to be considered a perturbation, then by using Fermi's Golden Rule, the transition rate is

$$\begin{aligned} R_{i \rightarrow d\Omega} &= \frac{dP(\mathbf{p}_i \rightarrow d\Omega)}{dt} \\ &= \frac{2\pi}{\hbar} \left[\int_0^\infty |\langle \mathbf{p}_f | V | \mathbf{p}_i \rangle|^2 \delta \left(\frac{p_f^2}{2\mu} - \frac{p_i^2}{2\mu} \right) p_f^2 dp_f \right] d\Omega \end{aligned}$$

The differential cross-section turns out to be

$$\frac{d\sigma}{d\Omega} = \left| \frac{\mu}{2\pi\hbar^2} \int e^{-i(\mathbf{p}_f - \mathbf{p}_i) \cdot \mathbf{r}/\hbar} V(\mathbf{r}) d^3\mathbf{r} \right|^2$$

From the previous section's result of $d\sigma/d\Omega = |f(\theta, \phi)|^2$, we know the form of $f(\theta, \phi)$ after setting the arbitrary phase to -1 . This is the **Born approximation**: plane wave, potential as perturbation.

$$f(\theta, \phi) = -\frac{\mu}{2\pi\hbar^3} \int e^{-i(\mathbf{p}_f - \mathbf{p}_i) \cdot \mathbf{r}/\hbar} V(\mathbf{r}) d^3\mathbf{r}$$

15.3.2 Yukawa & Coulomb Potentials

For a spherically symmetric potential, the above scattering amplitude can be simplified. Let $\mathbf{q} = (\mathbf{p}_f - \mathbf{p}_i)/\hbar$.

$$\begin{aligned} f(\theta, \phi) &= -\frac{\mu}{2\pi\hbar^3} \int e^{-iqr \cos \theta} V(r) d(\cos \theta) d\phi r^2 dr \\ &= -\frac{\mu}{\hbar^2} \int_0^\infty \frac{e^{-iqr \cos \theta}}{-iqr} \Big|_{-1}^{+1} r^2 dr \\ &= -\frac{\mu}{\hbar^2} \int_0^\infty \frac{\sin(qr)}{q} V(r) r dr \end{aligned}$$

For the **Yukawa potential** $V(r) = ge^{-\mu_0 r}/r$, the scattering amplitude is

$$\begin{aligned} f(\theta) &= -\frac{2\mu g}{\hbar^2 q} \int_0^\infty \frac{e^{iqr} - e^{-iqr}}{2i} e^{-\mu_0 r} dr \\ &= -\frac{2\mu g}{\hbar^2(\mu_0^2 + q^2)} \quad (\text{after algebra}) \end{aligned}$$

The Yukawa cross-section is therefore

$$\begin{aligned} \frac{d\sigma}{d\Omega} &= \frac{4\mu^2 g^2}{\hbar^4 [\mu_0^2 + 4k^2 \sin^2(\theta/2)]} \\ \mathbf{q} &= (\mathbf{p}_f - \mathbf{p}_i)/\hbar = \mathbf{k}_f - \mathbf{k}_i \\ |\mathbf{k}_f - \mathbf{k}_i|^2 &= 2k^2 - 2k^2 \cos \theta = 4k^2 \sin^2(\theta/2) \end{aligned}$$

For the **Coulomb potential**, set $g = Ze^2$ and $\mu_0 = 0$. The cross section (“Rutherford cross section”) is

$$\frac{d\sigma}{d\Omega} = \frac{Z^2 e^4}{16E^2 \sin^4(\theta/2)}, \quad E = p^2/2\mu$$

16 The Dirac Equation

16.1 Derivation

We seek to reconcile special relativity and quantum mechanics. The Dirac equation will be the starting point for QFT.

Start from the Schrodinger equation for free particles:

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = H |\psi\rangle = \frac{P^2}{2m} |\psi\rangle$$

16.1.1 Klein-Gordon Equation

To incorporate SR, we can try using $E^2 = (pc)^2 + (mc^2)^2$ as the eigenvalue of H . Observe that in the coordinate basis,

$$\begin{aligned} -\hbar^2 \frac{\partial^2 \psi}{\partial t^2} &= [(Pc)^2 + (mc^2)^2] \psi = [-c^2 \hbar^2 \nabla^2 + m^2 c^4] \psi \\ \left[\frac{1}{c} \frac{\partial^2}{\partial t^2} - \nabla^2 \right] \psi &= - \left(\frac{mc}{\hbar} \right)^2 \psi \end{aligned}$$

This is the Klein-Gordon equation. ψ is a scalar-valued function, thus suitable for zero-spin particles.

A possible wavefunction solution is $\psi(x) = e^{i(\mathbf{p} \cdot \mathbf{x} + Et)/\hbar}$. This results in a particle with energy $E^2 = (pc)^2 + (mc^2)^2$. Now the energy of a particle is $E = \hbar\omega = \hbar \frac{\partial \psi}{\partial t}$. Yet this particle results in a negative energy $-E$.

In real life, particles tend to the state with lowest energy and emit photons while at it. If the Klein-Gordon equation were correct, we’d see a ton of particles fall to an infinitely negative energy state while emitting infinitely many photons. But we don’t. We conclude that this equation is incorrect. (In QFT, it will reappear in a valid operator form.)

16.1.2 Dirac Equation

Dirac sought to fix the problems of the Klein-Gordon equation by linearizing the $(pc)^2 + (mc^2)^2$ term. He introduced matrices $\mathbf{a}$, β such that $E^2 = (cp)^2 + (mc^2)^2 = (\mathbf{ca} \cdot \mathbf{P} + \beta mc^2)^2$.

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = [\mathbf{ca} \cdot \mathbf{P} + \beta mc^2] |\psi\rangle$$

$$i\hbar \frac{\partial}{\partial t} \psi = [\mathbf{ca} \cdot (-i\hbar \nabla) + \beta mc^2] \psi$$

From now on, we use the 4-dimensional matrices:

$$\mathbf{a} = \begin{pmatrix} 0 & \vec{\sigma} \\ \vec{\sigma} & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$$

The Dirac equation is:

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = [\mathbf{ca} \cdot \mathbf{P} + \beta mc^2] |\psi\rangle$$

Note that $|\psi\rangle$ is a 4-dimensional vector “spinor.” This makes it suitable to describe vector-valued wavefunctions, e.g. particles with spin.

16.2 Positive/Negative Energy Solutions

Though Dirac sought to remove the negative-energy solutions, the Dirac equation still has such. For example, in the free-particle Dirac equation, we can attempt $\psi = \psi_0 e^{i(\mathbf{p} \cdot \mathbf{r} - Et)/\hbar}$.

$$i\hbar \frac{\partial \psi}{\partial t} = [\mathbf{ca} \cdot \mathbf{P} + \beta mc^2] \psi$$

Plugging in with $\psi_0 = \begin{pmatrix} \chi & \Phi \end{pmatrix}^T$ gives:

$$E\psi_0 = [\mathbf{ca} \cdot \mathbf{p} + \beta mc^2] \psi_0$$

$$E\chi = mc^2\chi + c(\vec{\sigma} \cdot \vec{p})\Phi \longrightarrow \chi = \frac{c\vec{\sigma} \cdot \vec{p}}{E - mc^2}\Phi$$

$$E\Phi = c(\vec{\sigma} \cdot \vec{p})\chi - mc^2\Phi \longrightarrow \Phi = \frac{c\vec{\sigma} \cdot \vec{p}}{E + mc^2}\chi$$

By setting $\vec{p} = 0$, we see that Φ corresponds to a negative-energy solution with $E = -mc^2$. A similar result holds for $\vec{p} \neq 0$.

Dirac sought to explain this result by proposing a “sea” of negative-energy particles. Fermions, so that positive-energy particles couldn’t lose energy to become a particle in the negative-energy sea. Then a “hole” in the sea would manifest as a positive-energy particle with characteristics of a negative-energy particle – an antiparticle.

Feynman gave an alternative interpretation that may be simpler to remember: negative-energy particles only travel *back* in time. I.e. negative-energy particles are positive-energy antiparticles traveling back in time. This is the way Feynman diagrams are written, for example.

In both cases, we have a need for a quantum theory that explains the annihilation/creation of particles. This is **Quantum Field Theory (QFT)**.

17 Path Integrals: Part II

A survey of some interesting results using (but not limited to) path integrals.

17.1 Different Path Integrals

The following resolutions of identity are useful:

$$I = \int_{-\infty}^{+\infty} |x\rangle \langle x| dx$$

$$I = \frac{1}{2\pi\hbar} \int_{-\infty}^{+\infty} |p\rangle \langle p| dx \quad \text{with} \quad \langle x|p\rangle = e^{ipx/\hbar}$$

We use the following Hamiltonian and propagator:

$$H = \frac{P^2}{2m} + V(X)$$

$$U(x, t; x_0, t) \equiv U(x, x_0, t) = \langle x | e^{-iHt/\hbar} | x_0 \rangle$$

Define $\varepsilon = t/N$ as the infinitesimal time with $N \rightarrow \infty$. Then using the identity $e^A e^B = e^{A+B+1/2[A,B]+\dots}$ and $\varepsilon \rightarrow 0$, we have

$$\exp[-iHt/\hbar] = \exp\left[-\frac{i\varepsilon}{\hbar} \left(\frac{P^2}{2m} + V(X)\right)\right]^N \approx \left[\exp\left(-\frac{i\varepsilon}{2m\hbar} P^2\right) \exp\left(-\frac{i\varepsilon}{\hbar} V(X)\right)\right]^N$$

17.1.1 Configuration Space Path Integral

Configuration space describes **where a particle is**. This is suited for Lagrangian mechanics.

We use the identity $I = \int |x\rangle \langle x| dx$. E.g. for $N = 3$,

$$U(x, x_0, t) = \int \langle x_3 | \exp\left(-\frac{i\varepsilon}{2m\hbar} P^2\right) \exp\left(-\frac{i\varepsilon}{\hbar} V(X)\right) | x_2 \rangle \times$$

$$\langle x_2 | \exp\left(-\frac{i\varepsilon}{2m\hbar} P^2\right) \exp\left(-\frac{i\varepsilon}{\hbar} V(X)\right) | x_1 \rangle \times$$

$$\langle x_1 | \exp\left(-\frac{i\varepsilon}{2m\hbar} P^2\right) \exp\left(-\frac{i\varepsilon}{\hbar} V(X)\right) | x_0 \rangle dx_1 dx_2$$

Using $\langle x|p\rangle = e^{ipx/\hbar}$ as a plane wave, we know that:

$$\begin{aligned}
\langle x_n|\exp\left(-\frac{i\varepsilon}{2m\hbar}P^2\right)\exp\left(-\frac{i\varepsilon}{\hbar}V(X)\right)|x_{n-1}\rangle &= \langle x_n|\exp\left(-\frac{i\varepsilon}{2m\hbar}P^2\right)|x_{n-1}\rangle \exp\left(-\frac{i\varepsilon}{\hbar}V(x_{n-1})\right) \\
&= \int_{-\infty}^{+\infty} \frac{1}{2\pi\hbar} \langle x_n|\exp\left(-\frac{i\varepsilon}{2m\hbar}P^2\right)|p\rangle \langle p|x_{n-1}\rangle dp \cdot \exp\left(-\frac{i\varepsilon}{\hbar}V(x_{n-1})\right) \\
&= \int_{-\infty}^{+\infty} \frac{1}{2\pi\hbar} \exp\left(\frac{i}{\hbar}p(x_n - x_{n-1}) - \frac{i\varepsilon}{2m\hbar}p^2\right) dp \cdot \exp\left(-\frac{i\varepsilon}{\hbar}V(x_{n-1})\right) \\
&= \frac{1}{2\pi\hbar} \sqrt{\frac{2m\pi\hbar}{i\varepsilon}} \exp\left(\frac{(-i(x_n - x_{n-1})/\hbar)^2}{2i\varepsilon/m\hbar}\right) \cdot \exp\left(-\frac{i\varepsilon}{\hbar}V(x_{n-1})\right) \\
&= \sqrt{\frac{m}{2\pi i\varepsilon\hbar}} \exp\left[\frac{im(x_n - x_{n-1})^2}{2\hbar\varepsilon}\right] \cdot \exp\left(-\frac{i\varepsilon}{\hbar}V(x_{n-1})\right)
\end{aligned}$$

Overall, our propagator for general N is

$$U(x, x_0, t) = \left(\frac{m}{2\pi i\varepsilon\hbar}\right)^{3/2} \int \exp\left(\sum_{n=1}^N \frac{im(x_n - x_{n-1})^2}{2\hbar\varepsilon} - \frac{i\varepsilon}{\hbar}V(x_{n-1})\right) dx_1 dx_2$$

Note that the integral is just a discretized product of $e^{iS/\hbar}$:

$$\begin{aligned}
\exp\left(\sum_{n=1}^N \frac{im(x_n - x_{n-1})^2}{2\hbar\varepsilon} - \frac{i\varepsilon}{\hbar}V(x_{n-1})\right) &= \exp\left(\frac{i}{\hbar}\varepsilon \sum_{n=1}^N \left(\frac{m(x_n - x_{n-1})^2}{2\varepsilon^2} - V(x_{n-1})\right)\right) \\
&= \exp\left(\frac{i}{\hbar}\varepsilon \sum_{n=1}^N (T - V)\right)
\end{aligned}$$

With the continuum notation involving $\mathcal{D}x$, the propagator and configuration-space path integral is:

$$\begin{aligned}
U(x, x', t) &= \int [\mathcal{D}x] \exp\left[\frac{i}{\hbar} \int_0^t \mathcal{L}(x, \dot{x}) dt\right] \\
\int [\mathcal{D}x] &= \lim_{N \rightarrow \infty} \left(\frac{m}{2\pi i\hbar\varepsilon}\right)^{3/2} \int dx_1 dx_2 \cdots dx_{N-1}
\end{aligned}$$

17.1.2 Phase Space Path Integral

Phase space describes **where a particle is and how it's moving**. This is suited for Hamiltonian mechanics.

We use the alternate resolution of identity $I = \frac{1}{2\pi\hbar} \int |p\rangle \langle p| dp$. E.g. for $N = 3$,

$$\begin{aligned}
U(x_3, x_0, t) &= \int [\mathcal{D}p \mathcal{D}x] \langle x_3|\exp\left(-\frac{i\varepsilon}{2m\hbar}P^2\right)|p_3\rangle \times \\
&\quad \langle p_3|\exp\left(-\frac{i\varepsilon}{\hbar}V(X)\right)|x_2\rangle \langle x_2|\exp\left(-\frac{i\varepsilon}{2m\hbar}P^2\right)|p_2\rangle \times \\
&\quad \langle p_2|\exp\left(-\frac{i\varepsilon}{\hbar}V(X)\right)|x_1\rangle \langle x_1|\exp\left(-\frac{i\varepsilon}{2m\hbar}P^2\right)|p_1\rangle \times \\
&\quad \langle p_1|\exp\left(-\frac{i\varepsilon}{\hbar}V(X)\right)|x_0\rangle
\end{aligned}$$

here the continuum notation just sets the appropriate factors.

$$\int [\mathcal{D}p\mathcal{D}x] \equiv \underbrace{\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty}}_{2N+1 \text{ times}} \prod_{n=1}^N \frac{dp_n}{2\pi\hbar} \prod_{n=1}^{N-1} dx_n$$

Thus for general N , the propagator and phase-space path integral is

$$\boxed{\begin{aligned} U(x, x', t) &= \int [\mathcal{D}p\mathcal{D}x] \exp \left[\sum_{n=1}^N \frac{-i\varepsilon}{2m\hbar} p_n^2 + \frac{i}{\hbar} p_n (x_n - x_{n-1}) - \frac{i\varepsilon}{\hbar} V(x_{n-1}) \right] \\ U(x, x', t) &= \int [\mathcal{D}p\mathcal{D}x] \exp \left[\frac{i}{\hbar} \int_0^t p \dot{x} - \mathcal{H}(x, p) dt \right] \text{ as } N \rightarrow \infty, \varepsilon \rightarrow 0 \end{aligned}}$$

17.2 Landau Levels

Landau levels show that a particle in a magnetic field has quantized energy levels, and moreover that each level is highly degenerate (accommodates multiple states).

Consider a particle of mass μ and charge q in the x - y plane with B along $\hat{z}$. The vector potential is $A = \frac{B}{2}(-y\hat{i} + x\hat{j})$ and the Hamiltonian with Coulomb gauge $\phi = 0$ is

$$\begin{aligned} H &= \frac{1}{2\mu} (\mathbf{p} - q\mathbf{A}/c)^2 \\ &= \frac{1}{2\mu} [P_x + qYB/2c]^2 + \frac{1}{2\mu} [P_y - qXB/2c]^2 \end{aligned}$$

We can use the following canonical variables that obey $[Q, P] = i\hbar$:

$$Q = \frac{(cP_x + qYB/2)}{qB}, \quad P = (P_y - qXB/2c)$$

The Hamiltonian simplifies to a simple harmonic oscillator. We've reduced a 2D problem to a 1D one, but this should be the first sign that there's degeneracy involved in this problem.

$$\begin{aligned} H &= \frac{1}{2\mu} P^2 + \frac{q^2 B^2}{2\mu c^2} Q^2 \equiv \frac{1}{2\mu} P^2 + \frac{1}{2} \mu \omega_0^2 Q^2 \\ \omega_0 &= \frac{qB}{\mu c} \end{aligned}$$

We can even define creation/annihilation operators $a^\dagger$ and a such that

$$\begin{aligned} a &= \sqrt{\frac{\mu\omega_0}{2\hbar}} Q + i\sqrt{\frac{1}{2\mu\omega_0\hbar}} P \\ H &= \left(a^\dagger a + \frac{1}{2} \right) \hbar\omega_0 \end{aligned}$$

To see the degeneracy in each energy level, consider the ground state (Lowest Landau Level, LLL). This obeys $a|0\rangle = |0\rangle$.

$$\left[\frac{\partial}{\partial \bar{z}} + \frac{qB}{4\hbar c} \right] \psi_0(z, \bar{z}) = 0, \quad \begin{cases} z = x + iy \\ \bar{z} = x - iy \end{cases}$$

A reasonable solution is

$$\psi_0(z, \bar{z}) = \exp \left[-\frac{qB}{4\hbar c} z \bar{z} \right] u(z, \bar{z})$$

If we plug this in, we find that $\frac{\partial}{\partial \bar{z}} u(z, \bar{z}) = 0$, which if we recall from Complex Analysis, is just the requirement for an analytic function (from the Cauchy-Riemann equations). So any analytic function in u can form a ground state, thus the ground state is infinitely degenerate. We could guess a superposition of the linearly independent functions z^m .

$$\psi_0 = \sum_m c_m \psi_{0,m}, \quad \psi_{0,m} = z^m \exp \left[-\frac{qB}{4\hbar c} z \bar{z} \right]$$

For a state with all LLLs filled (1 electron in each LLL), the ground state is

$$\psi = z_1^0 z_2^1 \cdots z_N^{N-1} \exp \left[-\frac{qB}{4\hbar c} \sum_i \bar{z}_i z_i \right] \equiv u_p \exp \left[-\frac{qB}{4\hbar c} \sum_i \bar{z}_i z_i \right]$$

After antisymmetrization for fermions, the factor in front is

$$u_p = \prod_{i=1}^N \prod_{j=1}^{i-1} (z_i - z_j)$$

For states that aren't fully filled – say νN electrons instead of N , where $\nu \leq 1$ – then there's more than 1 ground state. Laughlin's proposal is

$$u_p = \prod_{i=1}^{\nu N} \prod_{j=1}^{i-1} (z_i - z_j)^{1/\nu}$$

This will be useful later in the Quantum Hall Effect.

p.s. this was the theme of the 2015 APhO. Interesting problem that doesn't require a ton of quantum mechanics: https://apho.olimpicos.net/pdf/APhO_2015_Q1.pdf

17.3 The Berry Phase

17.3.1 Accumulation of Geometric Phase

Say a particle is in a box with length L , expanding slowly according to $L(t)$. If a particle is initially in $|n(L(0))\rangle$ at $t = 0$, then by the adiabatic theorem, it'll be in $|n(L(t))\rangle$ after time t given slow evolution.

As the solution to the Schrodinger equation, we might guess that the state picks up a phase

$$|\psi(t)\rangle = \exp\left(-\frac{i}{\hbar} \int_0^t E_n(t') dt'\right) |n(t)\rangle$$

$$H(t) |n(t)\rangle = E_n(t) |n(t)\rangle$$

There's a slight inadequacy in this description. If this state is to satisfy the Schrodinger equation, we find that $\frac{d}{dt} |n(t)\rangle = 0$, which conflicts with the overall change $|n(L(0))\rangle \rightarrow |n(L(t))\rangle$. We may modify the state with a factor $c(t)$.

$$|\psi(t)\rangle = c(t) \exp\left(-\frac{i}{\hbar} \int_0^t E_n(t') dt'\right) |n(t)\rangle$$

Applying the Schrodinger equation, we find:

$$\left[i\hbar \frac{\partial}{\partial t} - H(t) \right] |\psi(t)\rangle = 0$$

$$\dot{c}(t) = -c(t) \langle n(t) | \frac{d}{dt} |n(t)\rangle$$

$$c(t) = c(0) \exp\left(-\int_0^t \langle n(t') | \frac{d}{dt'} |n(t')\rangle dt'\right)$$

Or $c(t) = c(0)e^{i\gamma}$ with γ as the **Berry phase**.

$$\gamma = i \int_0^t \langle n(t') | \frac{d}{dt'} |n(t')\rangle dt'$$

17.3.2 Vector and Scalar Berry Potentials

That's all well and good. But we've previously seen that phases don't particularly matter for states. What makes the Berry phase special is that **it has a physical effect**, similar to **A** in the Aharonov-Bohm effect.

Consider the "box" to be the potential of a nucleus. R is the coordinate of the nucleus and r is the coordinate of the electron. If the nucleus moves slowly, the electron stays in the same instantaneous eigenstate.

Note that

$$\begin{aligned} e^{i\gamma} &= \exp\left(-\int_0^t \langle n(t') | \frac{d}{dt'} |n(t')\rangle dt'\right) = \exp\left(\frac{i}{\hbar} \int_0^t \langle n(t') | \frac{d}{dt'} |n(t')\rangle dt'\right) \\ &= \exp\left(\frac{i}{\hbar} \int_0^t i\hbar \langle n(R(t')) | \frac{d}{dR} |n(R(t'))\rangle \frac{dR}{dt'} dt'\right) \\ &= \exp\left(\frac{i}{\hbar} \int_0^t A^n(R) \frac{dR}{dt'} dt'\right) \end{aligned}$$

where $A^n(R) = i\hbar \langle n(R) | \frac{d}{dR} |n(R)\rangle$ is the **vector Berry potential**. We've recast the phase γ as a coupling between the vector Berry potential and the velocity.

By constructing a path integral, we also see a **scalar Berry potential**. Use the resolution of identity $I = \int \sum_n |R, n(R)\rangle \langle R, n(R)| dR$. For an adiabatic process, we assume the energy level stays at the same n , thus

$$I \approx \int |R, n(R)\rangle \langle R, n(R)| dR$$

Let H_e be the instantaneous electronic Hamiltonian such that $H_e(R, r, p) |R, n(R)\rangle = E_n(R) |R, n(R)\rangle$. If we construct the path integral (see prev. section), we'll get factors of

$$\begin{aligned} & \langle n(R(t+\varepsilon)), R(t+\varepsilon) | \exp \left[-\frac{i\varepsilon}{\hbar} H(R, P) \right] \exp \left[-\frac{i\varepsilon}{\hbar} H_e(R, r, p) \right] | n(R(t)), R(t) \rangle \\ &= \langle n(R(t+\varepsilon)), R(t+\varepsilon) | \exp \left[-\frac{i\varepsilon}{\hbar} H(R, P) \right] | n(R(t)), R(t) \rangle \cdot \exp \left[-\frac{i\varepsilon}{\hbar} E_n(R) \right] \\ &= \sqrt{\frac{m}{2\pi\hbar i\varepsilon}} \exp \left[\frac{i\varepsilon}{\hbar} \left(\frac{m(R(t+\varepsilon) - R(t))^2}{2\varepsilon^2} - V(R) \right) \right] \cdot \exp \left[-\frac{i\varepsilon}{\hbar} E_n(R) \right] \cdot \langle n(R(t+\varepsilon)) | n(R(t)) \rangle \end{aligned}$$

(See Shankar p. 597 for details) By Taylor-expanding the difference $R(t+\varepsilon) - R(t) \equiv R' - R$, we can reconstruct the following Berry potentials and Hamiltonian. $|\partial n\rangle$ means $\frac{d}{dR'} |n\rangle$.

$$\begin{aligned} H &= \frac{1}{2m} (P - A^n)^2 + \Phi^n \\ A^n &= i\hbar \langle n | \partial n \rangle \\ \Phi^n &= \frac{\hbar^2}{2m} [\langle \partial n | \partial n \rangle - \langle \partial n | n \rangle \langle n | \partial n \rangle] \end{aligned}$$

17.3.3 The Magnetic Monopole Argument

Consider a sphere of radius $R = 1$ surrounding a magnetic monopole of charge g . The magnetic field, similar to the electric field with point source, falls off as $\mathbf{B} = \frac{g}{r^2} \hat{r}$. Yet Maxwell's equation $\nabla \cdot \mathbf{B} = 0$ is not satisfied since in spherical coordinates,

$$\begin{aligned} \nabla \cdot \mathbf{B} &= \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \cdot \frac{g}{r^2} \hat{r} \right] \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} [g\hat{r}] \end{aligned}$$

This is 0 for $r > 0$, but at $r = 0$ we have a singularity and the **distributional derivative** is $4\pi g \delta^3(\mathbf{r})$. Thus it's impossible to define a singularity-free $\mathbf{A}$ in the entire parameter space such that $\mathbf{B} = \nabla \times \mathbf{A}$.

We could define $\mathbf{A}$ such that it yields the correct $\mathbf{B}$ everywhere except $r = 0$.

$$\mathbf{A} = \frac{g(1 - \cos \theta)}{r \sin \theta} \hat{\phi}$$

Note the singularity along $\hat{z}$ from the origin to $-\infty$. This is called a “Dirac string” and the endpoint is the magnetic monopole. By Stokes' theorem, the line integral around the potential is the total flux of $\nabla \times \mathbf{A} = \mathbf{B}$, thus $4\pi g$.

By similar reasoning to the Aharonov-Bohm effect, we see that the particle picks up a phase of $e^{iq\Phi/\hbar c} = e^{4\pi i q g / \hbar c}$.

- In QM, $\mathbf{A}$ is a physical entity (A-B effect shows this clearly).
- The location of the string across $\hat{z}$ was arbitrary, thus the phase must be unobservable.
- Then $e^{4\pi i q g / \hbar c} = 1$, or

$$q = \frac{\hbar c}{2g} n, \quad n \in \mathbb{Z}$$

The existence of even a single magnetic monopole forces all charges to be multiples of $\hbar c / 2g$. This explains, e.g. why the proton and electron happen to have the same magnitude of charge. It's an attractive argument, but whether it's true is moot.

Magnetic monopoles haven't been observed. If they exist, they must exist in very extreme conditions – the early universe, maybe, or the inside of black holes.

17.4 Imaginary-Time Path Integrals

17.4.1 Formalism

What if we tried to remove the i in path integrals? Set $\tau = it$, the imaginary time, so that

$$U(\tau) = \exp\left(-\frac{1}{\hbar} H \tau\right)$$

Note that $U(\tau) = \sum_n |n\rangle \langle n| e^{-E_n \tau / \hbar}$ where $H |n\rangle = E_n |n\rangle$. We've removed the $e^{-iE_n t / \hbar}$ factor in favor of $e^{-H_n \tau / \hbar}$. The immediate benefit is that time & space are on equal footing now – no arbitrary imaginary factor for time. Later, another benefit will be the similarity to classical statistical mechanics.

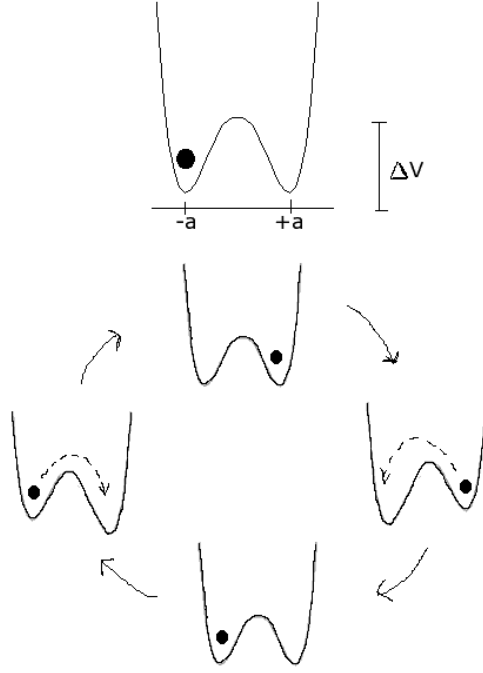
Now consider writing a path integral:

$$\begin{aligned} \langle x | U(\tau) | x' \rangle &= \int [\mathcal{D}x] \exp\left[-\frac{1}{\hbar} \int_0^\tau \mathcal{L}_E(x, \dot{x}) d\tau\right] \\ \int [\mathcal{D}x] &= \lim_{N \rightarrow \infty} \sqrt{\frac{m}{2\pi\hbar\varepsilon}} \prod_{i=0}^{N-1} \sqrt{\frac{m}{2\pi\hbar\varepsilon}} dx_i \\ \mathcal{L}_E &= \frac{m}{2} \left(\frac{dx}{d\tau}\right)^2 + V(x) \end{aligned}$$

$\mathcal{L}_E$ is the **Euclidean Lagrangian**. Note that the potential is reversed. This is exploited in the next section.

17.4.2 Spontaneous Symmetry Breaking

Consider a double-well potential, e.g. $V(x) = A^2(x^2 - a^2)^2$.



Double Well Potential (https://en.wikipedia.org/wiki/Double-well_potential)

Classically, the minima are at $\pm a$. Say the ground states are $|\pm a\rangle$. Both are likely. In the quantum version, tunneling between $|-a\rangle$ and $|a\rangle$ through the middle “bump” is a possibility. This has a probability density of

$$\langle a|U(\tau)|-a\rangle = \langle a|\exp\left[-\frac{1}{\hbar}H\tau\right]|-a\rangle \sim \exp\left[-\frac{1}{\hbar}S_{\text{cl}}\right]$$

We seek a classical path. In the normal real-time problem, this would be difficult to find. But in the imaginary-time formalism, the potential is reversed, so finding a classical path is easier. Imagine the above diagram reversed; the particle at the top of the middle bump can just fall into $|-a\rangle$ or $|a\rangle$.

The energy $E = \frac{m}{2} \left(\frac{dx}{d\tau}\right)^2 - V(x)$ is conserved. Assume just enough energy to tunnel, $E = 0$. The trajectory is found by solving

$$\int_{x_1}^{x_2} \sqrt{\frac{m}{2(E+V)}} dx = \int_{t_1}^{t_2} d\tau \quad \longrightarrow \quad x(\tau) = a \tanh\left[\sqrt{\frac{2}{m}} A\tau\right]$$

With $E = 0$, the Euclidean action and tunneling amplitude are

$$S_{\text{cl}} = \int T + V d\tau = \int 2T d\tau = \int_{-a}^a p(x) \frac{dx}{d\tau} d\tau = \int_{-a}^a \sqrt{2mV(x)} dx$$

$$\langle a|U|-a\rangle \approx \exp\left[-\frac{1}{\hbar} \int_{-a}^a \sqrt{2mV(x)} dx\right]$$

The double-well potential is useful to discuss other concepts, like spontaneous symmetry-breaking.

- Classically, the particle breaks symmetry by being in 1 of the 2 wells.
- In QM, tunneling allows us to form a superposition of $|a\rangle$ and $|-a\rangle$, maintaining the symmetry.

17.5 Quantum Statistical Mechanics

17.5.1 The Classical Limit

This is largely an application of imaginary-time QM. We use the partition function $Z = \sum_n e^{-\beta E_n} = \text{Tr}(e^{-\beta H})$ in the E -basis. The trace is invariant under a unitary change of basis, thus switch to the x -basis.

$$Z = \int_{-\infty}^{\infty} \langle x | e^{-\beta H} | x \rangle dx = \int_{-\infty}^{\infty} \langle x | \exp \left[-\frac{1}{\hbar} \beta \hbar H \right] | x \rangle dx$$

$$\langle x | \exp \left[-\frac{1}{\hbar} \beta \hbar H \right] | x \rangle = U(x, x, \beta \hbar)$$

We could interpret Z as the sum over the amplitudes of closed paths from x to x in imaginary time $\tau = \beta \hbar$.

The classical limit: consider a particle of mass m in potential $V(x)$.

$$Z(\beta) = \int dx \underbrace{\int_x^x [\mathcal{D}x] \exp \left[-\frac{1}{\hbar} \int_0^{\beta \hbar} \frac{m}{2} \left(\frac{dx}{d\tau} \right)^2 + V(x(\tau)) d\tau \right]}_{\text{Amplitude to go } x \rightarrow x \text{ in time } \tau = \beta \hbar}$$

We consider the limit $\beta \hbar \rightarrow 0$ at high temperatures and/or the classical limit $\hbar \rightarrow 0$. Thus the particle doesn't have much "time" to wander.

- If traveling a distance Δx , we expect a kinetic energy $\sim m(\Delta x / \beta \hbar)^2$ and the path is suppressed by an exponential factor

$$\sim \exp \left[-\frac{1}{\hbar} m \left(\frac{\Delta x}{\beta \hbar} \right)^2 \cdot \beta \hbar \right]$$

- To be confidently suppressed, we find $\Delta x \sim \sqrt{\frac{\beta}{m}} \hbar$, the **thermal wavelength**.
- If the potential doesn't vary much along the thermal wavelength, we can approximate it as a constant $V(x_0)$ near the x_0 in question.

$$Z(\beta) = \int dx e^{-\beta V(x)} \int_x^x [\mathcal{D}x] \exp \left[-\frac{1}{\hbar} \int_0^{\beta \hbar} \frac{m}{2} \left(\frac{dx}{d\tau} \right)^2 d\tau \right]$$

$$= \int dx e^{-\beta V(x)} \sqrt{\frac{m}{2\pi \hbar \cdot \beta \hbar}}$$

17.5.2 Coupled Oscillators

For a more specific problem, consider $N + 1$ nodes with degree of freedom x_i . But x_0, x_N are fixed. The partition function is

$$Z = \int_{-\infty}^{\infty} \exp \left[-\frac{1}{\beta} E(x_0, \dots, x_N) \right] dx_1 dx_2 \cdots dx_{N-1}$$

Say E has the following form:

$$E = \sum_{n=1}^{N-1} \left[K_1 (x_n - x_{n-1})^2 + K_2 x_n^2 \right]$$

The first term is like a spring coupling between neighboring nodes. The second term prevents nodes from drifting too far past $x_i = 0$.

Recall the path integral for a quantum oscillator:

$$U(x_N, x_0, t) = \int_{-\infty}^{\infty} \exp \left[-\frac{1}{\hbar} \sum_{n=1}^{N-1} \varepsilon \left(\frac{m(x_n - x_{n-1})^2}{2\varepsilon^2} + \frac{m\omega^2}{2} x_n^2 \right) \right] dx_1 dx_2 \cdots dx_{N-1}$$

We see a rough correspondence.

- Action in Feynman path integral is like E in classical oscillator model.
- Almost same structure with

$$\beta K_1 = \frac{m}{2\hbar\varepsilon}, \quad \beta K_2 = \frac{m\omega^2\varepsilon}{2\hbar}$$

- Role of $\hbar$ is played by T (up to factor since $\beta = kT$)

17.5.3 The Ising Model

(A brief description) The previous similarity leads us to try to apply the same approach to the Ising model.

Consider an Ising model in 1D with Ising spin $s = \pm 1$.

$$Z = \sum_{s_i = \pm 1} \exp \left(\sum_{i=0}^{N-1} K (s_i s_{i+1} - 1) \right), \quad K > 0$$

Large weight when neighboring cells agree, small weight when they disagree. Some features we can find classically:

1. Let $t_i = s_i s_{i+1}$ so that

$$Z = 2 \sum_{t_i} \exp \left[\sum_{i=0}^{N-1} K (t_i - 1) \right] = 2 \sum_{t_i} \prod_i e^{K(t_i - 1)}$$

The factor of 2 is for the 2 ways to form $t_i = \pm 1$. With polynomial expansion, the partition function over all states becomes

$$Z = 2(1 + e^{-2K})^N$$

2. We can define the correlation function to measure if two cells are in the same state:

$$\langle s_i s_j \rangle = \frac{\sum_{s_k} s_i s_j \exp [\sum_k K (s_k s_{k+1} - 1)]}{Z}$$

Since $s_i^2 = 1$, $s_i s_j = s_i s_{i+1} s_{i+1} s_{i+2} \cdots s_{j-1} s_j = t_i t_{i+1} \cdots t_{j-1}$. Thus $\langle s_i s_j \rangle = \langle t_i \rangle \langle t_{i+1} \rangle \cdots \langle t_{j-1} \rangle$. For each $\langle t \rangle$, we see that

$$\langle t \rangle = \frac{2 \cdot (e^{0K} - e^{-2K})}{2 \cdot (1 + e^{-2K})} = \tanh(K)$$

Thus $\langle s_i s_j \rangle = (\tanh K)^{j-i}$ and the correlation only depends on the difference in coordinates.

3. The correlation length ξ is defined as

$$\lim_{j-i \rightarrow \infty} \langle s_i s_j \rangle \rightarrow e^{-(j-i)\xi}$$

Here we have $\xi = -1/\ln(\tanh K)$

Now we try to recast the problem in QM. Specifically, a fermionic (spin- $\frac{1}{2}$) model, to account for the 2 choices in spin.

Rewrite Z as the product of 2×2 matrices T , called the **transfer matrix**.

$$Z = \sum_{s_i} \prod_i e^{K(s_i s_{i+1} - 1)} \rightarrow \sum_{s_i} T_{s_N s_{N-1}} \cdots T_{s_2 s_1} T_{s_1 s_0}$$

We use $T = I + e^{-2K} \sigma_1$.

- If s_0, s_N are fixed, $Z = \langle s_N | T^N | s_0 \rangle$.
- If s_0, s_N are not fixed, $Z = \sum_{s_0, s_N} \langle s_N | T^N | s_0 \rangle$

We could also rewrite T using an auxiliary variable K^* , with

$$\begin{aligned} e^{K^* \sigma_1} &= \cosh K^* + \sinh K^* \sigma_1 \\ &= \cosh K^* (1 + \tanh K^* \sigma_1) \end{aligned}$$

Choose $\tanh K^* = e^{-2K}$, and then up to a factor of $\cosh K^*$, we can write $T = e^{K^* \sigma_1}$.

Comparing to $e^{-H\tau/\hbar}$, we see that T can be interpreted as something like an evolution operator for 1 unit $\hbar$ in the imaginary time direction with Hamiltonian $H = -K^* \sigma_1$.

- The Ising model has further interesting applications to correlation and spontaneous symmetry breaking, clustering, etc.

17.6 Path Integrals with Spin

17.6.1 A Brief Introduction

Previously, the energy levels in the SHO were evenly spaced with $E_n = n\hbar\omega$. This allowed us to interpret the n -th energy level as containing n quanta of particles.

This interpretation works for bosons, which can clump. But fermions obey the Pauli Exclusion Principle, so a different approach is needed.

We guess a similar Hamiltonian $H_0 = \Psi^\dagger \Psi \Omega_0$. The anticommutation relations must be satisfied, since no two particles can occupy the same state.

$$\begin{aligned} \{\Psi^\dagger, \Psi\} &= 1 \\ \{\Psi, \Psi\} &= 0 \longrightarrow \Psi^2 = (\Psi^\dagger)^2 = 0 \end{aligned}$$

1. The number operator is $N = \Psi^\dagger \Psi$. We note that $N^2 = \Psi^\dagger \Psi \Psi^\dagger \Psi = \Psi^\dagger (1 - \Psi^\dagger \Psi) \Psi = N$. Thus N can only have eigenvalues 0 and 1, which makes sense – max. 1 particle in each level.
2. $\Psi^\dagger$ and Ψ are shown to be the raising/lowering operators.

$$\Psi^\dagger |0\rangle = |1\rangle$$

$$\Psi |1\rangle = |0\rangle$$

3. For future reference, we use $H = H_0 - \mu N$, offset by the **chemical potential** μ . Then

$$H = (\Omega_0 - \mu) \Psi^\dagger \Psi$$

4. The **grand partition function** (like partition function with different quanta considered) is

$$Z = \text{Tr} \left[e^{-\beta(H_0 - \mu N)} \right]$$

In the N basis, $Z = 1 + e^{-\beta(\Omega_0 - \mu)}$.

17.6.2 Grassman Variables and Berezin Integration

If we try to find a **coherent state** (eigenstate of creation/destruction operators), we run into a problem. E.g. $\Psi |\psi\rangle = \psi |\psi\rangle$ but $\Psi^2 |\psi\rangle = \psi^2 |\psi\rangle = 0$. So ψ is nonzero, but ψ^2 is 0.

We're forced to introduce some unintuitive concepts to fully pursue fermionic oscillators. Call ψ a **Grassmann variable**. Properties:

1. Anticommute with each other
2. Anticommute with fermionic creation operator $\Psi^\dagger$
3. Anticommute with fermionic destruction operator Ψ

The coherent states are $|\psi\rangle = |0\rangle - \psi |1\rangle$ and $\langle\bar{\psi}| = \langle 0| - \langle 1| \bar{\psi}$. The inner product is $\langle\bar{\psi}|\psi\rangle = e^{\bar{\psi}\psi}$.

- Due to $\psi^2 = 0$, all functions of Grassmann variables are of the form $F(\psi) = F_0 + F_1 \psi$.
- Taylor expansions also cut off at the ψ – there's no ψ^2 term.
- Integrals over Grassmann variables are like derivatives:

$$\int \psi d\psi = 1$$

$$\int 1 d\psi = 0$$

For path integrals, the following path integrals is useful:

$$I = \int |\psi\rangle \langle\bar{\psi}| e^{-\bar{\psi}\psi} d\bar{\psi} d\psi$$

For bosonic operators (even number of Fermi operators),

$$\text{Tr}(\Omega) = \int \langle -\bar{\psi} | \Omega | \psi \rangle e^{-\bar{\psi}\psi} d\bar{\psi} d\psi$$

17.6.3 The Fermionic Path Integral

Start off with $Z = \text{Tr} e^{-\beta H}$. Note that

$$e^{-\beta H} = \lim_{N \rightarrow \infty} \exp \left[-\frac{\beta}{N} H \right]^N = \underbrace{(1 - \varepsilon H) \cdots (1 - \varepsilon H)}_{N \text{ times}}, \quad \varepsilon = \beta/N$$

Take the trace using Grassmann variables as possible states. Use the resolution of identity.

$$\begin{aligned} Z = \int & \langle -\bar{\psi}_0 | (1 - \varepsilon H) | \psi_{N-1} \rangle e^{-\bar{\psi}_{N-1} \psi_{N-1}} \langle \bar{\psi}_{N-1} | (1 - \varepsilon H) | \psi_{N-2} \rangle e^{-\bar{\psi}_{N-2} \psi_{N-2}} \\ & \times \langle \bar{\psi}_{N-2} | \cdots \psi_1 \rangle e^{-\bar{\psi}_1 \psi_1} \langle \bar{\psi}_1 | (1 - \varepsilon H) | \psi_0 \rangle e^{-\bar{\psi}_0 \psi_0} \cdot \prod_{i=0}^{N-1} d\bar{\psi}_i d\psi_i \end{aligned}$$

1. Each factor in the integrand is

$$\begin{aligned} \langle \bar{\psi}_{i+1} | 1 - \varepsilon H(\Psi^\dagger, \Psi) | \psi_i \rangle &= \langle \bar{\psi}_{i+1} | 1 - \varepsilon H(\bar{\psi}_{i+1}, \psi_i) | \psi_i \rangle \\ &= \langle \bar{\psi}_{i+1} | \psi_i \rangle e^{-\varepsilon H(\bar{\psi}_{i+1}, \psi_i)} \\ &= e^{\bar{\psi}_{i+1} \psi_i} e^{-\varepsilon H(\bar{\psi}_{i+1}, \psi_i)}, \quad \varepsilon \rightarrow 0 \end{aligned}$$

2. Let $\langle -\bar{\psi}_0 | \equiv \langle \bar{\psi}_N |$. Now the path integral is

$$\begin{aligned} Z &= \int \prod_{i=0}^{N-1} e^{\bar{\psi}_{i+1} \psi_i} e^{-\varepsilon H(\bar{\psi}_{i+1}, \psi_i)} e^{-\bar{\psi}_i \psi_i} d\bar{\psi}_i d\psi_i \\ &= \int \prod_{i=0}^{N-1} \exp \left[\left(\frac{(\bar{\psi}_{i+1} - \bar{\psi}_i)}{\varepsilon} \psi_i - H(\bar{\psi}_{i+1}, \psi_i) \right) \varepsilon \right] d\bar{\psi}_i d\psi_i \end{aligned}$$

Caution is required since Grassmann variables aren't normal numbers, so there's no sense of $(\bar{\psi}_{i+1} - \bar{\psi}_i)$ being "large" or "small." But this is a useful starting-off point.